

Psyc 420 Introduction to Intelligence

Course transcript with lecture-scoped numbered paragraphs

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1. Defining Intelligence

At a glance

Opens the course by surveying the confused and contested state of defining intelligence, from strong identity claims (intelligence is X) to genuine scientific uncertainty. Introduces Spearman's general intelligence factor, g, and the overwhelming, still-growing body of evidence supporting it despite ongoing objections that it is merely a statistical artifact.

- 1.1 Welcome. We're going to explore the nature of **intelligence**, and ultimately, from a **cognitive science** perspective. You may not know what that means, but give me some time. Now, I'm going to start with a bit of a disclaimer that might say, well, what's the point of the course then? Because what I'm going to start with is, there's no consensus about what **intelligence** is. Contrary to the way people talk about it online, there's no clear consensus about it. Many researchers think that it's a mystery. They'll even use that word.
- 1.2 On the other hand, you have many researchers making strong identity claims about what **intelligence** is. **Intelligence** is X, **intelligence** is Y. That tells you the state that we're in. I should mention that this course will integrate very well with another course I've done for the Peterson Academy on **Intelligence**, Rationality, Wisdom, and Spirituality, and also another course called The Primacy of Beauty, which some of you were students of, and it's a great pleasure to see many of you back.
- 1.3 So, do we just stop then? Is that it? Well, no, we need to answer this question. It's really central to our understanding of who and what we are. It's becoming a pressing question with the advent, or at least it may be plausibly claimed to be the advent of artificial **general intelligence**, which we will talk about in this course. What is **intelligence**? What are we talking about? What do we mean by it? What am I going to do? I'm going to review some of the best theories and best empirical research and make the best argument I can for what I think **intelligence** is.
- 1.4 I'm going to spend a lot of time on formulating the problem today. As you will see from lecture three, formulating the problem well is a big part of what it is to be an intelligent agent. What do I mean in terms of this problem formulation by giving you my best argument? What do I mean by my best argument? Well, I'm going to give you a strong plausibility argument and a cognitive scientific argument.
- 1.5 First of all, what do I mean by plausibility? This is based on work that's going to come out this year on the nature of plausibility. There's

two meanings to plausibility. One is just a synonym for high probability. In that sense, I'm not talking about it. We have a good term for probability. It's called probability. I want to use plausibility, which is another term for another distinct meaning, which is when a claim makes good sense, it should be taken seriously. It's convinced me that I understand it well, and it seems like there could be truth there. It's reasonable to reflect upon it.

- 1.6 I want to talk about what I think plausibility is, and what kind of argument you make in order to make a plausible argument. Probability is our best estimation of where we will find truth. Plausibility is our best estimation of where we will find probability. That's how we're doing it. Now, one of the first books to really talk about this explicitly was Plausible Reasoning by Rescher from 1976. He made a very good point that one factor of plausibility is, we want convergence from multiple independent sources.
- 1.7 This is why you, or even infants, prefer multisensory information. You want convergence. If you can only see something, you're not very sure. It might be real, but it might be an illusion. But if you can see it and touch it, oh, right? If you can see it and touch it and hear it, then it's very probably real. And even from a young child, you're making use of that. Now, why does that make sense? Because if you have just one channel, the chance that your subjective bias is resulting in your conclusion, your model, quite high. But the chance that all of these share the same bias, much lower.
- 1.8 What convergence does is it gives you what Rescher calls trustworthiness. Trustworthiness is good reason to believe that the probability of the effect of subjective bias and distortion has been significantly reduced. Now, trustworthiness is not probability. You have to establish that empirically. And trustworthiness is certainly not a certainty. But trustworthiness is important. And that's why you see children, even very young children, using it. So you have convergence for trustworthiness. You're going to see me making these kinds of convergence moves, trustworthiness.
- 1.9 Now, if you take a look at the work of Kyle and other people from the psychology of explanation and other people from the philosophy of science, work on understanding, you get the fact that we actually like this, but we like something else, too, and we like to have the two together. We also like, and scientists will often refer to their theories this way, we want something that has elegance. We want our model to go into many different areas, find a formula, and solve problems for us.

- 1.10** This is why we like things like force equals mass times acceleration. We can use it to explain how a blue whale is floating. We can use it to explain the movement of the Earth around the sun. We can use it to explain why a balloon is floating. We can use it to explain what happens if you crash into a brick wall with your car and so on and so on and so forth. We can go into all of these on a superficial examination, disparate areas, and we can find and formulate and solve our problems there. It's a very powerful theory, which, of course, is why we like elegance. Elegance makes our theories powerful. It gives us a lot of explanatory power.
- 1.11** One of the ways we judge reality is, how much explanatory power does it have? We think this is a very real thing, $E = MC^2$, because we think it explains so much, and it gives us great power. With this, we can take a paperclip's worth of matter and we can smash a city to the ground. That's impressive. So elegance gives us power, and power is one of the things we also use to judge the realness.
- 1.12** Now, from Elijah Millgram's work on practical induction, he has a book by that name, and he has essays around this, he made a point that we actually want a balance between what he called these backward and our forward commitments. What if you just had a lot of convergence but very little elegance? What is that? That is to state something that people regard as very trustworthy but doesn't have much power.
- 1.13** Like, you know what? When people are hungry, they seek food. How many of you doubt that? How many of you are like, "Oh, wow, that's going to change my whole understanding." Does very little. It's not false, it's trivial. It's trivial. If you read through scientific articles, certain claims will be dismissed as not being false, but as being trivial. What about the other one? What about if I have very little convergence, but a lot of elegance? I've got a model that explains a lot, but you don't have a lot of trust in the model. These are our conspiracy theories.
- 1.14** One of my favorite I like to use is, if you just accept that the British Royal family are lizards from outer space, you can explain so much of their behavior. And it's like, yeah, that would actually explain a lot of their behavior. However, I don't trust your model. How did you come up with that? What you have there, this is triviality, this is far-fetched. This is far-fetched. What we want is, we want a balance between these.
- 1.15** Now, notice what plausibility is doing. It's helping us zero in and take seriously because it's helping us ignore triviality and ignore far-

fetches. OK, is that everything we need? No. There's important properties of the model. From Murphy and Medin, and ongoing, we've learned a lot about people's concepts, their internal models of the world, 1985. They pointed out something very important. We don't want a feature list. What do I mean by that?

- 1.16** It's interesting because people think that what they have in their head is a feature list. If you ask people what a bird is, they'll say, "Feathers, wings, a beak, flies." Oh, yeah, that's a bird. OK, so I'm going to put some feathers here, a couple of wings, a beak, gather it all together, throw it in the air. That's not a bird. It's a low-budget horror movie. OK, so what's missing? Well, what's missing is a structural, functional organization that shows how all the features hang together. So the bird operates as a causal whole, so you can predict and explain its behavior. It has to be coherent.
- 1.17** And that coherence allows you to rule out things that don't fit the structure and rule in things. Oh, yes, this belongs in, this fits in. And so you have coherence. And what that does is, it allows you to use the model predictively and in an explanatory fashion. What we want is, we're going to want—can we get a convergence of a lot of different theories and empirical findings to get to a coherent model of what **intelligence** is that has a lot of explanatory power that is balanced by a lot of convergence, trustworthiness?
- 1.18** That's the argument I'm going to make. It's going to be exactly that style of argument. I'm not just going to say strong argument. This is half of what I mean by a strong argument. I mean a plausibility argument. That's how we're going to do this. At times, it'll be like, what is he doing? Then you'll see, oh, he's doing this, and then, oh, and it keeps coming. Oh, right, the convergence. I see. Keep your eyes open for that. The next one is, I'm going to give you ultimately a cognitive scientific argument because I actually think that's one of the best ways of getting the most plausibility in a model of **intelligence**.
- 1.19** Now, what's **cognitive science**? **Cognitive science** is a discipline that took off in '70s, and, of course, is now a big deal. I'm currently the director of the **Cognitive Science** Program at the University of Toronto. Very proud of that. I've been working in that and for that program for, it was 1993. Wow, it's a long time. OK, anyways, I've been doing that. I'm very proud of it. Now, what is **cognitive science**? Well, you can open the calendar and you'll get the very dry, useless. Well, **cognitive science** is the interdisciplinary study of the mind, making use of blah, blah, blah, blah.

- 1.20** You go, "What? Who cares?" It sounds like you're going on a buffet. "I'll have a little bit of psychology and a little bit of philosophy." That's not what it is. At least that's not what I argue **cognitive science** is or should be. Instead, I want to go through something more carefully and make an argument for you. We're going to see some of this in this course, which is the point.
- 1.21** You have neuroscience. It's going to say what it thinks **intelligence** is, and what it's going to do is, it's going to study the brain. How's it going to study the brain? It's going to use EEG and fMRIs and some really new exciting technologies. We'll just put fMRI here because that's the one that people really like a lot, fMRI. Make sure, by the way, you're very careful about this. A quick aside, you take a study, you get published, you have a paper, you've written it up, you want to get it published, you submit it to a bunch of journals, you can't get it published.
- 1.22** You take the same study, you don't change it at all, and you put in some fMRI pictures, and then it gets published. People have an incorrect understanding of what these pictures are. They think they're snapshots of the brain. They are not like that. They are highly processed statistical averages of a lot of different measurements, etc. I'm not saying they're useless. I'm not saying that. They're important and they're relevant, but you have to be careful about them.
- 1.23** Second thing, there's been a lot of recent meta-analyses where people have gone back and looked at these studies, and many of them have too small of a number of participants in order to provide enough statistical power for the claims that are made. It doesn't mean they're false. It means we need to do more. Take the fMRI stuff as, well, it's not like the courtroom, where "I did it." It's not a confession. You have to take it as a part of information. It's a kind of evidence, but it has to be fit in with a lot of other stuff.
- 1.24** OK, let's go back. What do the neuroscientists talk about? Well, they talk about brain networks. They'll talk about two types. They'll talk about anatomical ones, and they'll talk about functional ones. They will talk about particular theories. One theory we're going to look at, of a kind of network, is called the P-FIT, the Parieto-Frontal Integration Theory. That's what you're doing. Oh, is that it? Well, no, because the problem is, we've got a whole other discipline that is talking about **intelligence** right now, and it's actually dominating the airwaves and the social media, and that's artificial **intelligence**.
- 1.25** If they're succeeding, they must be figuring out something about the nature of **intelligence**, we would think. Maybe, maybe not. Now,

artificial **intelligence**. John, I get that. What's this? Well, this is the project that people are gung-ho about right now. This is artificial **general intelligence**. They're trying to get an artificial **intelligence** that has what you have. We're going to come back to this again and again and again. You have **general intelligence**. You can solve a wide variety of problems in a wide variety of domains in a wide variety of ways.

- 1.26 Currently, we have no machine that is close to that, although the LLMs, the large language models like ChatGPT-4, seem to be getting closer to being able to be generally intelligent. Put a pin in that. We're going to come back to this idea. This is crucial. And notice, the fact that they are moving in this direction tells you that **general intelligence** is actually essential to understanding our kind of **intelligence**.
- 1.27 OK, and what do they do? Well, they don't study brains at all. They do learning algorithms, they generate simulations, they make models. And what do they say to us? Well, they'll say things like, maybe **intelligence** is features of deep learning. Some of you might have heard that with respect to ChatGPT. Deep learning is going on, and that's what makes it so intelligent, massive reinforcement. We're going to talk about that.
- 1.28 Notice if we just talked about this, we would be missing all of this. If we just talked about this, we would be missing all of this. Is that enough? No, no. There's a whole group of people that have been talking about **intelligence** for a much longer time than those other two. Those are the psychologists. The philosophers have been talking about it for a very long time, but we'll bring them in at another point. But you got psychology. Well, what is psychology doing? Well, it's studying behavior. Sometimes it's behavior and experience.
- 1.29 There's a lot of question about what psychology is actually studying. Of course, they run experiments and they do statistical analyses. Then what do they conclude? Well, they'll conclude things like, **intelligence** is working memory. WM, please remember, working memory. What's working memory? We'll get to that. But working memory isn't something you can just crack open the skull and say, "Oh, look, there's working memory." It's an abstract model. It's not the same thing as one of these things because it's not running on a computer. It's based on and derived from all the data that comes from these behavioral experiments.
- 1.30 Now, notice, this is like different countries. This is different levels of reality, different levels of analysis, different methods, and different

theoretical entities all talking about **intelligence**. We should try and get them, well, first of all, to converge. What if we don't try to get them to converge? What do we face? Well, we face the problem of equivocation because, if I ask you what **intelligence** is, you can give me three, four, five, not only different answers, but different kinds of answers. That means the same term has multiple different meanings.

- 1.31** The problem with that is human beings very readily fall into fallacious reasoning when they face equivocation, especially if they don't notice the equivocation. Here's one of my favorites. Nothing is better than long life and happiness. But a peanut butter and jelly sandwich is better than nothing. Ergo, a peanut butter jelly sandwich is better than long life and happiness. So, you should eat one and then kill yourself.
- 1.32** Now, it's like, wait, what's going on there? Well, what's going on is, I'm equivocating on the term nothing. I'm using two different meanings. Nothing from the set of things that makes life worth living, and then nothing from the set of things you eat. Very different meanings of nothing. But if I don't pay attention to the two meanings, I get a ridiculous fallacious argument. Equivocation is a big, big problem in the disciplines that study the mind.
- 1.33** One of the most equivocal words that's used in a lot of this, although a lot of people are also very careful to avoid the equivocation, so I want to be balanced here, is the word information. There's many different, a very technical mathematical meaning of information and an everyday meaning of information. We want to avoid equivocation. We want to avoid equivocation.
- 1.34** Is there anything else we should be worrying about here? Yeah, well, the problem with this is, it's really unlikely that these different levels of reality are not interacting with each other in significant ways. That's highly implausible. It's highly unlikely that they do not causally affect or constrain each other, condition each other, in important ways. Which means, if these disciplines remain isolated from each other, we are missing all of this important bottom-up, top-down, causation, constraint, conditioning. We're missing important information.
- 1.35** Of course, that means we're not getting the structural functional organization that holds all of these different kinds of features together. We're getting a very equivocal, very complete feature list. There's a good possibility that it's actually incoherent. We don't know how they all fit together. When you have that kind of incompleteness

and incoherence in your model, it's highly likely that it's false in very important ways.

- 1.36** You need to make make sure you are pursuing a way of integrating the work from these different disciplines. I call this synoptic. Synoptic means looking over the whole thing. Synoptic integration. You need synoptic integration. Now, what would be really good is if the synoptic integration got all three of these to converge in important ways, and then that convergent model was elegant and powerful, and fed back and found and solved problems in each one of these. That would be a very powerful thing to do.
- 1.37** That's exactly what we're going to be looking for. You're seeing how the plausibility and the cognitive scientific synoptic integration are feeding together? You may be wondering, well, where is philosophy in all of this? Philosophers have been talking about **intelligence** and its relationship to rationality and wisdom since, well, philosophy. The problem we have is philosophy has a bit of a bad rap. It's misrepresented.
- 1.38** Many people think it's just giving your opinion with a French accent and you smoke a cigarette or something like that. "It is not the being of consciousness. It is the consciousness of being. You understand, you understand," right? No, I mean, that's not a very good model of the role that philosophy plays in **cognitive science**. See, what philosophy does is philosophy is a practice of trying to come up with a shared conceptual vocabulary, a shared theoretical grammar, so that all of these disciplines can talk to each other in mutually transformative and insightful ways.
- 1.39** That's what we're going to be doing as well. Can we build a shared conceptual vocabulary? Can we build a shared theoretical grammar for how we put the concepts together that will allow all of these different disciplines to talk to each other? That will afford convergence. Hopefully, that synoptically integrated model with the new vocabulary and the new grammar will be elegantly powerful. If we could do all that, big if, big if, and I'm a little bit like, it's a big if. But if we do that, that would be a strong argument as to what **intelligence** is.
- 1.40** I'm not just going to make a strong argument. No, I've told you specifically how I'm going to make a strong argument because of what I mean by what a strong argument is. It's a plausible argument, and it's a cognitive scientific argument. Does that make sense? OK. That's exactly what we're going to be doing. Just to give you a quick overview of what we'll be doing in the course, we're going to look at

psychological theories, we're going to look at neuroscientific theories, we're going to look at these cognitive scientific theories that try to integrate those together.

- 1.41 When we do psychology, there's going to be two parts. There's going to be psychometrics. I'll tell you what all this means in a minute. There's going to be cognitive theories. There's two parts. We're going to go through the psychology, both parts, what they converge on, then neuroscience, what's going on between the psychology and the neuroscience. Then we go into the artificial **intelligence**, and then we try and draw it all together in the AI. We're going to go through all those levels.
- 1.42 Now, before we do that, first, I want to draw a couple of lessons from **cognitive science**. We've been doing **cognitive science** for a while, and we've learned some lessons from that about what we should be doing that are very relevant to talking about **intelligence**. One is, like all scientific theories, we want theories that generalize. Generalize means that I can apply. It's powerful. I can apply the same theory in many different domains. It applies generally. That makes it a very powerful theory. That's what science is after.
- 1.43 But we want a specific kind of power. We want it to generalize in a way that takes into account what's called multiple realizability. Multiple realizability. I will often just use MR. It doesn't mean Mister. It means multiple realizability. You're saying, well, why? Well, just think about what we've talked about. I want to do, I want to do **cognitive science** in the sense that I want to generate a theory that generalizes to human **intelligence**, obviously. What's human **intelligence**?
- 1.44 But are humans the only organisms that are intelligent? No. I need a theory of **intelligence** that should generalize also to chimps. It should generalize even to an octopus, and octopuses are like from outer space. They are weird. Their brain doesn't look the way ours does. They have a brain, but they have more neurons actually in their tentacles. It's like, what's it like to be an octopus? You see, they're highly intelligent. They can solve problems that even dogs can't solve, and they're really sneaky about it, too. I love octopus. Octopi, octopuses. I think it's octopuses, actually.
- 1.45 Now, what's the thing? Why am I saying this? Well, I can't get too specific about **intelligence** in the sense of saying it's a property of a specific human brain. Because maybe the chimp brain, and even more likely, maybe the octopus's brain doesn't have those structures, yet somehow they have very powerful **intelligence**. It's like, oh, wait, I

need a theory that's abstract enough that it will talk about humans and chimps and octopuses. It gets even, worse? No, even more demanding.

- 1.46** Because what about artificial **intelligence**? What's it running on? It's not running on anything like a brain at all. It's running on circuit boards using electricity, and yet, these LLMs are holding conversations with you. It's like, oh, I need a theory of **intelligence** that might not even—like, I want a theory of **intelligence** that isn't bound to talking about human brain structures because that won't allow me to understand how the LLMs and other AI is intelligent.
- 1.47** **Intelligence** is multiply realizable. It shows up in many different domains. You may think that's a really weird property. Well, no. Let's talk about this. We'll need this. There's a little bit of foreshadowing because this will also help us to understand, what do we mean by a computational model when we do the AI section? OK, so how many of you know what chess is? You all know what chess is. It's not some sort of weird thing.
- 1.48** You can tell when people don't know what something is. If you go in an archeological museum and they don't know what it is, they'll have the object there. They don't know what it is, so they'll say ceremonial object. That means they don't know what it is. Chess is not a ceremonial object. You know what it is. I'm playing chess with somebody. Let's say I'm playing chess with Andrew. We're playing the game, and I'm starting to lose. I pick up his queen and I snap it and throw it away. I go, "Aha." He goes, "No problem." He takes out a coin.
- 1.49** In Canada, we have, and we still do, we used to have pictures of Queen Elizabeth on the coin, the front of the coin, by the way, not the back of the coin. He says, "This will be my queen." He puts it on the board and he starts using it as a queen. What? I pick up the coin and throw it away. He goes, "OK." He says, "This rock." He picks up the rock and he puts it on there and he starts to, and I can't deal with his sorcery. It's too powerful. I go, "How does he do that?"
- 1.50** But I'm making a mistake. I'm making a mistake because you can play chess with pieces of wood, pieces of metal, pieces of plastic, pieces of stone. You can even get a whole bunch of helicopters together, put flags on them and get them moving around rooftops, and you can play helicopter chess. You can be on different sides of the world, and all you're doing is sending letters and numbers to each other through the Internet, and you're playing a game of chess. See, chess is multiply realizable. As long as you can play the game, as long as you can run

the functions, it doesn't matter what it's made out of. Does that make sense?

- 1.51** In fact, this would be a significant mistake. You're watching people playing chess, and you wait till they're done, and you get the pieces and you go home and you go, "I'm going to figure out chess. I'm going to do a chemical analysis on these pieces, and that'll tell me what chess is and how chess works." You're not getting chess. Chess is multiply realizable. We know, at least it's very plausible, that **intelligence** is multiply realizable. We have to be very, very careful to not just simply say **intelligence** is brain process X or human brain process Y or human brain structure Z.
- 1.52** We need a theory of **intelligence** that properly generalizes and takes into account multiple realizability. This is going to be important. This is going to be very important because we're going to see that some theories tend to get too bound to their specific level and forget that **intelligence** is existing at multiple levels and it is multiply realizable. We're going to need a very distinct kind of theory there. Is that OK? Does that make sense? OK, so remember that.
- 1.53** The second lesson, that we have to avoid what's called the homuncular fallacy. It sounds important. Latin. Must be important. The homuncular fallacy. Homunculus is a little man. Little man. What? Little man fallacy? What does that mean? OK, so when I was a kid, and it bothered me even as a kid, by the way, but I was a weird kid, so that's OK. I was watching a Disney film before Disney became whatever it is now. It was one of these shorts that Disney used to make trying to explain how things work. Science for Kids by Disney.
- 1.54** It was like, well, this is how vision works. There's a triangle out there in the world. It makes light. The light goes into the eye. It travels as information, and then they equivocate on information. It goes into your mind, and there's a little screen there, and there's a triangle on there, and the little man recognizes it, and he goes, "Triangle." And that's how you see a triangle. And you go, "Wait, that's not an explanation. You're telling me that how vision works is I have a little guy inside of me that has vision. How does the little guy see?" "Oh, well, inside his mind, there's an even littler."
- 1.55** You could just get an infinite regress. You don't get an explanation. It's like somebody gives you a check for \$1,000 and you go to the bank to cash it and all they give you is another check. And you keep going up from bank to bank to bank and it never cashes out. You never get what is it. You may think, "Nobody makes that mistake." Oh, yeah, they do. You'll see it. You'll see it. People will try to explain

- intelligence** in terms of processes that are themselves intelligent, cognitive agencies.
- 1.56** Now, that tells us something we have to do. We have to do the following. We must explain **intelligence**, ultimately, in terms of principles and processes that are not themselves intelligent or intelligent agents. And before you think, "Wow, can we do that?" Well, think about, this isn't exactly the same thing, but it shows you the principle I'm talking about. How did **intelligence** come to be here? Now, for a long time, people thought you had to have **intelligence** to make **intelligence**. You had to have an intelligent designer, aka God, and that is the being that was responsible for making **intelligence**.
- 1.57** And by the way, for a very long time, that's not a stupid argument. But then Darwin came along. And what Darwin showed is you could get a process that was not itself an intelligent agent, evolution, that could actually design and create intelligent beings. He's, "Oh, so it is possible to explain the emergence of **intelligence** from processes that are not themselves intelligent." Now, that's across speciation. But that means it is in principle possible to explain **intelligence** in terms of processes that are not themselves intelligent.
- 1.58** OK, so we want generalizabilities that include multiple realizability and avoid the homuncular fallacy. We're going to be doing all of that as we try to build a plausible cognitive scientific theory of **intelligence**. You're probably saying, "OK, can we get to it?" We had to do all this work. Why? Because remember what I said at the beginning. **Intelligence** is controversial. I don't mean in the public media sense. I mean, it's scientifically, intellectually controversial.
- 1.59** We have to build a very careful framework so we are properly responsible to that controversy, and we can give a strong and clear answer about what we propose **intelligence** is. As I said, we're going to start with the psychology, and there's two parts of it, the psychometric part and cognitive. Of course, cognitive is part of **cognitive science**, and that's why I'm here. Psychometric, metron, geometry, measuring the Earth. This is you're measuring the psyche. This is how we measure the psyche. There's lots of psychometric tests.
- 1.60** We can give you a psychometric test to determine your personality structure. Are you high in openness? I think you are, by the way. Are you high in openness? Are you high in conscientiousness? You've heard all of that. I can do psychometric tests to determine your attachment style. Are you securely attached in your relationships? Are you anxiously attached? Are you avoidantly attached? These are all powerful. We also have been doing, and the one we've been doing it

for the longest for, psychometric tests, since basically the end of the 19th century, in fact, we've been doing psychometric tests on **intelligence**.

- 1.61** Now, those tests are almost exclusively, although people have been doing some recent stuff that's interesting. I've seen one on dogs, and I'll talk about one on mice. But generally, the psychometric tests have been done on human beings because the concern with the psychometric test is to explain within-species variation. First of all, is it the case that she is more intelligent than him? I don't know. I'm not making any claims here, but we can ask that question. Then can we give a plausible, clear answer to that question?
- 1.62** Well, what would I need to do? I would need to measure her **intelligence**, his **intelligence**, with some good measure, and use that to compare their **intelligence**. That's the main goal. You might say, "Why do we want to do that? Isn't that going to lead to the oppression of human beings and things like that?" OK, there has been a bad history of IQ tests being used inappropriately. They've been used to enforce eugenics programs and racist—I don't know what you even call them. I don't want to dignify them with any other term—a racist project sounds like something legitimate.
- 1.63** But that has been the misuse of this. I wrote a textbook chapter on **intelligence** for a textbook, and I go through that history. I'm not going to do that because we're not doing the history of psychometric testing of **intelligence**. But what I want to say about that is the following. All of that is the case, but it's not the case now. What do I mean by that? The measurement of **intelligence** has taken into account those factors and repeatedly been self-correcting to try and reduce the impact of those factors on this testing. We're now even doing the testing cross-culturally in significant ways.
- 1.64** You get, OK, and here's what I want you to hear. There is no other measure anywhere that has gone through the same amount of self-correction. Those other measures I told you about, personality, little bit cross-cultural, attachment, not very much at all. So, yes, measurements of **intelligence** have been used for evil purposes, but that has precisely, strongly motivated reliable corrections to the measurement of **intelligence**. So I would say, you can make a good case that in comparison to many other of our psychometric measures, it is free from those biases.
- 1.65** Is anything ever completely free from them? No. But that can't be the standard we hold anything to, or we're not doing any science whatsoever. What I'm saying is, I acknowledge this evil, and all of the

good researchers acknowledge it as well. Then they point to the fact that we have had decades and decades, over a century now, of improving the tests to deal with those criticisms. Is that OK?

- 1.66** Putting aside evil, why else might you want to measure people's **intelligence**? Well, you might want to measure people's **intelligence** to give them guidance in their academic careers. In fact, we'll see this in a minute, that's the original motivation for the first test that had some predictive power to it, trying to help children in their academic choices. You might say that's unfair, but you could make an equal argument, and I'm not going to get into this argument. It's also unfair to not tell them what their **intelligence** might be, because then you're keeping from them information that's strongly predictive of their capacity for doing certain things.
- 1.67** For example, the US Army makes you take an IQ test, and if you're too low, you're not allowed in. Why? Because if your IQ is too low, you know what happens to you? You die. You know what a lot of people don't want to do? Die. You can't be simplistic about this. You have to take it with very nuanced and sophisticated framework. Here we're doing within-species, the human species variation. What are cognitive theories? Cognitive theories are not measurements, although there will be measurements, obviously in things like experimental data.
- 1.68** But what I mean is their project is not just to measure **intelligence**, their project is to explain **intelligence**. What is **intelligence**? What are the cognitive processes and principles that are at work when we make somebody intelligent? Because here's the thing, the psychometric tests don't really answer that question. They just say, "There's **intelligence** there, and he has more of it than she does." Well, then you say, "Well, what does he have?" And they'll say, "**Intelligence**." And you'll say, "But what is **intelligence**?" "What he has more of." And it goes in that circle.
- 1.69** You want to break out of that circle, obviously. And what the cognitive theories do is they attempt to explain what are the principles and processes at work in **intelligence**. OK, so you have the psychological approach, which comprises both of these. This is the psychometric and the cognitive. Now, this is in contrast to the categorical. This is based on an unpublished manuscript that I did with Zach Irving a while ago. I should try and get it published. You're always trying to do 17,000 things.
- 1.70** What does categorical mean? It means that I'm really taking seriously multiple realizability. I want to come up with an account that explains not differences between individuals within a species, but differences

between species, or allows me to talk about human **intelligence** and AI in a way that makes good comparative and principled sense. Now I'm trying to figure out, what are the characteristics that allow anything to go into the category intelligent? And there, AI and biology play a very significant role, because I want to know what's going on in the octopus. I want to know what's going on in ChatGPT-4.

- 1.71 How is that like and unlike what's going on in human beings? Now, of course, and this is the argument I've already made, we want an integrated account of all of this, because to miss any of it is to have a radically incomplete account of what **intelligence** is. We're going to do psychology. We're going to start with the psychometrics. We're going to go into the cognitive. Then we're going to do the neuroscientific, and then we'll move into artificial **intelligence**. That is how we will keep expanding the size of the category we're considering. Is that OK? Does that make sense?
- 1.72 People began trying to measure **intelligence** in the late 19th century. Started with a guy named Galton. He was a cousin of Charles Darwin. He thought people's ability to make sensory discriminations, like distinguish between the weights of things or the sounds of things, was how you measure **intelligence**. There's another guy by the name of Cattell. He was trying to see sensorimotor abilities and other abilities, and he thought by measuring them—he was influenced by Galton—you could measure **intelligence**.
- 1.73 Now, both of these were failures, because the various tests didn't predict each other, which means they're not measuring the same thing. They didn't predict how people would behave in areas that everybody agrees requires **intelligence**, like school. It's like, oh, so that failed, that project. Now, Binet and Simon in the late 1800s, and by 1906, when they published, they were doing something else. They, in fact, made a very important move that was going to be crucial to all of the psychometric testing on **intelligence**.
- 1.74 They said, "We're not going to look at those low-level factors because they're not getting at what **intelligence** is. What we want to do is we want to find those things that reliably predict how well kids are going to do in school." Because everybody agrees school is a place that requires **intelligence**. What we're going to do is, we're going to find measures that reliably predict how well people are going to do in schools. This is the first thing that set their psychometrics apart from Galton and Cattell, was that they were successful in producing measures that reliably predicted how kids would do in school.

- 1.75 What they did is they shifted, in order to do that, in order to get it to be predictively reliable, they shifted off these low-level measures to what they called skills of judgment, which is a much more abstract and quite vague notion. What did they mean by this? Directing thought to steps to complete a task, like directing, so you go through step-by-step, adopting a correct strategy for solving a problem, accurately monitoring your performance.
- 1.76 What they were actually doing was shifting **intelligence** testing to problem solving. How do people solve problems, abstract problems? This becomes the crucial thing for all psychometric testing. How do human beings solve these abstract problems? Whenever we talk about that, we face a problem. Sorry for the pun. I'm going to make a point now, and we're going to come back and do this point much more in detail when we do artificial **intelligence**, because this distinction really matters to artificial **intelligence**.
- 1.77 But there's two kinds of problems. Right now, we'll just do it intuitively. We'll come back and make it more specific. So, I want you to compare these two kinds of problems. A multiplication problem, like 33 times 25. Wow, that's a really bad three. How many of you could do that? How many of you can do that? I hope all of you. You can reliably predict that you'll probably get it right, yes? Now, what kind of problem is it? That's the thing. You know what kind of problem it is. It's a multiplication problem.
- 1.78 It tells you immediately what kind of problem it is and what kind of things you're supposed to do. You know, in fact, you probably didn't even consider it, that singing is irrelevant to solving this problem. You go, "Well, what I'll do is I'll sing." That's not—what's that going to do? That's not going to do anything. Not relevant. You also know what the goal is going to look like. You know what's going to end up here. You know it's not going to be the best picture of a penguin that anybody has ever made. That doesn't answer the problem.
- 1.79 What has to go here? There has to be a number. There has to be a number that's bigger than these two numbers. I can take this number and divide it by this number and get this number. You have all these very clear constraints and guidance as to what you're supposed to do. Does that make sense? This is called a well-defined problem. Other people call it an analytic problem. There's many different namings for this. One of the problems in psychology, it's called the jingle-jangle problem. We give multiple names to the same thing.
- 1.80 Now, I want you to compare that to another problem, the one you're facing right now, taking good notes. What does it look like at the

beginning? I have a blank page. Is that very helpful? No, because a blank page is the beginning of a lot of different things. It's like, what should I do? Well, I should write stuff down. Which stuff? If you look at her notes and his notes and her notes, will they all look the same? No. And some of them will have non-words. They'll have arrows or stars or underlining or capitalization, or there'll be indentation.

- 1.81** What's going on here? What's the result look like? Well, good notes. What do those look like, given all the variation? Then you'll give me synonyms. You'll say, I'll have captured the key information, the crucial information. What you mean is, you've zeroed in on the relevant information. Now, notice, you don't know. When you're starting, you don't really know. You don't have much guidance from your starting point what you should be doing. You're not really clear what the goal state looks like. You don't know what the operations are.
- 1.82** Maybe singing in your head is actually good for taking notes. I don't know. When I was studying, I used to play music for certain things because then when I was taking the exam, I would remember that song, and it would help me remember it. There you go. That was good. Taking notes. These are ill-defined problems. Here's my favorite. Well, it's my favorite example, although I hate going through it, of an ill-defined problem. I'm much older, so there's been multiple times in my life when my relationships have ended, and I find myself single. And then after the appropriate grieving, I go back into the wonderful world of dating.
- 1.83** And I hate dating because you're either rejecting or being rejected. And human beings hate both of those. I hate both of them pretty much equally. Here's the problem. Go on a successful first date. You know it's an ill-defined problem because your friends give you useless advice that sounds important. I'm straight, and I'm just going to do it from that orientation because that's what I am. Look into her eyes, but not too much. Try to be funny, but not too much. Ask questions, but not too many. Lean in, but not too often.
- 1.84** I realized that, no matter what, they could say to me later, I told you so. Now, here's the important point. In your education in school—and what are Binet and Simon training people for? School—you are given a lot of well-defined problems. What's the world made up of mostly? Ill-defined problems. This is a problem that haunts psychometry of **intelligence**, which is, we test people on well-defined problems to try and predict how well they will do on ill-defined problems. Now, we've had success at this. I'm not saying it's all a waste, but we're going to come back to this issue again and again and again.

- 1.85** Now, why are Binet and Simon using well-defined problems? What's the reason? They are eminently measurable. You can measure these because they're so well-defined, you can measure. That was the right answer. That was the wrong answer. That took you too much time, et cetera. Does that work for ill-defined problems? How much time are you supposed to take on a first date? Well, less than a year. OK. How do you know if you got it right or wrong? In fact, that's one of the most—like, even if the person says they want to see you again, that doesn't mean your first date was successful. I learned that.
- 1.86** Oh, somebody's just being kind. OK. So two reasons. They're testing kids on the problems, the kinds of problems they're going to meet in school, well-defined problems, and those well-defined problems give them what they need. They're highly measurable. They're highly measurable. This is why well-defined problems are given significant priority from the very beginning in the psychometrics of **intelligence**. There are some important researchers on **intelligence**, Robert Sternberg is one of them, who emphasize, "We need to revise our tests so that we include more ill-defined problems."
- 1.87** He talks about successful **intelligence**, which would include the ability to not only solve well-defined problems, but to solve ill-defined problems. I think this is important. He's taken seriously, but I don't know if his work is strong enough to garner any significant consensus. He's making the point well. People acknowledge it, but I don't think he's actually significantly solved the problem.
- 1.88** OK, so let's go back now to what came out of all of this research. Binet and Simon are doing it, and the tests are getting revised. Remember, I said there's this long history of revision. It keeps going, by the way. We got to the work of somebody who is going to give us a central phenomena for **intelligence**, and this is **Spearman**. So **Spearman** is publishing in 1904, 1923, 1930. And what did he find? He found something that's really interesting, and it goes a lot against a lot of our current cultural tropes.
- 1.89** He found that how kids were doing in any one subject at school was strongly predictive of how they would do on all the other subjects. And we have tropes that, if you're good at sports, you can't be good at art. And if you're good at art, you can't be good at math. Of course, people do have talents. Nobody's going to deny that. But that claim is actually false. When I'm sparring with somebody as a martial artist, I don't want somebody who's really smart. They're deadly. They're dangerous.

- 1.90** So he found this, and what he started to do is he started to give more specific tests, and he kept getting evidence for this strong positive manifold. How I do on A is predictive of how I do on B, which is predictive of how I do on A. This is predictive of how I do on C. So is A. C is predictive of B. Do you see what I'm talking about? Each part predicts how you're going to do on all the others. You have a strong positive manifold. This became known as **g**, the general factor of **intelligence**. There seems to be some general factor at work.
- 1.91** Now, of course, you have **intelligence** that's specific to riding a bike and things like that. But the idea is, oh, but there's a general factor. Here's something to think about. Maybe this general factor is responsible for the fact that you are a general domain problem-solver. You have **general intelligence**. Remember, you can solve a wide variety of problems in a wide variety of domains. Well, that's what's happening here.
- 1.92** **G** is showing kids, and eventually adults, because the testing spread, are capable of solving a wide variety of problems in a wide variety of manners, plausibly using the same something to do it. Oh, that's really interesting. That's really interesting. So he understood **g** as reflecting reasoning.
- 1.93** Now, we have a problem with this term because it's equivocal. Oh, I know what reasoning is. Do you? Because there's a difference between well-defined and ill-defined reasoning. What's well-defined? Well-defined reasoning is deduction and induction. Deduction is what we do in logic. All men are mortal. Socrates is a man, therefore Socrates is mortal. There you go. That's deduction. We have a really clear formal system for what that is. Induction. We have a very clear formal system for that. It's called probability theory. Great. These are well-defined forms of reasoning.
- 1.94** Is that the reasoning you mostly do? No, it's not. We've talked about a great scientist here, Darwin, and how Darwin beat out Paley. Paley was the guy who argued intelligent design. Now, how did Paley argue that creatures have their design based on an intelligent designer? Well, he said, "Well, here's the animal." I'm not going to try and draw an animal. I don't claim to be an artist. That's an animal. Just accept it. That's a goat. What do you note about goats? They're really well-designed for where they live and how they live.
- 1.95** It's like, well, what could make a goat? Could erosion make a goat? The way the Grand Canyon is made by erosion? No, that doesn't seem very plausible. So, we generate an explanation. Explanation one, nah, not that. What about lightning? Lightning hits things and it could—

lightning, zap some sand, and there's a goat. Nah, no. What you're doing is, he says, look what you're doing. Notice how all of these would fail to explain a watch. And he says, "Well, what would you do?" You'd say, "Oh, well, what explains a watch is an intelligent designer. I need that kind of explanation to explain the kind of complexity in design and fittedness to function of a watch."

- 1.96** And then what Paley says is, we need the same thing to explain the goat. Now, until Darwin, that's a good argument. Now, here's what you need to know about this. This is not an inductive argument. He didn't gather a bunch of goats together and then conclude from that God. That's not an—what? Goat, goat, goat, goat, goat, God. That doesn't work. It's not a deduction, God, therefore, goats. It doesn't work. This is called inference to the best explanation or, somewhat confusingly, abduction, which sounds like something that aliens do to you.
- 1.97** Then how did Darwin argue in *The Origin of Species*? He said, "Wait, I have another explanation, and it's better than this one." Inference to the best explanation, it's better than this one. Why is it better? Because it's based on entities I can see—reproduction, competition, variation. It explains not only bad design, not only good design, but bad design. You have a nerve that goes all the way down and all the way back up. It's really bad in giraffes because it goes all the way up and all the way down in their neck. It's like, what? Why is that?
- 1.98** It's because you're the result of evolution. Gould, the panda's thumb. Pandas, they grasp bamboo and they look like they have a thumb. It's just this bone has become a little bit bigger. Bad design. You're filled with lots of bad design. Your breathing tube and your food tube are the same tube. You know how people die? Eating their food. It's a really bad—having a baby. Until we had medicine, either the child or the mother would die in 25% of the times. We figured out doors, take child out, close the door.
- 1.99** Wheels. Wheels. Where are the wheels? OK, do you see the point? Darwin's theory explains both bad design and good design because Darwin's theory says, all you need is a design that beats the competition. You don't need perfect or good design. This is inference to the best explanation. This is how most of science works. It's not induction, it's not deduction. To make it deductive, you know what I have to do? I have to generate all possible explanation. The number of possible explanations is indeterminately large. I can't possibly do it.
- 1.100** We also make arguments by analogy. I compare a goat to a watch. It's not deductive or inductive. That's by analogy. I do argument by

eduction. I look at a bunch of things and I say, "Oh, there's the pattern in there." You have eduction, you have abduction, you have analogy, and we're doing this all the time to solve our problems. These are also considered reasoning, but they're all ill-defined forms of reasoning. Do you see?

- 1.101 We have to be very careful to not hear logic when we hear these researchers saying reasoning. It doesn't exclude logic. It doesn't exclude probability theory, but it's not identical to that. It's a much more, well, general ability of enabling you to do things with your mind to solve problems. Is that OK? A lot of people think that what's being talked about here is just logical ability. That's not the case.
- 1.102 What immediately happened when **Spearman** produced all the evidence for **g**, and by the way, the evidence for this keeps piling up. Year by year, we keep getting confirmation, more evidence for **g**. Nobody denies that **g** is a thing, but there was initially, in a debate, where some people said, "It's just a statistical artifact, like how what's the average number of kids? Three and a half. So I should be able to go into a home and find three and a half children." That's ridiculous. You know that's a statistical artifact. It's just a result of the way statistical reasoning works.
- 1.103 So some people are saying, "**G** is not a thing, it's just a statistical artifact." I'm not going to go into the history of that debate because there's overwhelming consensus. See Deary's book on **intelligence** from 2020, or Haier, Colom, and Hunt from 2024, **The Science of Intelligence**. Overwhelming consensus that **g** is a thing. It's not a statistical artifact. So it's pointing at something. I'm not going to go into that debate because I think that debate has largely been resolved by the scientific community.

2. Models of General Intelligence

At a glance

Contrasts two competing models of how *g* arises: a top-down model in which a common cause drives cognitive strategies, versus a common-effect (bottom-up) model in which many strategies converge on shared behavioral outcomes. Closes with a discussion of pain as a predictive signal about future relevance rather than the moment of harm itself, linking back to relevance realization.

- 2.1 Now, there's a second debate, and this debate is ongoing, so we have to look at it more carefully. There's a second debate, and this debate is ongoing because there's two models. There's a **top-down model** and a **bottom-up**. What does top-down mean? It's, there's a common cause. What **g** points to is a common cause, a common process or principle, and that goes down into many more specific strategies for solving problems. Common cause.
- 2.2 Then there's the common effect model. No, no, all of these different strategies go up and have a common effect on behavior. You see why one is called top-down and the other is called **bottom-up**? Common cause down to the individual skills, strategies? No, **bottom-up**. Is it a common cause or a common effect? Because that is important, because that'll change your theorizing about what **g** is.
- 2.3 So, a little bit of language here. Many people, and I think this is correct, say that **g** is talking about general intelligence. The question is, is general intelligence a thing that is a common cause for many different problem-solving behaviors? Or you have many different problem-solving behaviors, and they have a common effect that is general intelligence? Is that clear about the difference between the two? It really matters because if you think the first, you're going to go looking for the process. If you think the second, you're going to go looking at the processes and how do they work together.
- 2.4 It's a very different theoretical approach. Spearman argued for this. We have to pay attention because he discovered it, so we got to pay attention to what he thought. It doesn't mean he's right. We have to take him seriously. He thought that he was talking about one of two things. He had two proposals. One was, he thought that general intelligence was mental energy. People who are smarter have more mental energy. First of all, that's really vague. What's mental energy? As we'll see later, it's actually false.
- 2.5 People that are intelligent expend less metabolic energy than people who are less intelligent. The more intelligent you are, the less

metabolic energy you're expending. That part, that theory from Spearman, no. We'll come back to that whole idea. Why are you using less of your brain if you're smarter? What's going on there? What you might be thinking is, well, maybe smart people use their brains more efficiently. Yeah, we'll come back to that. We'll come back to that.

- 2.6** Then he had another theory. So let's put aside the mental energy theory. Let's come back to his second theory, which we should take more seriously. He thought that what was going on was a set of abilities that were in **g**, in your general intelligence. One he called apprehension. A good way of understanding apprehension is the work of Matson from his 1976 book, or Matson, I don't know how he pronounces it, called Sizing Up. This is a process by, out of all the information in the environment, I pick out some that are features, and then I foreground, background. Some of the features go into the foreground, some of them go into the background.
- 2.7** Then I figure out what those features are, how they hang together. Remember we talked about that? I figure out. That's such a good phrase. Think about what that phrase is telling you. I figure out. I find the structure that binds and holds them all together. I figure out, and then from that, I can frame a problem or formulate a problem I'm trying to solve. That's what's going on in apprehension. Of course, there's all kinds of feedback loops, et cetera. I'm not going to draw it all out. It's a very dynamic system. Is that OK?
- 2.8** Then he said, so that's one thing that's going on. Remember that. This is the way in which you're noticing the world, how you're forming a perspective on the world. Then, of course, you do that when you come to a problem. What are you noticing in the problem? What perspective are you taking on the problem? That's going to turn out to be, he's onto something there. That turns out to be, well, it's prescient. Then he talks about general intelligence.
- 2.9** When you're doing this apprehension, it allows for eduction. What's eduction mean? Remember this. It's really interesting about Spearman. He's got one really false theory, and then he's got this other theory that's going to be prescient. It turns out to predict a lot of the data we're going to look at. What's eduction? Education is you finding a pattern in things. You're finding a pattern in things. That's what an education is. It's part of reasoning. It's part of that ill-defined reasoning. When it has "duction" in it, deduction, induction, abduction, it's reasoning of some kind. But it's not the reasoning you make in a logical argument.

- 2.10 I'm not going to keep harping on that. Is that really landing? You're going to remember that from now on, right? OK, good. Then the last is, he talked about the eduction of— you make a distinction within eduction between relations and correlates. Relations and correlates. As a doctor is to his patient, a gardener is to, what? What would you think? What would you say? Land or garden or plants. What you did is, you found a relation and then you mapped it from one system to another. Correlates. You did something by analogy. Remember, that's also part of reasoning.
- 2.11 He thought these interlocking abilities of apprehension, eduction and the affordance of analogous thinking, was the key abilities behind general intelligence, because it seemed to him that no matter what you're doing, you've got to size up a situation, you got to draw out the patterns, and you have to see if the patterns transfer in order to be generally intelligent. It's a very plausible proposal. Because it's plausible, it turns out to have, I will argue, quite a bit of truth in it.
- 2.12 Now, that doesn't mean I've defeated the opposing view, the **bottom-up** view, because it's got a lot going for it, too. The most powerful current model of the **bottom-up** approach is known as process overlapping theory that has the very unfortunate acronym. Because you don't typically associate that with being smarter, but that's the acronym, process overlap theory. I don't think they realized that when they made the acronym. OK, so this is Kovacs and Conway. Kovacs with a K, Conway from 2016.
- 2.13 Their proposal is this. Instead of there being one process, there is a system of processes from which intelligence emerges. At one level, this has to be true. How intelligent do you think an individual neuron is? You don't see them voting, you don't see them getting married or driving cars. An individual neuron is pretty, pretty, pretty, pretty stupid. You get them together and you organize them in the right way, a brain, and you get intelligence. Again, there's initial important plausibility to this proposal. In some sense, it's got to be right, unless intelligence is magical goo or something in your head.
- 2.14 Notice what they're saying. They're saying it's not a system-wide function, but it is the functioning of a system. The top-down says, throughout the whole system, there's a function, a system-wide function. No, no, no. Process overlap theory says no, no, no. The system functions as a whole, and that's intelligence. You see the difference? In the first, intelligence is what glues the system together. In the second, intelligence is what emerges out of the system. Is that OK?

- 2.15 Now, you're probably thinking, well, can't they both be true? We'll play with that. That's good. Think that. Keep thinking that. That's a happy thought. But what we have to understand is that this debate has generated some really—how do I want to put it? It's generated empirical findings that are hard to get a clear theoretical integration about. One of the most prominent versions of this is that theory I mentioned, P-FIT theory. I mentioned it. It's this kind of theory.
- 2.16 It says there is—the parietal area of your brain, the frontal area, and the degree to which they're integrated, predicts how intelligent you are. It's an emerging function that comes out of how these different parts are interacting with each other. We'll see this, because we'll go this in more detail when we do the neuroscience. There is a lot of evidence for the P-FIT theory. A lot of evidence for it. Oh, then the **bottom-up** is right. Well, yeah, but you can also get other kinds of evidence.
- 2.17 So, this is Protzko from 2016. He says, well, look, if intelligence is emerging out of these, if I improve one of these by training it, that should improve intelligence, yes? Because it's **bottom-up**. And if I improve the bottom, it should affect what's coming out at the top. What you can do, he says, is you can train people, one of these underlying processes, we'll get back to this. Here it is again, working memory. You can do a lot of stuff to train working memory. People, they have these apps and the brain apps, and you do all this, and training your working memory and all that.
- 2.18 And what do we reliably find from this? It doesn't improve your general intelligence at all. You're getting ripped off by all these apps, by the way. There seems to be one thing that really modifies working intelligence and may have an impact on general intelligence, which is a lifetime of dedicated meditation. You might want to try that. Well, what's his point? His point is that this failure of training of the sub-processes to transfer to general intelligence really says the following, that, "A top-down causal structure makes upward causation from subfactors like working memory to general intelligence impossible."
- 2.19 If it's top-down, this shouldn't work, training working memory, because it's top-down, because you're not really changing general intelligence. He says, "The failures of transfer may be indicative of the correct causal direction between intelligence and its subprocesses." Namely, the fact that you're getting this failure is good evidence that it's top-down. So, now what do we do? We've got really good evidence that it's **bottom-up** and really good evidence that it's top-down. And yet the pieces of evidence seem to be in competition with each other.

- 2.20** You may be thinking, "Oh, wow, what do we do?" Well, let's take a look at a really important article that Gottfredson, that's a **G**, from 2016, and Gottfredson said, look, there's an actual conceptual problem going on here. She says, look, how do you determine if something is a subprocess or a single thing? Is your brain a single thing, or is it made up of a bunch of subprocesses? Is a table a single thing or is it made up of a bunch of atoms?
- 2.21** You go, "Oh, wait." Because even Spearman thought that general intelligence was made up of apprehension and education of relations and correlates. And Kovacs and Conway agree that general intelligence is what is behind your domain general reasoning. Is general intelligence a thing? Well, that depends. Do you think general reasoning is a thing? You say yes, then you're back to one thing. You say no, it's many things. How do you tell? Do you see the problem here?
- 2.22** What's going on here, and this is her point, is, you deciding whether or not it's a thing or a system of things is completely dependent on your level of analysis. Where did we see that? We saw that at the very beginning of the course, where I argued that if you don't pay attention to the levels of analysis, you'll get messed up when you try and think about what intelligence is. Here it's happening. Because we don't know. What's the correct level of analysis? Remember the board? What do you think? Is it the brain level? Is it the information processing level of AI? Is it the behavioral level? Is it the social level?
- 2.23** "Well, urg," right. Do you understand? The reason why we're getting this evidence is because we have a confusion. We can't decide the confusion by looking at the evidence, because the confusion has to be decided by, how do we properly relate these different levels of analysis to each other? So, here's what John Vervaeke proposed. You know how you do that? You do cognitive science. That's how you do that. We're going to do cognitive science.
- 2.24** That's another argument, a specific example, of why you need to do cognitive science, if you're going to try and answer the question, what is intelligence? Because, like, everybody belongs to their own church of this level. "I think it's the brain." "No, I think it's information process." "No, it's behavior." Do you understand? There isn't an answer in the data. We have to reformulate the problem by paying attention to the proper relationship between the levels of analysis by doing synoptic integration in a highly plausible manner. We have to do cognitive science.

- 2.25** That's why I'm here. We're going to do that by moving from the psychometric theories of intelligence to cognitive theories of intelligence. What are the underlying processes? Now, here, we're still in psychology, so we're going to be still concentrating on the psychological level, but hang in there, that's going to change over time. What do we mean by cognition? Cognition, it comes from the Latin for knowing. How do we know the world? All the processes that allow us to know the world, to have knowledge about the world. That's what cognition is.
- 2.26** Now, this is important, by the way. How do you get knowledge? The proposal is, the processes by which you get knowledge are intelligence. That's what intelligence is. When you exercise your intelligence, the result is knowledge. Now, you might want to refine that about bias, and then you talk about rationality. Right now, we're talking about intelligence. Now, one of the things you want to do, therefore, when you're talking about cognition, is you want to keep your theory of intelligence and your theory of knowledge separate from each other.
- 2.27** Because you want to be able to say what causes knowledge is intelligence. Because you don't want to say, you know what causes knowledge? Knowledge. You may think, well, of course. Put a pin in that because we're going to come to that. We're going to come to that. We're going to take a look at what's called the standard information processing model. This is the model that runs through all of cognitive psychology in how information is taken in and processed in order to give us knowledge. Is that OK?
- 2.28** Now, right away, by the way, notice what's happening here. Although we're doing psychology and we're talking about people's behavior, we're sliding in AI language. We're sliding in language from computer science. We're sliding in language about information processing because cognitive psychology is deeply influenced by the computational metaphor. OK, so let's go through it. This is from Haier, Colom, and Hunt, from 2024, second edition, *The Science of Intelligence*, Haier, H-A-I-E-R, one of the premier neuroscientists on intelligence. The two other people, important psychologists.
- 2.29** Nobody, I think, would object to their explanation. But I want to go through this carefully. I'm going to quote them in places because I want to show you something, a pattern. I'm going to educe a pattern out of what they're saying for you that there may not be foregrounding. OK, so how does it start? Step one, information is taken in from the environment and then "classified into progressively higher-order categories." So, at the beginning of this is categorization.

There's information coming in from your senses, but it starts with, it gets categorized.

- 2.30** Two, there's a quote. "Information from the environment together with relevant information retrieved from long-term memory." What's long-term memory? It's the memory you have for your whole life. Relevant information from that is then integrated with the information coming in, in working memory. Working memory is where you do things in your head. If I were to ask you right now to do that 22 times 33 without using any paper or pencil. Is that OK?
- 2.31** Now, notice the centrality that working memory plays in this. What's going on in working memory? Internal representation is formed. Then what's working memory do? It's establishing priorities for action, so it's prioritizing things. It has attentional processes that highlight. What does highlight mean? It's a metaphor. Make more salient to you, make it stand out for you, some pieces of information and suppress other pieces of information. It's prioritizing, it's making some stuff salient, it's making some stuff background or suppressed, it's doing all that stuff. It's noticing, like Spearman said. Sizing up. OK, it's plausible.
- 2.32** Next, there's the identification of goals and subgoals in problem-solving. This is all for problem-solving, which I've already said. It's all about problem-solving. Next, working memory acts as a blackboard. What? Why all these metaphors? That broadcasts information into long-term memory. We're going to come back to that. Remember that. Long-term memory processes. I want to do the direct quote because this direct quote is really interesting.
- 2.33** OK, so one way of saying it is, working memory is a blackboard in which you're figuring things out and then you broadcast it to the rest of the processes in your brain. Then they say, or you can think about it this way, "Long-term memory processes are actively watching working memory to see if their cue has been called, rather like actors on a stage production." What's wrong with that explanation? Long-term memory processes are watching and waiting to see if they're called and should come into action like people watching a stage production.
- 2.34** It's homuncular. What's going on in long-term memory to do that? Imagine if I try to—I'm going to make a machine that has long-term memory. What you need is somebody that can watch what's going on in working memory, notice when it's been called, and then it should behave, and then jump into action. Well, how do I do that? You need a little man. A little man that's judging what's the relevant information?

Am I relevant? Is this the relevant time? What's the relevant intervention? Then information from working memory is consolidated in long-term memory or the information in working memory goes into an action.

- 2.35** That's the standard model. I emphasized a lot of things there. I kept pointing out to things like relevant and highlighting and prioritizing, and it's at the core of problem solving, and it's homuncular because there's somebody that's making judgments about, "Am I relevant? Is this the relevant information? Should I leap into action?" We're going to see that the blackboard model is actually a version of what's called the global workspace theory. We'll come to that in a sec. But let's start. Let's start with categorization.
- 2.36** Why is categorization so basic? Because it is. Because if you don't categorize things, you're in trouble. What are you sitting on right now? Chair. Imagine if you had to meet each one of these as a raw individual. This is Agnes, and you have to get to know Agnes really well. Then when you go here, this is Tom, and you have to start all over again. But you didn't do that. You go, "Chair." You categorize. What it does is it allows you to really powerfully encode the information. It allows you to rapidly make inferences. I can sit on that, even though I have never sat on it before, because it is a chair.
- 2.37** It allows you to communicate. Imagine if you didn't have the concept chair, the category chair. I want you to go put your gluteus maximus on the flat part of the sticky-up thing. It takes too long. If you can't shout tiger, you're dead. Do you see how powerful categorization is? Everything you do, all your problem solving, seems to depend on categorization. It seems to be such a basic process.
- 2.38** Here's the thing. Categorization seems so obvious to you, and remember obviousness, that we think, "I know how it works." I don't know, you're too young. Did you still watch Sesame Street when you were growing up? Or is that not a thing anymore? Because TV is, what? TV? I don't do TV. I stream. It's really funny. When I grew up, there was four or five channels, and there was commercials, and then there was the cable revolution, and then there was the Internet, and then there were all the streaming services. And now what do we have? We have four channels and they have commercials, all the way back.
- 2.39** In Sesame Street, it was like, the kids had to play this game. One of these things is not like the others. Three of these things are kind of the same. There'd be a hammer, a saw, a screwdriver, and a dog. The kids would say, "Oh, the dog is the one that doesn't fit." Sometimes

Tommy would say, "The screwdriver," and then you're worried about Tommy. Oh, no, Tommy's doomed. It looks like, oh, this is how we do it. What we do is, we notice the things that are kind of the same. What's the word for kind of the same? Partial identity. Similarity. We notice that things are similar.

- 2.40** We notice that they're similar, and then we group them together mentally. We bring them together in working memory, and we group them together, and we form a category. It's so easy. It's so obvious. Now, here's where the philosophers come in, because what philosophers do is take things you think are obvious and make them deeply problematic, so that you wonder, "Was I born four seconds ago? Maybe the whole universe has only existed for five minutes. How would I know? Go away from me, philosopher." Now, let's do it in a more useful fashion.
- 2.41** Nelson Goodman. What a great name for a philosopher. Goodman. There's one philosopher who has an even better name. His name was John Wisdom. I'd love to have that name. OK, so Nelson Goodman in 1972, he said there's a problem. What we mean by similarity is partial identity, which means two things share a lot of properties. They don't share all properties because if they shared all properties, they'd be identical. They share some properties, and that's what makes them similar. Yes?
- 2.42** And he said, OK, what is a property? A property is if you can truly attribute something to something else, like dogs are mammals, or something like that. Is that OK? Any true attribution is a property of a thing. Is that OK? All right, so let's pick two things, a plum and a lawnmower. Well, they both contain carbon. Both have shiny curved surfaces. Both are found in North America. Neither one existed 300 million years ago. Neither one is a particularly good weapon. Both of them have a distinctive odor. How many shared properties do you think I can generate for a plum and a lawnmower? How many? Combinatorially explosive.
- 2.43** You know what I can do? I can do that with any two things. So everything is similar to everything else. Well, if everything is logically similar to everything else, you know what it doesn't help you to do? Categorize things. It's like, oh, oh, oh. Then what you should say to me, and it's right to say it, but it's not helpful, you should say, "No, no, it's not all properties. It's only the relevant ones." I have to zero in on, out of all the possible properties, I quickly zero in on the relevant ones. The thing is, you can do that really fast.

- 2.44 Let me give you an example. This is from the work of Barsalou. Your spouse, your kids, a dog, works of art, gasoline, explosive material. Things you care about when there's a fire. Oh, there you go. Here you go. What's at the core of categorization is **relevance realization**, realizing what the relevant information is. Even more so, if I'm going to categorize, I have to be able to do something before I have the category, right? Or else I'm born with all my categories, and that's ridiculous. Do you believe in door knobs? Do you think you were born with the concept door knob? Do you think your ancestors, as they're loping across the savannah, are evolving door knob, so you have it ready to go? That doesn't make a lot of sense, does it?
- 2.45 OK. So, how do I categorize? Well, I have to get a bunch of things and we're putting aside now the similarity because your brain is going to go, "Similarity, similarity." Put that aside. But I have to do the following. This, this, this, this, this. I have to do this thing. This is called demonstrative reference. This doesn't pick out any concept or category. Watch. This, this, this, this, this, this, this. Do you understand? It doesn't pick anything out. All it's doing is making things salient, perceptually relevant to you. That's all it's doing. So that you can gather them together to form a category.
- 2.46 Before categorization can even get going, and Zenon Pylyshyn has shown this experimentally with multiple object tracking experience, you have to be able to tag your experience, "That's relevant. I don't even know what it is, but it's relevant." Do you see that? By making it salient. That's all you're doing. Salient, touching it. Don't know what it is. I'm touching it with my mind. It's salient to me. It's relevant. Categorization completely depends on **relevance realization**.
- 2.47 And notice, and then they went and they say, "Retrieval of relevant information from long-term memory." That's **relevance realization**. How much memory do you have in long-term memory? How much do you think? What about, not only what's in there, but possible combinations? The Titanic, the British Empire, Australia. There you go. Did you do that before? But none of you went, "I don't understand." So you've got a titanic amount of information in your mind, your long-term memory, and even more possible ways you could combine it. This is combinatorially explosive. It's overwhelming.
- 2.48 And this is what you do. "That, I need that." **Relevance realization**. You bring it into working memory. When what are you doing there? You're prioritizing this. That's more relevant than that. You're making something, you're highlighting this, and you're suppressing that. This is relevant, that's irrelevant. What about this idea that working

memory is like a blackboard or a whiteboard? This is part of the global workspace theory. What we need to understand here is, when we're talking about things like working memory, it overlaps with consciousness, with attention, with your online cognition.

- 2.49** Even when we're talking about working memory or consciousness, whenever we're talking about one of these core four, we're always implicitly talking about the other three. Just keep that in mind, OK? Because the theory that I'm going to talk about, although it's basically a significant theory for working memory, it uses the same model, in fact, even the same metaphor of actors on a stage—when you read about it, it'll be called a theory of consciousness. This is called the global workspace theory. This is the idea that what consciousness is, is it's like a workspace, like the whiteboard.
- 2.50** What I do is I have a bunch of unconscious files. Now, think of your computer and your desktop. Your desktop is your workspace. What I do is, I activate these files and I make them co-present in the workspace. I can manipulate them, compare them, do things with them. Then I can broadcast it back. Remember, they talked about broadcasting it back. I can broadcast what I've done in working memory back to any part of my long-term memory. Is that OK? "Hey, there is a connection between the Titanic and Australia. I didn't think about that," and now it goes back into your files. Does that make sense?
- 2.51** The guy who came up with this, Baars, B-A-A-R-S, used the stage metaphor. What is consciousness? Well, here's the stage, it's working memory, and you shine the light of attention on it. Remember, they were talking about that, and that's what consciousness is. The unconscious things are the people in the audience looking to see if they are connecting to. And what they can do is, they can stand up and say, "I'm relevant." They literally say this, by the way, in their theory. "I'm relevant to this." OK, so you say, OK, so what?
- 2.52** Well, if you look at their account of what's the function of the global workspace, and the paper from there is Shanahan and Baars. I need you to remember Shanahan because we'll come back to him, 2005. They say the function of this is to solve the frame problem. We're going to go into the frame problem in detail when we do artificial intelligence. But the frame problem is this problem. According to Shanahan, it's the relevance problem. How do you frame your cognition, so you only include the relevant, and so you exclude the irrelevant?

- 2.53 You see what's happening here? What keeps coming out to be the function? **Relevance realization, relevance realization, relevance realization.** What we see is a general requirement to have to do **relevance realization** in order to do cognition. Now, surprisingly, at least to me, and this has been until about 10 years ago, a bit of a tragedy of my life, this idea that **relevance realization** is central to intelligence, which seems to be just there, was not explicit. It wasn't foregrounded. People weren't talking about it.
- 2.54 I kept making, I did this way back in my thesis, in 1997. My committee also, "This is great. This is fantastic." Yeah, but—**relevance realization** is very implicit in all of these theories. I'm just showing you it here now, and yet it's not explicit and foregrounded. It's like, why does this keep happening? Because it bothered me. Because first of all, I thought, maybe I'm just insane. Or maybe I'm just obsessed. Now, thankfully, and I'm glad my career is going this way, it's turning around. But why is it so difficult? Because I think it's this.
- 2.55 You know what **relevance realization** does is it makes things obvious to you. It makes things obvious to you. It's obvious I should sit on the chair. It's obvious I should look towards John, not towards over there. It's obvious. It's obvious. It's obvious. You know what's the problem with obvious? In everyday experience, it's what you assume to go on to do all of your problem-solving. But here's what we need to do, I would argue. We need to explain how your brain makes things obvious such that you are a general problem solver.
- 2.56 Can you tell me how to do that? When I refrain the question like that, people are, why? No, no, no. Your brain is making obviousness, but you haven't given me an explanation of how it does it. Obviousness is not a physical property of the environment, 17 grams of obviousness. Here's the thing that particularly bothered me. To the degree to which you leave **relevance realization** backgrounded and implicit, you know what fallacy you're committing? The homuncular fallacy, because you've got somebody in there, Bob, who's doing the **relevance realization.**
- 2.57 Now, we're only just starting, but you see some convergence happening here? Let's take a look at working memory. One of the most powerful theories of general intelligence is that general intelligence is working memory function. That's what general intelligence, when we're measuring **g**, what we're measuring is how well his and her, how your working memory is functioning. Now, why is this plausible? It's plausible because working memory is like the

bottleneck that everything is coming into and everything is going out of. It's very plausible that it could have a domain general function.

- 2.58** You're going to be relying on that no matter what you're doing because it holds everything else together. Oh, that's a good argument. That's a good argument. There's a lot of empirical evidence that shows working memory function is highly correlated with measures of general intelligence. A lot of evidence on this. Now, you have to be careful, again. When we're talking about working memory, we're really talking about consciousness, attention, and online cognition. But even with those caveats, wow, I measure working memory capacity, I measure your general intelligence, and very highly correlated with each other.
- 2.59** How do you test working memory? You do digit spans. I'll say a bunch of letters or numbers, and then you have to repeat them back to me. How much can you hold in working memory? Make it a more challenging, I'll do this. I'll say a bunch of things, and then you have to repeat it back to me in the opposite order that I gave it to you. You have to be able to manipulate and hold information in the right way. These things reliably predict how well people's working memory is functioning.
- 2.60** The person who came up with this notion of working memory, by doing these kinds of experimental measures, was Baddeley, in 1986. He argued that people need to hold pieces of information even in mind, though they are not immediately relevant to the task at hand. This is a quote from him. "Attention"—notice how he automatically gets attention involved. Like I said, attention, consciousness, working memory, online cognition, all bound up together—"Attention has to be focused on information relevant"—these are his words, not mine—"information relevant to the task at hand and irrelevant information must be suppressed."
- 2.61** Now, within working memory, there's what's called the central executive. What does the central executive do? So in addition to the holding space, there's the central executive, and I'm doing this for a reason, looking at this. The central executive is a process that focuses attention on some information in working memory to make it available for processing, while at the same time preventing loss of information that is not currently the focus of attention, but that has been accessed in the past and may be needed in the future.
- 2.62** These are all judgments. I may need this information. What's doing that? This is homuncular. It's homuncular. It's homuncular **relevance realization**. Maybe we can at least start to dissolve the homunculus a

bit. This goes to the work of Lynn Hasher. She's retired now, but she was a colleague of mine at the University of Toronto. I met her in a really wonderful way. Sorry, this is a little bit self-congratulatory. She's a very important theorist. She knocked on my door and comes in my office and she said, "Are you John Vervaeke?" And I said, "Yes." And she said, "Your cognitive science students are always some of the best students in my courses." And I thought, "Yay."

- 2.63** She made a very specific proposal, and she put a lot of evidence behind it, but I'll be able to give you a good intuitive argument for it. She argued that, and this is what she explicitly says, working memory isn't just a holding space, it's a relevance filter. She did this by making use of the phenomena known as chunking. I'm just going to carry my book over here for a second. T, C. OK, so I want you to look at that very quickly. Now, look away. Don't look at it. Tell me what's there. There's a **C**. OK, now I'm going to replace it. Same letters. Look away. Now tell me what the letters are. Can you do it?
- 2.64** This is known as chunking. What you've done is, you made the letters relevant to each other insofar as they're relevant to you. That wouldn't work for somebody who speaks German. You have to make them relevant to each other. They're bound together in a way that makes it relevant to you. This is why working memory is a relevance filter. Working memory is looking for, how much have you chunked the information that is coming in from the world or coming in from long-term memory to be processed? How much chunking has gone on?
- 2.65** Why is it doing this? It's doing higher-order **relevance realization**. You've got some basic **relevance realization** going on unconsciously, and then you're doing **relevance realization** on previous **relevance realization**. This is called recursive **relevance realization**. Recursive is when you apply a process to itself. Now, think about it. What problems are you solving when you need to be conscious of them? You ever been driving and you experienced highway hypnotism. You realize, "Oh, I haven't been paying attention for the last 10 minutes," and you realize the zombie has been doing it, because you're off.
- 2.66** But you weren't like that when you were first driving. You can't really carry on a conversation that way, although I think some politicians do. The problems that require you to be conscious of them and to really be engaging your working memory and your intention are problems that are novel, have a lot of novelty in them, like this lecture, I hope, a lot of complexity, like this lecture, and often have a lot of ill-defined

stuff in it, where we're trying to figure things out. You can't do that without consciousness, without working memory.

- 2.67** See, there are problems in which you have to ramp up your **relevance realization** because the problems—remember, remember real-world problems?—are complex, have a lot of novelty in them, and are very ill-defined. You need to really ramp up your **relevance realization**. It's plausibly about what working memory is doing. Chuderski, C-H-U-D-E-R-S-K I, in 2019, what he did was, he was giving people tasks and he was varying the amount of chunking needed to perform the task. So he could reduce the amount of chunking. You'd have to do a lot of chunking, you have to do less and less chunking in order to complete task. Is that OK?
- 2.68** And then he was doing that, and he was taking measures of general fluid intelligence. OK, what's general fluid intelligence? OK, so a distinction that was made in literature is between fluid intelligence and crystallized intelligence. Fluid intelligence is the one you're using online right now, situations of high novelty, high complexity, etc. What's crystallized intelligence? Well, it's your use of your knowledge. You can solve problems like math because you've learned how to do multiplication.
- 2.69** Now, a lot of people don't like this distinction. I'm one of them. Because you know what crystallized intelligence sounds like to me? Knowledge. Now I'm getting close to saying, you know what causes knowledge? Knowledge. Well, how do I get the knowledge that's in my crystallized intelligence? Well, I use my intelligence. What is your intelligence? It's my knowledge. That's not an explanation. That's a circle. Now, what you might say is, well, it's how I make use of my knowledge. But that's exactly what fluid intelligence is.
- 2.70** In fact, if you look, over a lifetime, this is unfortunately what happens to everyone. Crystallized intelligence and fluid intelligence tracking, and then fluid intelligence goes down, and crystallized intelligence just keeps going up. They pull apart. When you do measures of fluid intelligence, they are highly, highly, highly correlated with measures of general intelligence. Many people, I think correctly, conclude that fluid intelligence just is general intelligence, and crystallized intelligence just is knowledge, and we should keep those two separate.
- 2.71** OK, is that all right? I'm going to talk as if all these experiments are just about general intelligence, which is how they're actually taken up to be. What he found is that, even very simple chunking was highly correlated, 0.5. All correlations go between zero and one. Even simple

chunking of 0.5 was highly correlated with measures of general intelligence. 0.5, that's a big correlation. That's half of the variance in your intelligence, is just your ability to chunk.

- 2.72 He followed that up in 2022 to argue on the basis of a lot of findings in cognitive neuroscience. He's doing cognitive science, by the way, psychology and computational modeling. He's doing synoptic integration. He came that what's going on in working memory and reasoning is that fluid intelligence is "representing in the mind the key relations for the task at hand." What is that, by the way? Representing the key relations for the task at hand. That's **relevance realization**.
- 2.73 It looks like it's the **relevance realization** in working memory. The recursive **relevance realization** is the fluid intelligence, which is probably the same thing as general intelligence. Kyllonen and Christal did a very famous study, 1990. They were doing a study for the armed services. They were making use of the Armed Services Vocational Aptitude Battery, very powerful IQ test. Remember what I said? The military needs to know your IQ very well. They gave people, in addition, a bunch of working memory tasks that were highly, what are called **g**-loaded. They require a lot of general intelligence to do these tests.
- 2.74 First of all, they found, there was a common factor with all of these working memory tests. Then what they found is that that common factor in all of the working memory tests, the common function, was almost perfectly identical with measures of general intelligence. A lot of people, and this has been backed up by people like Colom, Oberauer, and, in neuroscience, by Bowren in 2020, a lot of converging evidence on this. A lot of people, yeah. Some people even just say this, "General intelligence is working memory function." Very bold.
- 2.75 Now, Deary, D-E-A-R-Y, in 2000, said, well— this is what he said. He said, well, he did more than say. He complained at the fact. He complained that explaining **g** with working memory was explaining one mystery with another. What? Like, "Oh, yeah, great. **G** is working memory." Well, what's working memory? Because it was not clear what the relevant functions in working memory was for researchers. Of course, what am I doing? I'm arguing that the central recursive **relevance realization** function of working memory, like all **relevance realization**, was left in the background. It was presupposed and not explicated and explained.

2.76 Again, general obviousness. It's obvious that that's what's going on in working memory. I bet you, if you could explain how working memory makes things obvious in that way, you'd get an explanation of the main function of working memory, rather than it being a mystery. See, I can simultaneously clear up the mystery of intelligence and the mystery of working memory, elegance, by proposing **relevance realization**.

2.77 Now, I think it's not, therefore, correct to say that working memory per se is general intelligence, but here's a quote from Haier, Colom, and Hunt, which is 2024. Very important textbook on intelligence, the science of intelligence. "Crucial feature, working memory's crucial feature, may be the temporary and reliable storage of pertinent"—it's a little bit, synonym for relevant—"pertinent information. Strong overlap between intelligence and working memory may be explained by a common capacity to build and keep relevant connections in the short-term memory space."

2.78 They actually reference Chuderski's work that we just talked about, about the chunking. They then cite Jolly and Dell, from 2020, "Reasoning is dependent on the ability to first hold task-relevant information in working memory." So, I'm not making this stuff up. They keep saying it again and again and again and again. It seems clear that recursive **relevance realization** is the working memory function that is being held responsible for general intelligence.

2.79 What are some of the other candidates that are proposed by cognitive psychologists? One is processing speed. We say things like, smart people are quick or they're sharp, and what does a sharp thing does? It cuts things quickly. A lot of people wanted this to be true. It turns out, there's a correlation between processing speed and **g**, but those results have been mixed. Some find it, some people don't find it very much. Even when confirmed, the correlation is often not that strong.

2.80 Now, why? Well, think about speed. If I speed you up, you know what I'm doing? I'm also speeding up the rate at which you make mistakes. That's not being intelligent. There's a speed-accuracy trade-off for a lot of things. In order to do something accurately, you have to do it more slowly. If you speed it up, you make more errors. If I just speed you up, you know what you're going to do? You're going to make a lot of errors. You know what you are when you make a lot of errors? Stupid.

2.81 It's like, oh, wait. This is where you're probably thinking, "Well, maybe it's not speed, maybe it's efficiency." Speed is part of efficiency, but you're balancing speed off against accuracy and getting the best

balance between speed and accuracy, and that's efficiency. That's maybe what brains are doing. Maybe smart people have more efficient brains. This is known as the neural efficiency hypothesis or the brain efficiency hypothesis. We're going to take a look at this in the next lecture because those are neuroscientific theories about the brain.

- 2.82** One last thing, visuospatial ability. This is your ability to manipulate objects mentally or visually. You can either do it in your head or you can do it with an actual object. Visuospatial manipulation. So you can detect hidden forms in a larger display. Where's Waldo? Oh, there's Waldo. You can rotate, how fast you can rotate a mental image in your head. Speed of recognition. How quickly can you recognize something? Change in orientation. What would the marker look like if I held it horizontally? What would it look like if I pointed it this way to you?
- 2.83** You could do all that, yes? And your memory for shapes. What was the shape of X? These are all the things we can measure and test to determine your visuospatial ability. Now, there's a general factor behind all these tests, something, a general factor, strong positive manifold, and that general factor correlates quite strongly with **g**. Now, of course, there's a problem with this right away that you should be seeing, but that overlaps with working memory. Yeah, it does. It overlaps with working memory. But there might be something more.
- 2.84** There's some really good work done by Manuel Juan-Espinosa, E-S-P-I-N-O-S-A, in 2000. This is really, really important work. They studied wayfinding abilities, how you find your way, wayfinding abilities. What are the things you have to do? You have to properly orient yourself. You have to represent the route to yourself. You have to survey your representations, and you have to update your position. What you can do is, you can measure all of these abilities, and these components correlate very well with visuospatial ability, and they correlate very well with measures of **g**.
- 2.85** Well, what's going on here? Well, it looks like imagery is helping us with anticipatory navigation of the world. I want to be over there and I need to get there, and I need to orient, I need to represent, I need to survey, and I need to navigate to get there. I have to wayfind in order to get there. Now, first of all, we're going to come back to this. But notice that that's a very adaptive feature that we're going to see next time, that this is a theory of how general intelligence actually evolved, this wayfaring ability.
- 2.86** The idea is, well, you know one of the things your ancestors were doing on their savannah? Yes, they're not thinking about door knobs,

but you know what they're doing? They're doing wayfinding. They're doing a lot of it because if they don't do it well, they die. If they do it well, they live. What it looks like we're doing is, we take that capacity for sensorimotor wayfinding, and this is the work of Barbara Tversky and a whole bunch of other people, and we'll talk about this more when we talk about exaptation. That ability has been transferred from sensorimotor navigation up into moving through an abstract space. Moving through a different abstract space.

- 2.87** In fact, the word abstract means to lift up, to move into a different space. What you've got here is, you've got this anticipatory, orienting, and navigation function. That's going to overlap a lot with predictive processing models of what it is to be intelligent. Now, that doesn't mean anything to you now. I need you to put a pin in it. But the visuospatial ability is your ability to take orientation and navigation, and exapt it up, and use it in abstract ways to think about possible worlds. Not only possibilities in this world, but imaginary worlds that you could move through.
- 2.88** You watch the Lord of the Rings and you go on this long journey through a place that doesn't exist. Don't tell some people that, by the way. I think there's an important clue here in the visuospatial ability. It's pointing to exaptation. We'll come back to that. It's pointing to the fact there's this ability in addition to **relevance realization**, there's ability to anticipate the world and to orient so you can navigate. I'm going to propose that those two abilities are actually interlocking, because the more I have to anticipate the world, the deeper I can anticipate it, that is a way of determining how intelligent something is.
- 2.89** But the more I do that, the more information and possibilities I have to consider. **Relevance realization** goes up as I increase the depth of my anticipation. So anticipatory **relevance realization** could be a key function for making you generally intelligent. With that, we're done the psychology part of intelligence. We'll now move to Q&A.
- 2.90** OK. So in terms of **relevance realization**, what about **relevance realization** being dependent? To me, I hypothesize that it's dependent on personal affinities, personal interests, what you care about, and the impact of these on trainability of an individual. For example, you can see this in people with ADHD who also happen to have issues with working memory. So that doesn't necessarily impact, for example, these people might have a harder time being able to focus on certain well-defined problem, subject areas, but that doesn't necessarily impact their ability to solve ill-defined problems.

- 2.91** Whereas on the other hand, I've met plenty of people who are really, really good in school, and they're really good at solving all sorts of well-defined problems, but that doesn't necessarily translate to them solving ill-defined problems, like street-smarts. You'll find plenty of people who are really smart in school, but in a street, they'll walk into a road. It's terrible. So how does that all work? Because it seems to me that personal interests, affinities, make, are what help people have the tools and skill sets in different problems. But is it actually even really possible, because of our personal affinities and cares, to be generally well-rounded in a lot of different areas?
- 2.92** OK, this is excellent. Let's do the ADHD first. It's like, maybe these people are good at well-defined problems. Here's a current, and it's a well-respected theory of what's actually going on in ADHD. ADHD is not actually a disorder or a disease. You know what you're really good at, if you have the kind of cognitive style that somebody with ADHD has? You're really good at wayfinding. You're really good at navigating. You'd be a really good hunter and gatherer.
- 2.93** People in the evolutionary environment in which human beings evolved that had this style would actually have been high-status individuals. But in our world, as you rightly pointed out, that's sedentary and symbolic and focused on well-defined problems, they seem to suffer. So yes, which means we need a theory of intelligence that can reach into ill-defined problems and not be locked to what's going on in well-defined problems. That's what I'm trying to offer you.
- 2.94** Second point, can you be well-rounded? There are things you can do to improve using your **relevance realization**. You can, for example, when you're studying, you could do something really stupid. You could study by merely rehearsing the material. I'll read it over to myself multiple times. Or you could do this. You could do a lot of chunking and then chunk your chunks, and then you'll remember a lot more. There's ways in which you can play along with your **relevance realization** and improve your functioning in many areas where you may not initially be able to function that well by tapping into the general processes that are at work in **relevance realization**. Is that OK?
- 2.95** Now, the third point, well, isn't it based on my personal affinities? Yes and no. I mean, you're right. **Relevance realization** is not cold calculation. It's how you care about information. This is how you're different from a computer. This is how you're still different from the LLMs. They don't care about the information they're processing. You do. This is really a significant, important difference. Now, the thing is,

is that just personal in the sense of subjective? Can you solve your problems according to subjective whim? No. It has to be a fitting between your personal goals and desires and the objective demands of the world. It's a fitting of the subjective and the objective together.

- 2.96** And so, yes, the personal affinity. Now, what does that mean? Will everybody make the same judgments of relevance? Of course not. That's why we disagree. But although the products are different, the same process is at work. **Relevance realization** is like evolution. Evolution is a universal process. It doesn't make homogenous results. It makes exactly the opposite. It makes an increasing variety of different organisms. But we still know that all of those different things are due to the same universal process. So it's operating. Evolution is operating generally, even though sharks can only operate in the ocean. Is that OK? All right. Another question.
- 2.97** General intelligence is the correlate that predicts success in not just one field, but multiple fields. And so this led me to wonder about what we can learn from exceptions to this rule. I'm thinking in autism, where you have people that are very proficient in one field, but that are not so good in other fields. This leads me to see it as maybe a different model, or what do we learn from this?
- 2.98** I would say it's the same model. In fact, in 2022, with Brett Anderson and Mark Miller, I published a paper on applying **relevance realization** theory to the psychosis-autism spectrum. By the way, we're all on that spectrum. What the argument is, and we'll get more into this, but with **relevance realization**, you're always dealing with trade-off relationships, like the speed-accuracy trade-off or things like that. What you can see is, people vary in to where do they prioritize how they're—like, there's two poles in a trade-off, and people can vary in which pole they're prioritizing.
- 2.99** There are people that will prioritize, for example, do you want to pay attention to the features or to the gestalt? What you can find for many people who are diagnosed with autism, autism is a spectrum disorder, but many of them have what's called weak central coherence. They tend to be very good at features and very poor at gestalts. That's why they're very good at debugging or editing. Other people who are schizotypal, more prone to psychosis, they see gestalts everywhere. That's a problem when it goes too far because they come up with conspiracy theories and weird connections. The model actually explains the variations quite well.
- 2.100** OK, so I'm not sure if this question is going to make too much sense. I feel like I understand essentially what you're trying to do as—

2.101 Oh, good.

2.102 Part of me is thinking, OK, it's a question of what intelligence is, but the other part of me is thinking, why something like evolution would create something like intelligence? So there's the why of intelligence. With a lot of the stuff that you've been saying, what I find surprising is that you're not touching a lot on what Jordan Peterson talks about in regards to the reality of pain and how relevant that is to intelligence and how we act and react to intelligence. I'm going to ramble for one quick second.

2.103 I just want to chunk this. The first thing is, why did it evolve? The second is, what relationships does it bear to pain? Is that OK? Am I getting you? OK, what's the next? I'll try and get what I can out of the rabble. I'll try and deduce the right—

2.104 For sure. With that notion of pain, and so some of the notes that I have here is just in regards to the top-down, **bottom-up**, and all of the reasons that we have these reactions is, I would argue, to pain. Now, the problem with the homunculus problem and issue with that is, how relevant do you think it is to connect that homunculus person to a pain person, where you have a pain person here?

2.105 OK, so let's do those in the right order. First of all, why did general intelligence evolve? General intelligence evolved because our ancestors adopted a nomadic strategy. A nomadic strategy is, you move around a lot. When you move around a lot, you don't want to get the organism really locked into a particular niche.

2.106 To avoid pain, to avoid—

2.107 To avoid dying off as a species. A lot of organisms don't adopt those strategies. A lot of organisms are very tightly adapted to a very specific environment. Now, what do they gain from that? They don't have to rely on a huge amount of learning because they can come really ready to go by evolution. What's the price they pay for that? If that environment goes away, they go away. There's a trade-off there. The answer is, our ancestors, they didn't choose. But what happened is, they evolved to be nomads because that is often a very good strategy when the environment is volatile and changing, and the environment in Africa at that time was very volatile. There's the reason.

2.108 The second one, pain. Jordan Peterson, my friend, talking about pain a lot. The problem with pain is, it's not very discriminatorily helpful to you. What you may say, what's the function of pain? Well, the function of pain is to tell me when things are dangerous. Because the point of

pain isn't the moment of pain because that's useless. You're just in pain. What's the point of pain? The point of pain is for you to predict things that are dangerous so that you avoid those in the future. Are we agreeing with the function of pain?

2.109 OK, so all we need now are the universal features of dangerous things. What do all dangerous things share? Well, some of them have sharp teeth, some of them don't. Some of them are like poison. They're not even organisms. Some of them are killer bees, some of them are cats, some of them are holes in the ground, some of them are too much water. And so, guess what you need to be able to do, to predict what to avoid in the world? You have to anticipate the world and do **relevance realization**.

2.110 Because the point of pain is not the moment of pain. The point of pain is predicting what's going to be relevant to you in the future. Pain is a signal that you've made a mistake. The point is, mistakes are not a part of the physical universe. The universe isn't making mistakes. Mistakes are events that are relevant to you because of the goals that are relevant to you. Is that OK?

3. The Neuroscientific Perspective

At a glance

Emphasizes epistemic humility given how mysterious and controversial the nature of intelligence remains, while building toward a cognitive scientific model developed across the lectures. Frames the instructor's approach as advocating the strongest available arguments and evidence rather than claiming final authority.

- 3.1 Last time, we were opening this topic, and I opened with something that I think is really important to remember throughout. The nature of intelligence is very mysterious, very controversial. You get a wide variety of answers. "We don't know what it is," or "I know it's X," in the literature. We have to be really, really careful. We have to spend some time on how we're going to set this up so we'll be very responsible to how the science is actually existing right now.
- 3.2 I'm going to try and give you what I think are my selection and my argument for what are the best theories, the best arguments, the best evidence, and try to make my best argument, strongest argument, as to what the nature of general intelligence is. I gave you a proposal about what that is. I'm going to do a plausible cognitive scientific account, and we went through what plausibility is. Remember that convergence for trustworthiness, elegance for power, balance between them, and you want a model that is coherent, that has a structural functional organization that rules out alternatives and rules in what the model can actually explain and predict and all of that.
- 3.3 I'm building that, and I hope you saw that was taking shape in lecture one. Then we need a **cognitive scientific model**. Why? Because when we're talking about intelligence, we're talking about something at many different levels of analysis, different levels of reality, neuroscience on the brain, artificial intelligence and learning algorithms, psychology and behavior. And so, if we don't pay attention to these levels, we can fall into equivocation. What we need to do is consider the very plausible proposal that there's causal interaction between these levels of reality.
- 3.4 If we don't get them properly integrated, we will miss out on that causation, that mutual constraint. So not only will we have equivocation, we'll be missing important information, we'll have an incoherent picture. And incoherence and incompleteness with equivocation makes a good chance our theory or our idea or our conclusions about intelligence will be false. We want to avoid that. We want a plausible cognitive scientific theory. We're going to try and do synoptic integration.

- 3.5** Now, we took two lessons from cognitive science. One is we want a theory that will generalize in a way that will incorporate multiple realizability, that intelligence is realized in multiple species and perhaps also in things that aren't even organic at all, like artificial intelligence. We also want to not ever engage in the homuncular fallacy in which we presuppose intelligent agents in any model that we're using to try and explain intelligence. Then I know that seems like nobody would fall into that, but we saw a couple of instances where this looks pretty homuncular all the way through.
- 3.6** We then took a look at the overview for the course. I won't go over that again, but we started with the psychological domain. There are two components to that. There's the psychometric, which is the measuring of intelligence, and the cognitive was an attempt to explain what are the cognitive processes that work in intelligence. They come together and give us our psychological theory of intelligence. But we have to keep in mind that we also need a categorical theory. We need a theory that does not only apply to human beings, that could apply to dogs, octopus, ChatGPT-4, et cetera.
- 3.7** What we're going to do is follow that. We're going to take a look in the course at the psychometric and the cognitive, which we did last time, and then we're going to move into the neuroscientific today. We took a look at the history of the psychometric approach to intelligence, and we took a look at the first significant model, which was from Binet and Simon, significant in that it hung together well and it was predictive in a powerful way. Remember, they moved the test from very low-level to testing skills of judgment and problem solving, and that turned out to be very prescient.
- 3.8** But I reminded us at that point that we got to be careful about problem solving because there are two types of problems. There are the types of problems that were used in the Binet and Simon, and are continuing to be used in many different psychometrics. These are well-defined problems, and they're very measurable, but they are not the same as a lot of real-life problems, which are ill-defined problems. We keep coming back to that again and again. We did throughout the first lecture, we're going to keep bumping into that. You asked an excellent question about that.
- 3.9** What seems to be going on in both of these are some notion of reasoning. It's supposed to be the ability that gives us the capacity to be intelligent in a wide variety of domains in a wide variety of ways. But there are two versions also of reasoning, and we don't want to equivocate between them. There are well-defined, like induction with probability theory, and deduction with logical theory, compared to ill-

defined notions of reasoning that are nevertheless important, like abduction, inference to the best explanation, eduction, like Spearman talked about, and analogy, which Spearman also talked about.

- 3.10** Speaking of Spearman, we moved on to Spearman and his great discovery, which really has been seminal for the whole psychometric approach to intelligence, the discovery of the positive manifold and the postulation of the g factor of intelligence. I'll often refer to this as general intelligence because many people do. There's some people that want to split it. I don't think much is gained by trying to split that particular hair.
- 3.11** First initial question that came up in history is, is this a statistical artifact or is it something real? There was a lot of ink spilled on this, but the consensus now is that g is pointing to something real. It's not a statistical artifact. Nevertheless, there are two competing models about what is being pointed to. One is a top-down common cause. This is Spearman, and he had two proposals. One is mental energy. That's turned out to largely be a nonstarter. The other one, very prescient, that the general ability is the ability to apprehend a situation, size it up, as we talked about, and the eduction of relations and correlates, do pattern detection, pattern completion, and analogically project that pattern to other areas.
- 3.12** We talked, or maybe I didn't, but one of the things I can mention is, one of the most powerful measures for general intelligence is exactly that. It's called Raven's Progressive Matrices. What it does is it gives you a bunch of geometrical patterns, and you look at the patterns, and you have to predict what is coming next in the pattern. It's very pure because it has nothing to do with your culture, your knowledge, or anything like that. People take it to be a very good measure. It's an indication of something I said. There's been a lot of work done to try and make these tests free from cultural bias, historical bias, et cetera. That's a good example of it.
- 3.13** That's a top-down common source, common cause. Then there's bottom-up, which is common effect, and this is the idea, you have a bunch of factors, and they all work together. When they're working together, that's what gives you the general reasoning that is constitutive of general intelligence. We noted Process Overlap Theory with its unfortunate acronym by Kovacs and Conway is one of the most prominent versions of this theory. We'll take a look at P-FIT, which is a particularly powerful instance of a bottom-up common effect model. What these people are proposing, instead of there being a system-wide function, there is a system that then functions to generate general intelligence.

- 3.14** As I pointed out, there's quite a lot of evidence for the P-FIT theory, and we'll get into that today. But also there's strong evidence for a failure of the training in components, like working memory, to transfer to general intelligence, which seems to indicate that it should be a top-down approach rather than a bottom-up. We can keep going back and forth on this, which people do, by the way. But I think Gottfredson is right that this is an issue about the levels of analysis. When you shift the level of analysis, you move between a single thing and a system of component parts.
- 3.15** She points out that Kovacs and Conway even think that general intelligence is captured by domain-general reasoning, and why isn't that a process? You can go on and on and up about this. My point is, we're in the levels of analysis problems, and this is exactly where we need to do cognitive science, about getting the levels of analysis, the different disciplines, to talk to each other in a more coherent way. We basically ended the psychometric approach there, and we moved to doing that cognitive science by picking up cognitive psychology and looking at cognitive models, where cognition has to do with how we know the world.
- 3.16** Then we looked at the standard information processing model of cognition. And notice again, the computer metaphor being smuggled into the psychology. We talked about categorization, and I argued that categorization requires relevance realization. We looked at prioritization of information, foregrounding and backgrounding of information that requires relevance realization. Problem-solving requires relevance realization. We took a look at working memory and the fact that it works like a blackboard or a global workspace.
- 3.17** The global workspace theory is that the function of the global workspace is to solve the frame problem, which is to solve the problem of relevance realization. Already, we see that cognition is dripping with presupposing relevance realization. I'm going to, as you can no doubt guess, or perhaps better predict, I'm going to argue that that is what general intelligence is. I'm going to keep building that argument. It's like, well, if it's so pervasive, why isn't it made an explicit foregrounded theory? I argued a lot of that has to do with the fact that relevance realization generates obviousness, and people tend to work from obviousness, rather than trying to get behind it and explain how obviousness emerges.
- 3.18** I would like to know, as a scientist, how your brain generates obviousness in such a way that you can solve a wide variety of problems in a wide variety of domains in a wide variety of manners. We then took a look at the idea that maybe what's going on in general

intelligence is just the function of working memory. This is plausible. Working memory is really central. We took a look at Baddeley, and we saw about how he's really presupposing relevance realization in what working memory is doing. We've got the central executive in there, but the central executive is really homuncular and just presupposes relevance realization.

- 3.19** We went to Lynn Hasher's work about, well, working memory isn't a holding space, it's a relevance filter, and we talked about the importance of chunking. Then we took a look at the work of Chuderski, and even simple chunking is quite significantly correlated with measures of general intelligence, like 0.5. More recent work that he did in 2022, that general intelligence represents in the mind the key relations for the task at hand. It's doing recursive relevance realization, which is the thesis I'm arguing for.
- 3.20** There was a series of studies showing a high correlation between working memory and g. But then Deary complained, "Oh, right, great. You're explaining the mystery of intelligence with the mystery of working memory." Well, what's mysterious about working memory is exactly, I think, the fact that it ultimately relies on relevance realization, and that is not being foregrounded and explained. I gave you a series of quotes around that. Then we started taking a look at the idea that it's not working memory, it's the central function, and the quotes pointed to the fact that the central function is recursive relevance realization.
- 3.21** Then we looked at alternative candidates in addition to working memory function. Well, what might general intelligence be? We looked at processing speed. That was problematic, and it actually folds into the idea that intelligence is something like neural efficiency or brain efficiency, the efficiency model. Then we took a look at visuospatial ability, and that turned out to be much more powerful. There's plausibility in that because it seems to be related to wayfinding and the sensorimotor orientation and navigation. This is very adaptive, and we'll see it has to do with the evolution of general intelligence.
- 3.22** The idea is, when wayfaring, we get anticipatory navigation. We've got that anticipation and the interlocking, recursive relevance realization going on, and that gets exapted, we'll talk more about that today, into our imaginal capacity for anticipating and navigating worlds, abstract worlds or possible worlds other than our own. We, therefore, are adding to the proposal that general intelligence is not simply a recursive relevance realization, it's anticipatory recursive

relevance realization. That was the review. So new material, and now we're moving squarely into neuroscience.

- 3.23** Neuroscientific theories talk about brain structures. They talk about anatomical networks and functions. Obviously, it's based on evidence that's drawn from the brain, not through behavior, and then inferred about the brain, but in which, in some ways, we are directly observing the brain, EEG, fMRIs, a bunch of new technologies that I won't go into, but the Haier et al book from 2024 has a nice section going over a lot of these emergent technologies, and they're helping a lot.
- 3.24** Now, in keeping with synoptic integration, although I'll be talking about neuroscience, I'm going to be toggling back and forth, as many people do, by the way, in cognitive neuroscience, between cognitive psychology and neuroscience. We will be toggling back and forth between those. Now, although that happens a lot, it's typically not methodologically foregrounded. People don't stand back and talk about, well, how do you do this well? How do you toggle well? How do you toggle poorly? How do we evaluate where you should toggle and when you should toggle? This is largely done—and I'm making a methodological criticism—it's largely done intuitively and implicitly.
- 3.25** Why? Let's do it reflectively, explicitly. That would mean we could be more careful in the toggling. One of the things we want to be as good scientists is we want to be careful. Now, the most impressive neuroscientific theory is the P-FIT theory. Remember, that stands for parietal, this area of the brain, frontal integration, degree to which you're integrating what's happening in the frontal areas of the brain with the parietal areas. I'm actually technically pointing towards the prefrontal, but I can't literally—it's there, trust me.
- 3.26** The idea here is, when I'm measuring differences in intelligence, general intelligence, what I'm measuring is differences in how well people dynamically integrate. Why do I say dynamic? Because it's not just [noise]. It's [noise], back and forth. It's dynamic. It's constantly self-organizing and correcting. It's a flowing integration, not a static. But what I'm measuring is the dynamic integration of their frontal and parietal areas of the brain. That's what we're measuring. Some people are better at integrating, and that makes them more intelligent.
- 3.27** General intelligence is the dynamical integration of the parietal and frontal areas of the brain. Haier. I don't know how his name is pronounced. H-A-I-E-R. I've been referring to him repeatedly. He's the first author on the book on the science of intelligence. He has a book from 2017 called *The Neuroscience of Intelligence*, where he makes a very strong case. He masses a lot of evidence across a lot of time,

pointing to the P-FIT theory as the one that the neuroscience is providing strong and convergent evidence for.

- 3.28** Then, of course, the recent book that he is doing with Colom and Hunt from 2024, *The Science of Human Intelligence*, second edition, they also go through gathering all of this evidence together. There's a lot of publications. So, P-FIT, I wouldn't say it's absolute consensus, but it's a really, really well-respected theory of what the neuroscience of intelligence is. There's a lot of careful and thorough convergence of evidence. It's a highly plausible theory, convergence. It explains a lot, it's quite elegant, and there's a lot of empirical evidence for it, so it's also quite probable. It's highly plausible and highly probable.
- 3.29** It doesn't mean it's true because we've had highly plausible, highly probable theories before— Newton's theories of physics, that turned out to be in some ways, fundamentally false. But nevertheless, we work from highly plausible, highly probable models in science. I'm not here to criticize or critique this theory in terms of whether or not it's true. I think it's largely true. But that's not enough for a good theory. I want to criticize aspects of this theory that are the case, even if it turns out to be true.
- 3.30** It's important to do that in science. People think that science is about gathering facts, gathering disconnected truths. That's crap. This isn't good science. I wonder if there's a correlation between the number of black socks in my upper drawer and the paper clips in my bookcase, and I wonder if that tells me anything about the history of tin mining in Albania. Is that good science? Do you care about it? No, because you want truths that explain the world. Science is about explanation. It's not just about prediction. Science answers questions.
- 3.31** Now, that brings me to my first issue with the P-FIT theory. What you don't want is what I would call a cartographic theory. You know what cartography is? You make a map. Here's the United States, here's Canada, things like that. You don't want this. Well, here's this area of the brain, and here's this area of the brain, and they're active when you're intelligent. It's like, OK, you're telling me this is where it happens, but you haven't told me what is happening. What's happening there?
- 3.32** By the way, this is one of the problems with fMRIs. People go, "Oh, look, when he's thinking of his grandmother, this area of the brain lights up." Big deal. You need to have an explanatory function, but not where, but what's going on there. So, what are the function, or functions—what is the function or what are the functions that explain how people solve a wide variety of problems in a wide variety of

domains in a wide variety of manner, wide variety of ways? Now, of course, I'm making an argument for some integration of anticipation, which will be covered with predictive processing theory and relevance realization. But let's put that aside for now. That's more where we're going.

- 3.33** If you take a look, what seems to be going on, although you won't get anybody explicitly saying this, this is me culling from a lot of material, is this possibility. There's a self-organizing system of selection and variation going on. What happens is, the frontal area of the brain, and this seems to be one of its jobs, is a selection. What it does is, its job is to select information, sculpting the response space, as some people say. Whereas the parietal gives you a variety of functions. The parietal function is integrating a lot of different functions together.
- 3.34** So, what you've got is, you've got a source of variety, of variation, and then you have a selection process. This is very analogous to evolution. In evolution, you have a variety between the members of a population, and then you have selective pressure on it. Many of the organisms die off, and the ones that survive, they then reproduce, and the variation now opens up again, and then you get selection and variation and selection. It's an evolutionary model. What you would get is, if that's right and you make it work really tightly, you could use the fact that evolution is a kind of theory that explains how you can get intelligence out of things, out of processes that are not themselves intelligent. We'll have to play with that, but that's important.
- 3.35** Now, I want to pick up on a promise I made—why what I just said is plausible. You'd look at one of the Genovisia from 2017, a proposal about, well, how did this general intelligence evolve? And notice how this will be convergent with what we talked about with the wayfinding. The basic argument is that general intelligence evolved from a foraging capacity, a capacity to do this exploratory moving around in the world and navigating in the world. In fact, there have been instances where people have taken foraging strategies from living organisms, foraging and/or hunting strategies, and they have then translated them into search functions for artificial intelligence, often quite successfully.
- 3.36** The idea is that we evolve this capacity for foraging, and then it gets exapted into our capacity for general intelligence. She has a really—she or he, I can't remember, I'm sorry—they have a really nice way of putting it. "The evolution from foraging to foresight," which is a nice turn of phrase. You're doing foraging, and then you use that to do this much more developed anticipation and relevance realization in a much more abstract way. That's a really nice turn of phrase. I mean,

scientists should make an attempt to give things really good names, really good expressions.

- 3.37** What you've got here is, you've got this idea that this sensorimotor capacity gets exapted. I keep invoking this notion. Let's quickly talk about it. I'll use a standard example, my favorite example because it's really non-controversial. I'm using my tongue to speak. Without my tongue [noise], it doesn't work. I really need my tongue. Now, there are many organisms with tongues. My cat has a tongue, my dog has a tongue. You know what they don't do, except in nightmares or something? They don't talk to me. They can't speak.
- 3.38** So tongues did not originally evolve for speech. They evolved to move food around in the mouth. That's called mastication. They evolved, so they're really flexible. Then they evolved for the detection of poison and sugar concentration, taste buds. So very highly sensitive, lots of nerves, very highly flexible. Now, that turns out to be a really good organ for the articulation of sounds because the tongue happens, for evolutionary reasons, to also be in the air pathway. We, of course, have vocal cords like other organisms do, so we can make noises.
- 3.39** Now we've got, if I can make noises and I have my tongue, I almost have what I need to make a speaking machine. I don't—evolution, I'm sorry, I have to speak anthropomorphically here. Evolution doesn't have to design a speaking machine from scratch. It can repurpose the tongue for speech. It's been exapted for speech. The idea is, work by Michael Anderson and Dehaene and a whole bunch of other people, is this is not only happening across species, it's happening across stages of your development. You are building sensorimotor machines in your head, and then they're getting exapted to do more abstract things.
- 3.40** The idea is, in evolution, we had foraging, and then that machinery got exapted across species into foresight. But it's also the idea that that happened developmentally. At one point, and this is no insult to you, you were a forager. You're a little kid, and all you do is you forage. That's it. You're just going around. "Can I put this in my mouth? Is this good? Can I eat this? Can I play with this? Is this candy? Is this a toy?" That's all you're doing. You're foraging. The adults are running around making sure the universe doesn't kill you while you're doing that.
- 3.41** It's like when you're a parent. I've raised two children and four stepchildren. It's like, for the first year of your life, for the first year of their life, all you're doing is keeping them alive because, like, wow. It doesn't seem this way from the outside, but when you're a parent, you get this weird mindset where everything's trying to kill my kid, and

you're just trying. Because all they want to do is forage. But the idea is, you don't remain a forager in the sensorimotor. You exapt that up. You use that machinery to explore abstract ideas. We are foraging foresightfully right now together. And notice that we call it a course that we're making our way through, and we want to make sure we're on track as we're doing it. You hear all the language? OK.

- 3.42** Now, this converges with a lot of evidence for what people think, what's going on between the prefrontal cortex and the parietal cortex. The idea is that here you've got this selection process and this variation process, and you're evolving. What are you evolving? The idea is, it's almost like these areas of the brain are self-organizing different programs for how to behave. These are called configurations. The brain, certain areas, are engaged and activated and coordinated. That's a configuration. Then what's going on in this loop is constant reconfiguration. Does that make sense? It's almost like a computer that was constantly writing programs, not just applying them.
- 3.43** Is that OK? That's reconfiguration, and you can actually measure that. There's a lot of powerful ways of measuring that. And so, if you take a look, for example, at Scholz and Kohl from 2016, they are wrestling with a question here. They're wrestling with a question that goes like this. The brain exceeds computers, even still right now, by being very domain general with a very little amount of energy. Your brain uses the energy of a 60- to 100-watt light bulb. And yet [noise].
- 3.44** Now, the machines that are the best approximation to that are the LLMs, ChatGPT. They require a tremendous amount of energy to train them. Tremendous amount of energy. It's something like the kilowatts for running a city like Toronto for a couple of weeks. It's huge. This is one of the reasons why we won't have a billion of these robots running around automatically. I was talking at lunch. If you take a look inside the guts of these people who are predicting, "There's going to be these robots everywhere, like iRobot or something like that," they'll say, "And inside every robot is like a little miniature fusion reactor." And I go, "Wait, slow down. When did we get fusion reactors? And when did we get them so they were miniature?"
- 3.45** So, this is actually a significant engineering issue. So your brain is really efficient. Remember this proposal, and we'll come back to this. Now, what might it be efficient in? This is the idea that maybe what your brain is efficient at is the process of reconfiguration, constantly evolving, right? You are programmed to fit a changing world. Is that OK? Is that landing? Now, how would you go about investigating that? Well, there's ways of measuring your resting state of your networks.

You can look at patterns of functional connectivity, how neurons are firing together.

- 3.46** Now, what happens is, when you go from resting state, you're not doing anything. You're just—then I give you a task. What you can do is you can watch how the connectivity patterns are modified as you take up a task. Does that make sense? Of course. What you seem to be doing in the task is fitting your cognition to the specific task. You've got this resting state, and then you fit it to the specific task. Now, there's two things that might be happening when you're doing that fitting. That refitting is the reconfiguration process.
- 3.47** There's two possibilities that might be happening here. Large changes in configuration. There's a lot going on. You're making a huge change. Now, why would you make a huge change? Because you really need to fit to this circumstance. You really need to tightly fit to it. Or you could make small changes that reflect an optimization. What does optimization mean? Optimization is not the same thing as maximization. Maximization is, I get what I want by just taking some process to its maximum.
- 3.48** So people try to maximize their income. They don't say, "No, I want, no, more." But you can't do maximization, and that's why I've been pointing these out, where you have trade-off relationships. Because if I maximize in a trade-off relationship, I will lose the other side in the trade-off relationship too much. Let's do one from foraging. Here's a trade-off relationship. One of the things I can do is explore, obviously. But you know what I don't want to do? Just explore because, oh, there are the berries, there are the nuts. I keep walking. No, you got to stop and you have to exploit. But now I'm stopping. And while I'm exploiting, what's happening?
- 3.49** Well, I'm starting to lose resources because I'm eating these up and I'm missing opportunities because just over the hill—this is a meager berry bush. But over that hill, unbeknownst to me, is the most glorious berry bush of all time. And if I just went over the hill. And so what do you do? Well, I explore all the time. But if I never stop and eat, I starve. But if I stop and eat too much, I starve. So if I just maximize exploration, I starve. If I just maximize exploitation, I starve. So what I want to do? I want to optimize between exploring and exploiting.
- 3.50** Now, that tells you right away that, even in navigation, we're dealing with a trade-off relationship of an important kind. Is that OK? Now, think about this. Why is the brain optimizing? Because there are times when you need to be really specifically fitted to the situation. That's

the only way you're going to achieve your goal. You got to be really fitted to that situation. But if you overfit to this situation, that's like staying in one place and exploiting that one place, and you start to lose the ability to explore other places. Does that make sense? You're trying to optimize in some way.

- 3.51** Now, what they found in this trade-off relationship between a domain-general application, I can do very little reconfiguration because I'm trying to find a way that will generalize as much as possible. I do a lot of changes in order to get a specific fit. What did they find in that trade—how the brain handles that trade-off? They obtained evidence that individuals with higher intelligence show smaller reconfiguration moves. They don't reconfigure as much. They're somehow really good at optimizing so that they can exploit when they need to, but they generalize well. They're really efficient at getting that trade-off working.
- 3.52** Now, just to sneak it in, when you have a process that's trading between exploration and exploitation, it's doing—that's one of the ways in which you do relevance realization. We'll come back to that. There's prioritization for domain-general performance. That's putting selective pressure. There's variation that would allow you more specific models, but that's constrained. What you do is you evolve the kind of stance you have towards the world.
- 3.53** Now, what do I mean by that? I'm a martial artist. For example, if I'm doing tai chi, you do a fighting stance like this. There's variations on it, sometimes a little deeper, whatever, because it's a center line style. Now, you don't fight with this. This doesn't do anything. That doesn't do anything. Why do I stand like that? Why does any tai chi martial artist stand like that? What's the reason? Well, what they're doing is, they're trying to find a way of standing which involves the smallest amount of reconfiguration to get to this move or to get to this move.
- 3.54** Do you see? They're trying to find the stance that's the optimal place so I can do the minimal amount of reconfiguration to get to more specific things I need. Does that make sense? Does that make sense as an explanation? Because you're not going to find that language in the literature. This is me trying to give you a way of understanding it from something that's a little bit more concrete. So, your cognitive stance, which is kind of your forward-facing framing of the world, orientation that prepares you for navigation, what's happening in people with higher intelligence is that they're evolving their cognitive stance really well.

- 3.55** Moment to moment, they're evolving much better so they can do small amounts of reconfiguration and solve all the problems they need to solve. That's pretty powerful, eh? I want you to notice something, that we can actually gain a lot of traction not only by paying attention to information processing models drawn from computers, we can get a lot of traction on general intelligence by paying attention to biological models and sensorimotor models. That's another argument for doing cognitive science.
- 3.56** This idea is, intelligent people are really good at evolving their mental set, their forward-facing framing for anticipatory navigation, for foraging into foresight. This idea about being really good at reconfiguration was independently confirmed by Thiele et al, T-H-I-E-L-E, in 2022. There's more and more people are talking about this idea of what's going on in P-FIT is this optimization of reconfiguration, the evolving of your cognitive stance. "Ceteris paribus," that's Latin for everything else being equal, if she's better at a cognitive stance than her, you're going to solve more problems more effectively than she does. That would make you more generally intelligent.
- 3.57** Just like, who do you want to be with when you're trying to navigate the environment? The person who's really good at orienting and helps navigate you through, so you're not getting ensnarled in the brambles and tripping over rocks, or the person who's just like—and you have to make huge course changes, and you want to be with the first person rather than with the second. OK. Now, this would also help to converge this whole line of evidence with another proposal, which is the one we've talked about a couple of times, the idea that more intelligent brains are more efficient, called the brain efficiency hypothesis or the neural efficiency hypothesis.
- 3.58** Well, because what we should ask is, efficient at what? Here's a specific proposal. Efficient at this dynamical evolution, not biological evolution, information processing evolution, of your cognitive stance. You're optimal at reconfiguration. Now, this idea about evolving your cognitive stance converges with a lot of work that's being done right now. I mean right now, but has a long history to it, too.
- 3.59** Well, long in the history of psychology. The history of psychology goes back about—when I'm doing psychological work, you're not supposed to cite stuff more than 10 years old unless there's good reason for it because it's probably been overturned by more recent material. When I say it's gone back more than 10 years, I don't mean it goes back to the Stone Age or something like that. All right.

- 3.60** A lot of research, both cutting-edge and going back, pointing to deep connections between general intelligence and cognitive flexibility. Cognitive flexibility. Let's talk about what cognitive flexibility is. I'm going to put up something you've seen before. If you were in the Primacy of Beauty or if you were in Intelligence, Rationality, Wisdom, and Spirituality, which is the nine-dot problem.
- 3.61** One time I came in to lecture, and there was a box of Rice Krispies sitting on my lectern. I thought, "This is a really ineffective bribe." I thought, why? They said, "Turn the box around." I turned the box around. You know what was on the back of the box? The nine-dot problem. I was driving in northern Ontario. I don't mean like Arctic northern, but northern enough so that when you stop for food, the name of the restaurant is Diner. That's it. I went in, I sat down at the table. Have you ever been in the diners and they have the paper placemats? And guess what was on there for me to pass the time with? The nine-dot problem. It's following me. It's following me.
- 3.62** OK, what is the nine-dot problem? Have you seen it before? Don't say anything because all that shows is you have memory. OK, so you have to join all nine dots with four straight lines, and you have to start the next line from the terminus of the previous line. People reliably think this is a very easy problem. They go, one, two, three, one, two, three, four. It missed the center dot. Right. One, two, three. No, one, two. Oh, oh, oh. The statistical solution rate for the spontaneous solution of this, that means not given extra time or some hints, is not distinguishable from zero. People find this a very hard problem.
- 3.63** It shouldn't be. Here's one line, here's another line, and four lines. When I was younger as a professor and cruel, I used to get people to come to the board and do this. Then I realized when I started to become more mature, don't do that. That's really not nice. But even when I do it, people will often shout out, "You're cheating." Why? I didn't break any rules. Four straight lines. They said, "Because you went outside the square."
- 3.64** By the way, this is where that comes from. "Think outside the box." Weisberg and Alba gave people nine dots, the nine-dot problem, to solve it. They couldn't solve it, and then they gave them this clue. "Think outside the box. Go beyond the square." This is great. You know what their experiment showed? That saying to people, "Think outside the box," does not help them to solve this problem. Telling people, "Think outside the box," means you haven't paid attention to the actual experiment from which the expression is drawn. That means there's something else going on in this. It's not just a matter of changing your belief.

- 3.65** I know that I need to go outside the box, but that doesn't mean I need to know how or what it would be like to see the problem in the way so I find the right things relevant to solve it. There's a lot of things that are needed beyond changing your beliefs. What's happening? This is called an insight. If people get this, they go, "Aha," and they get that flash and that super-salience, and they feel really connected to the problem. Insight is predicted by cognitive flexibility. People who are better at cognitive flexibility or who do things to get their cognitive flexibility going, like mindfulness practices, are better at insight.
- 3.66** There's an important relationship between cognitive flexibility and insight. What's going on in cognitive flexibility? What did you have to do to solve this problem? You had to break an inappropriate frame. You were framing it as a connect-the-dot problem, a standard connect-the-dot problem. And notice I broke the connect-the-dot problem rule. I turned where there was no dot. You're framing it as a standard connect-the-dot problem, and the square is relevant and important. So you said there's a really relevant rule, no changes of directions without dots, and there's a really relevant shape, the square, and I have to bind myself to that, and therefore I can't solve the problem.
- 3.67** And what you have to do to be cognitive and flexible is you have to break that. And notice that this is about relevance judgments. You have to break that frame. But breaking the frame, just telling you to go outside the box, is insufficient. You also have to make a new frame. You have to change what you're foregrounding and backgrounding. Cognitive flexibility is made up by frame-breaking and frame-making. This shows up in how we test for cognitive flexibility.
- 3.68** One of the things we do is we give people a pattern of dots, and we say, "You have one minute and you can use up to five lines. Draw as many different figures there as you possibly can." What happens is, initially, people think it's easy and they start drawing, and then they have to come up with a new one, and they have to break it and make it and break it and make it. People who are better at cognitive flexibility are better at insight. Another thing you can do is you can give people an object and say, "Other than its standard use, like a brick, tell me other things that I could do with this that other people would appreciate."
- 3.69** Not just other things. "Well, I could throw it at a tree." I don't mean that. Something that other people will regard useful. The more alternative use you can come up with for something, the more cognitive flexibility. You have to break the frame, come up with a new one, then break that frame and come up with a new one. Is that OK? Cognitive flexibility is measuring frame-breaking and frame-making,

which is predictive of insight. And of course, as I've been trying to show you, this is all dripping with relevance realization in an important way.

- 3.70** Colzato et al, in 2006, did some work on cognitive flexibility, and what they were doing is, they were basically measuring frame-breaking, the speed at which people update, quote, "binding a task-relevant," their language, not mine, "task-relevant feature and the response." So the basic idea here is, I give you the stimulus, sometimes it might be embedded and you have to break it out of that and link it to the appropriate response. Does that make sense? You have to break the frame so you can appropriately use the stimulus. Is that OK? Yeah.
- 3.71** What that means is that you give people a bunch of things and you can vary how embedded in there, I may say an equilateral triangle, and you, "Oh, there it is, right there." Something like that. Is it OK? Notice that that ability to do that with patterns is exactly what's going on in Raven's Progressive Matrices. I give you a pattern, and then you have to look at the next pattern, and then you have to see what's changed and what's relevant between, and you have to keep doing that.
- 3.72** What happens is, it gets harder and harder to decide what the next pattern should be because you get typically, well, this is the hypothesis. People tend to get bound into more automatic ways of looking at things. At least that's one plausible explanation. That's exactly what they used. They gave people these tasks where they had to break frame, and then they had them do the Raven's Progressive Matrices, which is a test, a very good test of general intelligence. And what did they find? They found that people high in the Raven's measures were less impaired by mismatches between the stimulus and response.
- 3.73** What I mean by that is they were more readily able of breaking the needed stimulus out of a version of it that didn't match the response, and reconfiguring, listen to the language, in a way that allows them to solve the problem. Does that make sense? Cognitive flexibility was predicting general intelligence. Cranford et al in 2002 did this experiment with mice. Now, usually you can't do these kinds of experiments because, well, "Mr. Mouse, here's this shape, here's this shape, what's the next shape?" "Meep-meep." That's all. It's not going to happen.
- 3.74** So what do you do with that? That's really interesting. They did something very clever. They first argue that what's going on in cognitive flexibility is selective attention, and they have to do what

they called attentional disengagement. That's frame-breaking. You have to disengage from the square. Is that OK? How are you going to vary that? What do you do is you compare young mice to old mice because guess what? Younger mice are more cognitively flexible than older mice, just like people, by the way. Your cognitive flexibility tends to go down as you get older, inevitably, inevitably, as we shrink and diminish until we're dead. But there's things you can do to really slow that down.

- 3.75** So same thing with mice. And then what you do is you give them two types of tests. You give them a simple discrimination test. They have to discriminate between a triangle and a circle. They have to push the lever that says which is the triangle. No big difference between the old and the young mice. Now, you give them the triangle embedded in a bunch of other things. I could make this more and more difficult. That's intuitively plausible to you, yeah? What you find is, the old mice get really bad at the discrimination task when it's embedded; the young mice are very good at it. Young mice are very good at it.
- 3.76** Younger mice are much better with the disengagement. They argued that this—now notice this. This is categorical intelligence. They're coming up with a way of measuring intelligence that applies to nonhuman species, which is very powerful. It's like, oh, in a very precise manner, we can tell how and why younger mice are more intelligent than older mice. Even though we intuitively knew that, now we can move beyond intuition to bonified explanation. Finally, Birney and Beckman. Birney is B-I-R-N-E-Y, and Beckman is just as it sounds, from 2022. I got to write the title of the article. That's literally the title. Like they're shouting at you. "General intelligence IS cognitive flexibility."
- 3.77** They review a lot, very comprehensive and careful argument for a couple of things. First of all, they bring up a problem with a lot of standard measurements, psychometric measurements. This is another problem, by the way. We measure people cross-sectionally. We measure them comparatively. We have Andrew and Michelle take the test, and then we compare them, and that's at that time. So I'm not tracking anybody through time. I'm just cross-sectionally at one time. It's a static cross-sectional. Now, of course, that's valuable or we wouldn't be here.
- 3.78** But they point out, but that's a problem because we need an account of intelligence that can explain how intelligence develops, not between people at the same time, but within a person across different times. And it's like one of those things that's like, of course. Right. Yeah. It's like, give these people a cookie because that is a great point

because, yeah, of course. If I just have a cross-sectional measure of intelligence, I'm leaving out the fact that intelligence is a property of a living thing that is growing and learning and maturing and et cetera.

- 3.79** What they did is, you take a careful look at all of that— longitudinal means we study it across time, like the longitude on a map. If you look at a within-person, and you look at within-person's dynamical developmental studies, you get a very clear case that general intelligence IS cognitive flexibility. It's just that ability to break frame and make frame, and you get better at that developmentally.
- 3.80** That's really interesting because we don't generally use flexibility metaphors for intelligence. We use quickness and sharpness and brightness metaphors, but they're not particularly helpful in a lot of ways. What does that tell you? That tells you that you have to be very careful with common sense and the way it builds things into our collective data set, our collective models of what something is. If all we're talking about is quickness and sharpness and brightness, and intelligence is much more about flexibility, we're actually, in a deep way, creating a shared model that is fundamentally misleading to us. It's why common sense has to be questioned by good science.
- 3.81** I'll be sort of personal for a minute here. This is sometimes fatiguing for me because—you're all seeing me in a context in which—see, by nature, by temperament, I'm a terrific introvert. I would like nothing better to leave now and go home and be in my room by myself. And I decided when I was young, I would develop the persona and the skills and the virtues that would compensate for that. So I became a teacher, which means, if I'm in the teaching situation or things that I can make relevantly similar to it, I can handle myself in social situations.
- 3.82** But if you take me out of that and put me in a context in which that's not relevant, like, I don't know, a party, I become indistinguishable from a potted plant because all I do is I try to find, where can I be less observed over in the corner? I'll be really happy if one person comes over and commits to talking to me for two hours. It's very important that we understand, there's a difference between the science and the common sense, because when I'm at these parties, I'll get people who want to talk to me, which terrifies me. They'll say, "Tell me what you do." I've learned this, by the way. "I'm a cognitive scientist and a philosopher." "What?" "I do a lot of psychology."
- 3.83** Then they want to tell me about their relationship with their mother. I say, "I'm not a psychotherapist. I'm a cognitive psychologist." Then they'll say, "What do you study?" I say, "Well, I study intelligence and

rationality." "Oh, I know what intelligence is." Everybody thinks they're an expert on intelligence. I want to say to them, "Do you think that about plumbing? Are you an expert at plumbing?" "Well, no, it's really—" "Do you think this might be really complicated and difficult to understand?" But the attitude, it's often like, "Who are you to tell me what intelligence is?" "Who do I have to be?" Why do you think a plumber gets to tell you about plumbing? Because they've built up the expertise and the evidence, et cetera, et cetera.

- 3.84** I'm not coming here as an authority figure. I'm coming speaking to you on behalf of what the entire community of science is trying to do about this. What you're giving me, as claiming to be equal, is your commonsense, uneducated, intuitive judgments about what intelligence is. Because what are they doing, my students? They're confusing the obviousness of their intelligence to them with being able to explain what it is.
- 3.85** We have to be really careful about this. We have to stand back with continual **epistemic humility**, and that's why I started this course with it. Let's start with the idea that we actually don't know. Then work our careful best way towards, what's our best argument right now? That's science.

4. Brain Dynamics

At a glance

Argues that cognitive flexibility is a strong candidate for explaining general intelligence because it is especially suited to solving real-life, ill-defined problems. Connects this to anticipatory, recursive relevance realization as the underlying mechanism that makes such flexible problem-solving possible.

- 4.1 **Cognitive flexibility** is a really good candidate because you know what kind of problems **cognitive flexibility** is good for? **Ill-defined problems**, the real-life problems. That's a good account of why they're showing up developmentally. Because in development, you're constantly encountering problems that are at least initially ill-defined to you, and you have to figure them out to the point where they become regularly well-defined for you. That's important.
- 4.2 See, **cognitive flexibility** points towards anticipatory, **recursive relevance realization** being really important to real-life problem-solving. Let's gather this all back together, I've argued, making use of other people's theoretical arguments and their empirical work, that the P-FIT theory points to the evolution of our cognitive stance that affords our **cognitive flexibility**. Highly intelligent people do the P-FIT information processing evolution that gives them that stance that makes them cognitively flexible.
- 4.3 Now, that's an interesting answer, don't you think? That's at least an answer that you should take initially seriously because of how well it fits with a lot of the data and a lot of the argument. It tells you something really interesting that we probably had not thought about, about what goes into making somebody intelligent, when we're using our standard commonsense metaphors of sharp, quick, and bright.
- 4.4 OK, so what's happening in anticipatory **recursive relevance realization** is we have our forward-facing framing, that's our orientation, and we have the **cognitive flexibility**, that sequence of frame-breaking and frame-making, that we are doing in a really efficient manner, the reconfiguration. That allows us to navigate our world. We're orienting and navigating well. That plugs back into the foraging into foresight.
- 4.5 Now, one of the things we can do with working memory, now to bring it all together—so we've got that going on here, sensorimotor, and you do it first as a kid. You get really good at navigating the world, orienting and navigating the world in a sensorimotor fashion before you start thinking about more abstract problems. But the idea is, we

get them exapted up into working memory where we can explore possible worlds. What we can do is we can decouple that sensorimotor machinery from any immediate problem and, in working memory, apply it to possible problems that we're not yet currently encountering.

- 4.6 Why? Because you know what you want to do? You want to anticipate and avoid the tiger. You don't want to fight the tiger. You want to anticipate what it will be like to have your university degree. You don't want to wait till you need all that knowledge and go, "Well, I'm going to wing it." It's not going to work. You want to form a long-term relationship with somebody. You better be anticipatory. What's it going to be like to be with this person 10 years from now? Because you know what they're not going to be? Like they are right now. Do you have enough **cognitive flexibility** to be able to stay in relationship with them?
- 4.7 Now, this is important. Why? Because one of the things that g predicts is not just how well you're going to do in school. It predicts how well you're going to do in your relationships. Now, other things do, too, like attachment and personality. But g is really predictive. It's predictive of your longevity. If you're higher in g, you're more likely to live longer. So, g must be helping people do what I'm talking about here: solve developmental, **ill-defined problems** that require tremendously powerful cognitive orientation and **cognitive flexibility**.
- 4.8 If I had to get one thing from you, I'm only allowed to measure one thing about you, I want a measurement of your g, which is why most governments make it illegal for corporations to measure that, because it is such a powerful predictor. Now, there's moral arguments around that, and I'm not tackling that. I'm just talking as a scientist right now.
- 4.9 If I could get this about you—what's your measurement? Measuring your g, measuring how rational you are, because measurements on all the different rationalities tests also form a positive manifold that's distinct from g, your attachment style, and your personality functionality. If I've got those four, I can predict about 85% of your life. That doesn't mean you're a machine that I can just manipulate. I'm not saying that. I'm saying these are powerful. They're really powerful.
- 4.10 All right. What's the next major neuroscientific theory? We did the P-FIT theory, and I gave it a lot of time, because everybody's giving it a lot of time. It's a really important theory. I think once you do the critique I did of it and you bring other stuff in, you start to get a very

powerful explanatory theory, not just cartography. OK, great. What's the next one? Well, we already took a look at it. Spearman, mental energy. No, it's not that you're more efficient. I'm not going to spend too much time on this because we've already worked our way through it.

- 4.11 It has been reliably found that more intelligent people actually use less metabolic energy than less intelligent people. I've already mentioned that to you. There's a fairly strong correlation between efficiency and general intelligence, but that's not very explanatory. Remember? We talked about this. Efficient at what? What are you efficient at? We've already got a possibility. You're efficient at cognitive orientation, cognitive stance, cognitive navigation, **cognitive flexibility**. You're really good at that. You're really efficient at optimizing in that way.
- 4.12 As I said, maybe what you're efficient at is reconfiguration. But we've got to stop now and take a note of something. It's already implied in some of the things we talked about, like exploring and exploiting. See, the problem with efficiency is it doesn't point to one thing, contrary to how people use it. The time scale of efficiency is very relevant. You can be efficient right now, which means you tightly fit yourself to the situation, so I'm not engaging in any wasted effort or energy, and I'm not acting at cross purposes. So, I tightly fit to the situation, and that makes me very efficient right now.
- 4.13 What was the problem with that? If I overfit right now, make myself too efficient, that will limit my long-term efficiency, my ability to move effectively between many different environments. When you say efficiency, which do you mean? Do you mean local efficiency? Or do you mean global efficiency? Are you concerned about not wasting energy right now? Or are you concerned about being open to the variation that you're going to encounter in the world, which—
- 4.14 I'll sometimes use the word resiliency to mean global efficiency, because sometimes saying local and, this is how you work, priming—when I say local and global efficiency, your brain might treat them equivocally. "Oh, the same thing, because he's saying efficiency." Often I'll refer to global efficiency as resiliency. Is that OK? Other people do that. I'm not the only one. As we will see in the next lecture when we talk about artificial intelligence and artificial general intelligence, this is something really interesting. You often have to disrupt local efficiency in order to get better resiliency and become overall more intelligent in your processing. It's really interesting.

- 4.15** When we look at this, this might help to explain, or at least gives us some new explanations, about why you dream or why psychedelics work, or why moderate amounts of distraction make you more insightful, for which there's all kinds of evidence. Basically, what you do, and they have to do it even in the LLMs, is you're training a neural network on a bunch of data, and it will get very fixed on that. Then you have to throw in some noise into the network. You either turn off a lot of the noise or flood some static. Because you got to get it more flexible for the broader world.
- 4.16** That's how you actually make your neural network more intelligent. You have to break it out of local efficiency and get this more general resiliency. Here's an interesting idea. Like I said, we'll come back to this, because you'll want to talk about this. Let's not talk about it now. We'll talk about it then. But that's a proposal. That's what you're doing when you're dreaming. You're throwing noise into your cortex. It's doing crazy stuff, and it's increasing your overall **cognitive flexibility**.
- 4.17** Because we need a good explanation for why you dream, because you spend a lot of energy and time doing it, and you're really vulnerable when you're dreaming. You know when it's really easy to kill you? When you're dreaming. It must have some really powerful adaptive advantage. Now, that points towards something I need to talk about. We keep bumping into trade-offs and that we're trying to do optimization rather than maximize any one of the two poles. Yes? Very often, the way you do that, even biologically, is from a process called opponent processing.
- 4.18** What do you do in opponent processing? What I do is I get—this can be multidimensional and more complex. I'm just going to do it simply with two-dimensions because it's easier to understand. What I'll do is I have process one, and it's oriented to one of the poles, and process two, that's oriented to the other pole. Then I get these two locked to each other. They're in opponent processing. This one is constantly trying to constrain that one, and this one is constantly trying to constrain that one. They're competing against each other.
- 4.19** Opponent processing, not adversarial. One side isn't trying to kill the other, like in American politics. It's opponent processing. Is that OK? Now, let me give you a quick example and one that's, I'm going to say this, relevant to intelligence. Your selective intention—we call it selective, but that's only half. Because what's happening is, you have a brain system, maybe multiple, but it's what you're using to stay task-focused. That's very active when you're trying to stay on track, like

stick with this crazy professor when he's talking all this abstract stuff. Stick with it, hang in there.

- 4.20** Then there's a part of your brain that wants to daydream. No, no, no. Let's think about other things. When you meditate, you find that that thing is really powerful. You know what you're doing the whole time you're meditating? Is you're trying to focus on your breath. That's the simplest task in the world. Just keep your attention on your breath. Three seconds in, you're off thinking about your laundry. Oh, you bring it back. You realize, wow. Raichle and others, we spend a lot of energy in daydreaming and mind-wandering, which doesn't seem to be doing anything.
- 4.21** You know what's another good time to kill something? When it's daydreaming. What's going on? Well, here's the idea. And notice how this is an evolutionary model again. I've got a network that is being selective in trying to keep you on task. And then I got another network that is introducing variations. Here's other things you could pay attention to. Here's other things you could pay attention to. Then the task focus network kills off most of those alternatives and brings it back into the task. But sometimes, one of those variations is actually useful, and that's a little bit of an insight, a little bit of a good connection.
- 4.22** So, what are you doing? Variation and selection, variation and selection. And your attention is constantly evolving, and now I'm going to give you a term from Merleau-Ponty for this, your "optimal grip" on the situation. Optimal grip is when you found a way of gripping something that is balancing between trade-offs in a way that is relevant to the task at hand. It comes from the idea of actually grabbing something. Because I can grab something if I want to see the whole thing. I can grab something if I want to see the details. I can grab something if I want to see an alternative aspect. I can grab something in a way that prepares me for using it. You see?
- 4.23** And I'm balancing off between all of these possibilities, depending on what is relevant to the task at hand. That's optimal gripping. The idea is, what's going on is, in your attention, you're evolving your attentional optimal grip on the situation. Why would that be important? Because what you're doing is trading between a lot of trade-offs, like maybe explore and exploit, to help you zero in on, what's the relevant information right now? Does that work? OK.
- 4.24** So, saying all of that leads to some more cutting-edge and scientifically controversial—I don't mean it's controversial the way the idea that the British monarchy are lizard beings. I mean, these are

respectable theories, but there's a lot of people who say, "I don't agree." But there's a growing, growing number of people who think these two types of theories I'm going to talk about, neuroscientific theories, are important for understanding intelligence. We'll see that it involves this opponent processing in order to get optimal grip, in order to optimize.

- 4.25** OK, so two things you can talk about in brains, and I've talked about it here and there, is we can talk about firing. What are the patterns of how neurons are firing? You know that neurons, they fire. What are the patterns of firing? You can talk about firing and you can talk about wiring. How do those patterns wire up so that they form networks together? I can talk about the firing and wiring of the brain. Now, here's a question you should ask. Given that firing and wiring are so pervasive, maybe there are differences in patterns of firing and wiring that are predictive of differences in intelligence. It's a reasonable proposal.
- 4.26** Notice how, every time I introduce a proposal, I give you at least a very quick argument on why it might be plausible. Because I don't ever say, "I think general intelligence is due to leprechauns in your head." I say, "OK, here's an idea. Do you see why it makes good sense?" You go, "Oh, yeah, yeah, yeah." You're not saying it's true, but you're saying, "Yeah, we should spend a little bit of time on this. It's a good proposal. I should take it seriously."
- 4.27** The most prominent of these kinds of theories is the proposal that general intelligence is a measure of self-organizing criticality or nearness to criticality, sometimes called paracriticality. OK, what is self-organizing criticality? This came from some work done by Peuerbach in physics. Whether or not this generalizes to many physical things is a little bit questionable. It's not clear if it works well for all physics, but it really works well for a lot of systems, particularly dynamically self-organizing systems, like you. Living things, mental things. But let's use Peuerbach's original research project to explain it.
- 4.28** Usually, in physics, you get hundreds of billions of dollars of machines and you smash particles together and you do all this. He was working with sandpiles. It goes like this. Like in an hourglass, you have grains of sand falling. What happens when they fall? Well, initially, they bounce around, and you can't really say with a grain of sand it's going to land. You can only say in some very large range, it's going to land somewhere in there.

- 4.29** But if you keep going, what happens? Well, what happens is, you start to get a bed of grains of sand, and they have friction and shape, and they start to constrain where the other grains can land. What happens is, it starts to form, I hope this doesn't shock you, it starts to form a mound. We don't think there's any sort of mound genie at work. We think this is a completely self-organizing process. At least, I hope you do. Somebody gave me this as a gift. I think it was my son. I've got an hourglass and it's got, the grains of sand are like metal filings, but secretly, there's a magnet underneath, and the grains fall, and they're falling normal, and then they do weird stuff. It's great watching people's reaction to this. "What? What's just going on?" Because it suddenly takes on weird shapes.
- 4.30** All right, so you've got this happening. Now, that just goes on forever. It just keeps going higher and higher and higher. What actually happens? It gets so high that it becomes unstable and it avalanches. It breaks up. Now, when a structure is on the point of breaking up, it's critical. That's what criticality means. Now, criticality has two possible outcomes. One is the system just breaks. If I build a building too tall, what happens? It falls over and the building is broken. And that's it. The system self-destructs.
- 4.31** We want things that self-destruct. That's what bombs do and things like that. We want them to go critical and blow apart. But that's not what happens if you watch inside the hourglass. What happens is, it makes a mound, it goes up, and then it avalanches. That avalanche actually allows the grains of sand to go farther than they could before. That means there's now a bigger base. The bigger base means the mound can now go from a bigger base and therefore go higher. It self-organizes to the point where it breaks, but in the right way, not breaks apart, but breaks so that it gets a bigger base so that it can go higher.
- 4.32** I hope you're thinking of something. I was thinking about the nine dots, and I had to break that frame and get a more encompassing frame that allowed me to go higher or more deeper into the problem, and that's how I solved the problem. You may think, "I wonder if this is a good model for what's going on in insight." Then you'd be Stephen and Dixon, who did some paper and some work from 2009, showing that it looks like insight is a version of self-organizing criticality in the brain, which means it probably has something to do with **cognitive flexibility**, which means, oh, that might be relevant to intelligence, at least general intelligence.
- 4.33** Now, again, people are like, "No," because that's what good scientists do. "No, I need more. I need more argument. I need more evidence."

You're right. I've got to build a lot more plausibility. But Hesse, H-E-S-S-E and Gross, they do a review. This was in 2014, and there's a lot of convergence, they argue, I think carefully and well, not consensus. So take everything I'm saying now about self-organizing criticality with a significant grain of salt. But there's more in convergence toward the idea that criticality plays a critical role.

- 4.34** Why did I do that? Because they make this joke all the time. "Criticality and its critical role in the brain," and they think they're being so clever. I want to say to them, "I've seen this 27 times already. Stop making this pun. Get out more." I should say that. The idea is, criticality plays a critical role in cognition. Now, in a more recent, and this is from an important high-impact journal, Trends in Neuroscience. It's a high-impact journal because it does exactly what it's talking about. It's talking about trends in neuroscience.
- 4.35** This is O'Byrne and Jerbi from 2022, also say, "No, it looks like we're converging on the idea that self-organizing criticality is playing an important role in cognition, probably in intelligence." There's a really good book out from 2022 by one of the premier figures on this, Beggs, B-E-G-G-S, called The Cortex and the Critical Point. This is moving towards something that, not consensus, but more and more people are thinking it's going to become consensus. Not there yet, but maybe.
- 4.36** Meijers—I'll put this up because it's a little bit—Meijers et al from 2021 argue that near criticality, getting the system near to criticality, is really important because it optimizes your information processing. Why? Because, get this, it manages a trade-off, and they use that language. It manages a trade-off. What's the trade-off? Well, on one pole, you want to increase the frequency of independent input messages. What's that a way of saying? You want new information. You want new information.
- 4.37** But what do you want to balance that off against? The need to respond reliably to these changes. You can't constantly be updating. You have to do something, and you have to do something that reliably works. Let's do something that's maybe familiar to some of you. This goes back to one of the great developmental psychologists, Piaget, who was basically the most important theorist in cognitive development. Piaget talked about the fact that you have opponent processing between these two processes.
- 4.38** One is assimilation. Assimilation is you try to fit the information from the world to, what, your existing models. You distort it if you need to, in order to make it fit your existing model. Nine dots, square. Oh, yeah, I know what to do. I assimilated. Now, why do you do that?

Because you spent a lot of time building that model, and it's worked a lot. For you to just simply abandon it whenever there's any little new information would be stupid, inefficient.

- 4.39** But sometimes the model gets so distorting that you have to update it. You have to give priority—listen to the language—you have to give priority to the new information, and that's accommodation. The system is actually constantly toggling between assimilation and accommodation. What you get is what Piaget called equilibration. You get an optimal grip between those, and you're constantly evolving that, and you're getting better. What you can do is you can open yourself up to a little bit more accommodation because you can more readily assimilate it when needed.
- 4.40** You get better at doing both, and you get, guess what? More cognitively flexible. That's what you get when you become more mature in your cognitive development. Let's stick with this for a little bit. What is accommodation doing? It's responding quickly to change. What's assimilation doing? It's giving you a reliable skill to use. Now, what you want is a reliable skill that adapts to change. The problem is, those are in a trade-off relationship, so you have to do optimal gripping, opponent processing. That's what Piaget said that was going on.
- 4.41** Notice that what this does, assimilation integrates new information to your existing information integration. This differentiates, it makes you come up with a different differentiation, cognitive schema, to use his language. Some people will talk this language and they'll say integration and segregation, which is a loaded term. But you'll see that in the neuroscience. Your brain is concentrating between integration and segregation, integration and differentiation.
- 4.42** Now, why is that important? There's a term for when you're simultaneously doing both of those. I have a suspicion that I came up with the term, but it's now widely used, and I'm figuring out if I can claim royalties on it, but maybe not. You've heard the term simplification, I hope. This is the term complexification. When a system is simultaneously integrating and differentiating. What is a complex system? There's a huge debate around that. I don't need a complete answer to that. All I need is this. A complex system is one that is highly integrated and highly differentiated.
- 4.43** You want a complex system? Look at your hand. You can do a lot of different things with your hand. Yes? You don't want this. I'm going to chop your hand off and put a hammer on it because a hammer is a real useful tool. No. I want my hands because I can do many different

things with them. Why? Because your hand is a complex machine. Highly differentiated but highly integrated. I can do many different things without falling apart. Complexification gives you new emergent abilities.

- 4.44** Patrick, you started out as a zygote. So did we all. You know what a zygote is? A fertilized egg. Now, you know what you'll notice? There's not going to be any zygotes in this course. They can't do anything. They're just—sometimes, wake up at 3 AM and remind yourself of that. "I was once a zygote." Then also think, "I'm just made up of trillions of little animals." Now, what happens with the zygote? Well, first, they just reproduce, the cells. But then what starts to happen very rapidly is the cells differentiate. They become different kinds of cells, liver cells and eye cells and lung cells.
- 4.45** Then what do they do? They literally, listen to the word, self-organize into lungs, heart, and then those organize into the circulatory respiratory system. The system is simultaneously differentiating and integrating, and that continues outside the womb. And as a system complexifies, it gets new abilities that it didn't have when it was simpler. That's why you can take this course, even though the zygote version of you couldn't.
- 4.46** Complexification. If you do this right, you can complexify the brain's functioning so that it's coupled to, that its complexification is tracking and going with the pace of the complexity of the world. So your complexification tracks reality. That would make you an intelligent problem solver. Sidney Pontes-Filho, P-O-N-T-E-S, has a thesis from 2023 where he just goes over all of this research about self-organizing criticality and its relationship to intelligence. Again, is this consensus? No. Is it highly plausible? Yeah, it explains a lot, very elegant, and a lot of data keeps converging on it.
- 4.47** Back in 2013, this guy named John Vervaeke published a paper with Leo Ferraro called Relevance Realization and the Neurodynamics and Neuroconnectivity of Intelligence. We talked about, maybe self-organizing criticality is doing relevance realization because it's doing opponent processing, and the opponent processing is the kind of opponent processing that would be at the work within general intelligence, given the idea that general intelligence is relevance realization theory.
- 4.48** Now, we were sticking our neck out in 2013. But things are going better for the proposal rather than worse. Hopefully, it'll stay that way until I retire. Of course, it'll eventually be overturned because, remember this, every theory in science gets overturned, including our

best ones now. That's their fate. Science is not about certainty, it's about reliable, self-improving, self-correction. It's about evolving our understanding.

- 4.49** Now, there was work right around that time by Thatcher and other people, and this has to do with, well, how—OK, I get a vague idea about how you're mapping self-organizing criticality into general intelligence, but can you fill it in a little bit? Well, so, yeah, I can. What happens is you need something that's analogous to the pile self-organizing, becoming a stable structure in the brain. Well, what that is, is when you get groups of neurons to fire in synchrony, they all fire together at the same time. There's mathematical reasons for when they fire together, they can do more than when they're firing independently.
- 4.50** You get something like data compression going on. Do you know what data compression is? We talked about this, I think, in the Primacy of Beauty. That's a scatterplot. You've all seen that, whatever it does, you get all your points of data. Then they taught you in high school, I hope, to do this. I drew that line deliberately because I made sure it didn't touch any line. It's called the line of best fit. Why do you do that? You do that, oh, because it looks pretty. No, that's not why you do that. You do that because it allows you to interpolate. It allows you to tell me what Y is when X is this, even though you don't have any data for that. It allows you to interpolate, that means within the data set, and extrapolate beyond it. It allows you to generalize your predictions.
- 4.51** We want science to do that. This is data compression. It's plausible that what's happening when things fire in synchrony is they're getting the gestalt and they're doing data compression and they're getting it. They're getting, oh, this is the line of best fit through the data. Now I've got the gestalt. Does that make sense? If I'm showing you a picture, have you ever seen those pictures that are blobs of black and white and you're looking at them and you're looking at them and you go, "Oh, it's a Dalmatian dog." Have you ever seen that? When you get that, you get synchrony. Have you ever noticed you can't go back to seeing it not as a dog? Because once your brain gets one of these, It goes, "Oh, you hang on to that," because that's going to predict things in the future. That's why it's so hard to go back.
- 4.52** OK, now what happens in the brain is you've got synchrony that's doing the integration. Is that OK? Then what happens is, and they even call it this, you get a neural avalanche. All of the neurons start to fire asynchronously. Like the grains of sand no longer held together, they go asynchronous. What does that do? It introduces variation. It

introduces new possibilities for the system. Then it comes back to a new synchrony and then goes into a new asynchrony. Do you see what it's doing? It's evolving.

- 4.53** What Thatcher showed, and it's only one or two studies, so don't hang your hat on it, but it's not nothing either, that higher intelligence people seem to be more flexible in self-organizing criticality. We thought, oh, look, **cognitive flexibility**, insight, opponent processing, it's going on in the brain. Maybe this is one of the ways in which the brain is implementing self-organizing criticality. Here's something that would help that argument. Self-organizing criticality seems to be scale-invariant in your brain. What does that mean?
- 4.54** At very low levels of analysis, you'll see self-organizing criticality happening. Even in your whole brain, you'll see. At all the levels of analysis, you see self-organizing criticality at work. That would be powerful, because you know where I need relevance realization to work? At all those levels at the same time in a coordinated fashion. Now, have I proven anything to you? No. No. Have I made a plausibility argument that maybe this aspect of the brain, how it fires, could implement relevance realization? I hope so.
- 4.55** Just like I made a case that, and that was an easier case to make because there's so much more evidence and argument, that P-FIT is implementing relevance realization. Remember how? Selection and variation. Selection and variation. Part of me telling you this, of course, is me being self-promotional and egocentric and a primate and trying to go up higher on the status hierarchy and all that thing. But part of it is, I'm a scientist and I made a prediction, and right now the evidence is accumulating towards it, and that's good science. This lines up with the theoretical moves that people have made.
- 4.56** When you're doing the synchrony, you're getting the capacity for generalization, which is really good. Predictive processing is going to talk about this. I want to generalize. Do you want to maximize generalization? Why not? Because sometimes I actually need to be more tightly fitted to the data. Mercado, in 2008, he said that what intelligence is, is the ability to differentiate information. What? That's the opposite of integration, differentiation. But think about it.
- 4.57** When I take off my glasses, and I've been wearing glasses since I was one, which tells you I need glasses. So what happens to me when I take off my glasses? Things blur together and my acuity goes down, my ability to differentiate. Now, do I get better at performing in the world or worse? Mercado is arguing that general intelligence is cognitive acuity. Other people are arguing general intelligence is

cognitive generalization. There's no answer to this because they're in a trade-off relationship. As I make something more discriminatory, I lose the capacity to generalize. As I make it more generalizable, I lose the capacity to discriminate. You face this problem in any experiment.

- 4.58** In order to make the experiment really generalizable, I have to control for a lot of variables. I have to bring it in the laboratory. I have to make a lot of things, and then I get a very simple thing, and that allows me to generalize, yes? What's the problem with that? What's inside the lab is less and less like the world. So what do I do? I'm constantly toggling between these things. That's why there's no single experiment in science that can decide anything. Because every experiment is a gamble on which configuration of the trade-off is going to be the one that tracks reality. You don't know that ahead of time.
- 4.59** The other thing is, you can be right, and then reality has this nasty habit of changing. John, you're being repetitive. No, I'm being convergent. I keep showing you, go here, go here, go here, go here, and it keeps converging on this same thing. We're getting more and more information, structural functional organization, of the model. We're ruling out and ruling in what's at that core of general intelligence.
- 4.60** Here's another high impact journal, not Trends in Neuroscience, but Trends in Cognitive Science. By the way, the trends journals are important. Pay attention to them. They do what they say. They're high impact because they pick up on the trends. They're good predictors of what's coming. Are they perfect predictors? There's no such thing as a perfect predictor. OK.
- 4.61** So, Barbey, from 2018, says something that's really important. Some of it you won't get, some of it you will, and then I'm going to bring it all together for you. "G emerges from the small-world topology." What's that? That's a pattern of wiring. We're going to come to that. So put a pin in it. "Small-world topology of brain networks"—now listen to the second half—"and dynamic reorganization of the community structure in the service of system-wide flexibility and adaptivity." So he's basically putting two things together. Flexibility systems, these systems that self-organize to give you that flexibility like SOC, self-organizing criticality. But did he also invoke something else? He invoked small-worlds topology.
- 4.62** Remember I said there's two cutting-edge, not totally consensus, types of theories? One is how the brain is firing. Now we're going to take a look at how the brain is wiring. That's topology. How is the

brain wiring? Are there patterns in how the brain is wiring that are predictive of intelligence? Well, you just heard there was one, small-worlds topology. What the heck is a small world?

- 4.63** OK, so first of all, what is a small world? OK. This is the truth. The name comes from the Disney song, It's a Small World After All. This is what I mean. Get a better joke. This was originally research done in social science. I used to do this when I was teaching a class and I'd say, "You're going to run a contest. You're allowed six connections, and the person wins the contest who can connect themselves through six connections to a famous person, and the most famous person, means you win the contest." They were thinking, "Oh," and they come back and they're, "I'm only six connections away from Queen Elizabeth. I'm only six connections away from Bono." They were all like, "Oh, this is a small world after all."
- 4.64** What people got really interested in is, well, how are things connected? Because we used to think the world, the natural world, was just connected randomly, higgledy-piggledy. Then the human world was connected in a very oriented way. It turns out both of those are wrong. Not individually, but on average. OK, so let's talk about what's called network theory or graph theory, not neural network theory, by the way. Graph theory is just the study of how components are integrated, connected together in a system. You can use this theory to talk about how cities are connected in an airline, or how neurons are connected in a brain, or how people are connected in a social system. It's that powerful a theory.
- 4.65** You can talk about basically three kinds. Now, it gets much more sophisticated with much more nuance, but we don't need that right now. You can talk about three kinds of networks. For pedagogical purposes, for purposes of explanation, I'm going to use the same number of lines and circles. Circles are nodes. Those are the things that are connected, and the lines are the connections. They're often called edges because they form usually the edge of a structure. It's not to claim that that's the case. I'm just doing it for ease of explanation. Is that OK?
- 4.66** We'll have a total of five. A line is a connection. If I have two lines, it's two connections, not a really thick pipe connection. Is that OK? All right, so you got these and they're connected like this. Just pretend that's right. All right. And then you've got these five. Sorry, these six connected like this. Yeah. It doesn't have to be symmetrical like that. I just did it so it's easy for you to see. Is that OK? Now this one. This is the hardest one to draw. Same number of nodes, same number of connections. Is that OK?

- 4.67** This is called a regular network. In a regular network, all of the connections are very close together. This tends to make a system that's very good at very limited things. We'll come back to how you can even measure this. Now, this is called a random or chaotic network because the connections are random. This one in the middle, this is a small-world network. Just to, how many of you actually have heard the Disney song? See? It's a small world after all. That's it.
- 4.68** Now, networks face, get this, a trade-off problem. They face a trade-off between efficiency and resiliency. How do you measure how efficient a network is? What you do is you take all the possible routes. OK, how many routes between this point and this point? 1, 2, 3, 4, 5, 6, 1. Then how many routes between this point and this point? You do all possible routes between all possible combinations of points. Do you see that? You get an average of that. That's called the mean path distance. On average, how long does it take you to get from one point to another? Your network is more efficient if you have a lower mean path distance. Is that OK?
- 4.69** How would you measure the resiliency of a network? But there's stuff around eigenvalues and there's a lot, and that's a little bit harder to make intuitive. But think about it this way. How robust is my network? How much damage can it suffer without ceasing to function, without some part getting disconnected? Is that OK? Now, here's the thing that is counterintuitive to people. Many people see this, and this is like your room. It's all really well ordered, so it must be really efficient, right? This is actually the most inefficient network. Very low inefficiency. The mean path distance in this is very low.
- 4.70** Why? Because all the path connections are short. Very low in efficiency. Now, because this is a trade-off relationship, you should predict, although it seems counterintuitive, this is actually the most efficient network. Why? Because it has all these long-term distance connections. Now, you may say, but in the brain, the long-distance connections are more expensive. We're not talking about metabolic stuff right now. Again, you can't cover everything with every theory.
- 4.71** Now, which one of these is more resilient? This one is. I can remove six connections from this, and this one will still stay totally functional. If I remove two or three, I can knock this system out. You know this, they build redundancy into the electrical grid. Why? Because redundancy makes it resilient to damage. This is very low in resiliency, and this is very high in resiliency. We have a trade-off relationship. What do we want to do? We want to optimize. We want to have something going on that will try to get the best of both worlds.

- 4.72** That's what this is. This is a system that is much more efficient than this. It's more efficient than this, and it's more resilient than that. This is more efficient than this, and this is more resilient, but this is optimal. It gives you the best overall balance between efficiency and resiliency. Right? Langer in 2012, 2013, showed that there, sorry, provided evidence that there's good correlation between how well somebody's brain is organized like a small-world network and how intelligent they are. Did that establish it? Of course not.
- 4.73** The problem with these fMRI studies is they have all the problems I mentioned to you earlier. So you get, "I found it, I didn't find it, we found it, we didn't find it." That's normal, by the way. But remember, I read you the review from Trends in Cognitive Science from Barbey, 2018. What did he say? "G emerges from the small-world topology of the human network." The more you're wiring like a small-world network, the more intelligent you are. Now, here's the promise. This is the elegance. What's the promise?
- 4.74** If this turns out to be right, if we can look at P-FIT, if we can look at SOC, self-organizing criticality, and if we can look at small-world networks, we can get a more clearly objective measure of intelligence. We can be absolutely sure this has nothing to do with a person's race, their gender, their cultural knowledge. We're just looking at their brain and how it's firing and wiring and what's the relationship between the frontal and the parietal cortex? This would be very, very powerful. It would be very explanatory, elegant, powerful, if this turns out to be correct.
- 4.75** Now, here's one additional thing, just to make this more tantalizing. As you may expect, firing and wiring aren't distinct from each other, independent from each other. How I'm wired is going to affect how I fire, and how I fire is going to change the way I'm wired. So the wiring and the firing constrain and affect each other. I hope I don't have to argue for that. I hope you can just see, yeah, of course, right. That's like, you eat when you're hungry. How could that be wrong?
- 4.76** But specifically, what does that mean? If a system is firing in self-organizing criticality, it will tend to wire as a small-world network. Why? Because it's building regular connections, and then it does that frame-breaking, and that throws out a long-distance connection and turns it into a small-world network. If it's a small-world network, it'll tend to fire as this. Why? Small-world network has a lot of local connections. It's doing this, and then something comes in from left field and makes it critical. If it fires this way, it wires this way, and if it wires this way, it fires this way. This is scale invariant at all levels of the brain. This is scale invariant at all levels of the brain.

- 4.77** That is a now quite powerful proposal. Is it consensus? One more time. No way. But is this a powerful proposal? Has it got trustworthiness, convergence to it? Yes. Does it explain a lot? Yes. Does it have a lot of promise? Yes. Good balance between the promise and the convergence? I think so. Does it line up with relevance realization theory for which we have all other convergent argument? Yeah, it does. Yeah, it does. Yeah, it does.
- 4.78** Now, you can tell when you're doing good science because it gets scary. Because when you get good science, you have the possibility that you could really intervene in something. We get the atom and we can make atomic bombs. We get genetics and we can do genetic manipulation. We get this. I mean, think about it. This is why educating is important, because education allows you to be more appropriately responsible to the possibilities that we might have to face as citizens. Do you want a machine that I can put on your head to read this? Do you want him, because he's wealthier than you, to have a machine that he can put on his head that makes him more intelligent than you?
- 4.79** Right now, we don't have any of that. But if this is right, and it might not be, but if it is, that goes from being science fiction to just over the horizon of where we could be. They're already talking about this. If this is right, could we? Haier, in his book, *The Holy Grail*, is can we do neuroscientific things to manipulate intelligence? It's like, well, right now, not very much, but maybe. Well, don't let them do that. Good luck with that. When have we ever been able to say, "Hey, don't open up that new power."
- 4.80** The only thing we can do is get deeply, properly educated in what intelligence is and how it's showing up in behavior, how it's showing up in the brain so that we are the most properly informed, so that we can make the most rationally responsible decisions. I want you to have this information, and I don't want it to be in the monopolistic hands of governments and corporations. That's why I've done a lot on YouTube and social media to try, not because I want everybody to love me, I do, but because I think deeply what I just said to you. This is not, oh, this is academic, who cares? You should care about this, because this matters.
- 4.81** That's going to lead us to our next topic because you know where the question of intelligence is really mattering in a way that has the potential to upset your world more than the printing press, more than electricity, more than atomic power? That's the advent of artificial general intelligence. And can we learn anything about general intelligence by looking at artificial intelligence? That's what we're

going to do in the next lecture. Thank you for today. OK, we're going to take some questions now.

- 4.82** Hi. Forgive me if my question doesn't make that sense, but—
- 4.83** Your questions have always made sense. I predict this one will, too.
- 4.84** When you talked about optimizing, when you talked about configurations and reconfigurations, when you talked about how measurements of general g can help you to predict 80%, 85% of life of someone, all of things make me question my—
- 4.85** I want to be careful. I said g with those other things added to it gives you 85%, not g alone. G alone will probably give you about 60% of your life. But go ahead.
- 4.86** G and attachment. After this session, I think that one of my role in regard of my intelligence is that I can learn its process, not perfectly, but I can learn how does it function, how it's processing things. But my other roles are being denied. I think that I have no part in my perception. Basically, my brain is understanding things automatically. I don't know if automatic is a correct term. It's just that my brain is receiving all of the things, all of the information concepts. I have no part in it. Because I was not even aware of the process. What is my part? Even my decisions that I take after this understanding, if, for example, if I decide to improve and enhance my cognition, even that my decisions are based on my previous perceptions.
- 4.87** To a degree. So I agree with you. You are not in complete top-down charge of your behavior. But it's also not true that it's completely unconscious bottom-up. I noticed throughout the lecture that you remained facing me. Why? Why didn't you turn somewhere else? Because you wanted to hear the lecture and you were aware of that, and you directed your attention. You participated in it. Now, it could be that you don't want to hear the lecture, and then you turn away or leave. You are playing an ongoing role in shaping your cognition as much as the world is playing an ongoing role in shaping your cognition. The two are interpenetrating, bottom-up and top-down, all the time.
- 4.88** Now, you're saying, but I'm not—well, let's try this. Is this a voluntary act? I want to be over there and I walk there. Is that a voluntary act? It seems to be. That seems to be very different to me than sneezing. It seems to me I can initiate it, I can interrupt it, I can stop it, I can correct it, I can speed it up, I can slow it down if I want. Do you grant me that?

- 4.89 I think so.
- 4.90 There's a lot of top-down control. But am I completely in control bottom-up? No, because your point, a lot of this is relying on unconscious processes about how muscle contraction is working, how my cerebellum is giving unconsciously evidence to my sensorimotor context and helping me keep my balance. But they're both happening. They're both. See, and we keep doing this, and that's OK, I'm not complaining, do you understand? We are told it's either one or the other.
- 4.91 John Locke. Your mind is a blank slate and the world just writes on it and you have no part in it. No, no. Jean-Jacques Rousseau. Forget that. You are like an artist. You express your will onto the world and the world is as you choose to see it. Both of those are wrong. It's not one or the other. They're both fundamentally wrong. You, and this is what the wisdom traditions actually keep saying, you are not a mere passive receiver. You are not the active maker. You are an ongoing participant in reality. You do play a part as you. But because you play a part, a part also belongs to something other than itself. That would be my answer to you.
- 4.92 OK. So in terms of flexibility, **cognitive flexibility**, is it possible to improve that by doing activities that you've never done before consistently? So if I've never drawn and I decide I'm going to start learning how to draw, and then you get reasonably good at it, and then you're like, OK, well, now I want to do something else, like racecar driving or escape rooms, if you consistently put yourselves in uncomfortable or new situations that forces yourself to find new patterns, can that improve your flexibility or is that just kind of—
- 4.93 Well, one thing we have evidence for is it can reliably reduce your loss of **cognitive flexibility**, your fluid intelligence, as you get older. That's becoming less and less controversial. Now, here's something that you can make at least a prima facie, meaning first pass at it, plausibility argument. We know that mindfulness practices, and I'll talk about two here, meditation and contemplation, and they're not the same thing, even though we use them as synonyms and we shouldn't. They don't even have the same etymological origin.
- 4.94 Mindfulness practices, which train attention, improve your **cognitive flexibility** and make you more prone to insight. Depending on the practice, how systemic the insight is. Sorry, how systemic it is, how much of your system is being changed, and systematic, how broadly it applies. So meditation is, you're frequently doing a lot of frame-breaking. You're trying to break up the gestalt and you're trying to

zero down and you're trying to step back and look at your frame. So that breaks it up. If you'll allow me a metaphor that's become a little bit of a meme, in meditation, this is what you're doing. I'm stepping back and looking at my framing. But if I were to just do that, I'm going to bump into things.

- 4.95** What do I need to do? I need to put them back on. And what? See if I can now see more clearly through them. Oh, there's dirt on here. Oh, wow. Now I can see more deeply, more clearly. That's what contemplation is. Listen to the word contemplation. It has the word temple in it. Contemplatio. Temple, you look deeply into things. It comes from the Greek word theoria, which means to look deeply into things.
- 4.96** What you do, for example, in, for example, a Buddhist practice, I'll do vipassana, in which I'm doing this, and then I'll do contemplative practices, in which I'm trying to see more deeply into the world, like I'm trying to realize that everything is impermanent and interconnected, not the way my common sense makes it look, permanent blobs of stuff. I'm trying to know, no, the truth is, it's actually impermanent and connect. So I contemplate and I get a little bit better, but I got to break up the framing that's habitual. I do this, then I do this, then I do this, and then I do this.
- 4.97** The proposal for these traditions, and more and more people are studying it, we have a whole academic journal dedicated to mindfulness, is that if you do that enough, you will massively increase your **cognitive flexibility** in the sense that you will get more systemic and systematic insight. If you do that comprehensively and coordinated, you'll become wiser. You'll become enlightened. A lot of people are investigating if these kinds of things that really emphasize frame-breaking and frame-making, practiced in a very disciplined and developed manner, can actually, reliably, permanently increase your **cognitive flexibility**.
- 4.98** We have good evidence that mindfulness, like the stuff you were proposing, slows down your loss of **cognitive flexibility**. But there are people who think, and I happen to argue that I think this is worth taking seriously, that these kinds of practices could reliably improve your **cognitive flexibility**. Then you get the question, wouldn't that make you more intelligent? Is that what we mean by wise and enlightened? We can do an entire course on that. Or wait a sec, I did. It's the course called Intelligence, Rationality, Wisdom, and Spirituality. That's where you can go to look for that.

- 4.99** I know. Yeah, I agree. I was just wondering because I've taken another lecture with Eagleman, and he seems to believe that you can continuously input new information and you'll just become more knowledgeable and smarter. And he has this argument that, oh, well, TikTok and new technology for younger generations just makes them smarter. But noticing how younger kids works, my—it doesn't seem like that's the case just based on their behavior.
- 4.100** Yeah. And filling people with knowledge isn't the same thing as making them more intelligent, or John Locke would have turned out to be right, and he turned out to be wrong. You're not tabula rasa. You're not a blank slate, and just loading more knowledge into you—does it give you more power? Yes, but—so they make a distinction in cognitive development between quantitative and qualitative. Quantitative is, I'm just taking in more information. Qualitative is, I'm taking in more kinds of information. And intelligence has to do not just with quantitative, but with qualitative development.
- 4.101** You describe the small-world neural network as an optimized—
- 4.102** It's not a neural network necessarily. It's just any kind of network.
- 4.103** Right. Got you. OK. In the context of a neural network as it applies to intelligence, is there a bias towards efficiency or resilience as it relates to being more intelligent for that network? Is it in the middle or is there a bias in one way or another?
- 4.104** Patrick, I just want to make sure. Are you asking me, what's showing up in the brain or what's showing up in AI?
- 4.105** Initially, I was curious about the brain, but now I'm curious about both.
- 4.106** There's very little—now I'll talk about this—concern about plasticity in neural network theory. That's where you're actually changing the architecture. We'll talk about some work that's been done on that and why it's important. There's much more concern with patterns of firing, patterns of activation. Not so much—I don't have enough to work with to answer the question when it comes to, what are neural network people doing? I think they should pay attention to this stuff, but not as much as one might hope. They're not doing enough cognitive science.
- 4.107** You may note that I seem to have this bias, that I'm advocating that a lot of the answers are going to be found by doing more cognitive science, and I'm completely impartial on that. About the brain, what's happening is, I presented it to you at—you know how in, sorry, I don't mean this as an insult to any of you. I hope you—please take it not as

an insult. They teach you the solar system model of the atom, and they get you to understand that. Then after that, you can go on to get the more complex and weird thing. I've given you the solar system model of network theory.

- 4.108** When you get into the more post-Bohr version, it gets much more nuanced and complex. There's some evidence, for example, that there's hub efficiency, which is an aspect of small-world topology, that has to do in the salience network, the part of the brain that makes things salient, that seems to be most highly correlated with *g*. Now, of course, I love that because, oh, right, salience network attention, relevance realization, that really plugs in well. I'm very excited about that. But one, two studies, we need more.
- 4.109** Is the brain biased? What I showed you about the reconfiguration is the brain seems *ceteris paribus*, and it's a very big *ceteris paribus*, it seems to err towards making you more of a general purpose machine than a special purpose machine. Because as I answered to Reuben, we evolved to be nomadic. If I don't know, and I have to guess, which is what you're asking me to do, I would say, given other things and given our evolutionary history, there's probably a bias towards making us more general purpose than special purpose.
- 4.110** But, of course, your particular idiosyncratic developmental history could really influence that and make you become much more specific because you've developed a lot of configurations for plumbing that other people haven't, things like that.
- 4.111** Yeah, and you'd probably have a distribution amongst a population.
- 4.112** Of course, and you'd want that because you don't want everybody to be a plumber, and you don't want everybody to be a cognitive scientist, contrary to what I seem to be arguing. You don't want everybody to be a cognitive scientist. Yeah, of course.
- 4.113** I assume the biggest reason the trade-off relationships exist is because we can't do both at the same time, and time is a finite resource.
- 4.114** It's not only at the same time, but as you gain on one, it undermines your ability to gain on the other.
- 4.115** Exactly. We don't want to go too much towards one because it's not—
- 4.116** Then we lose too much.
- 4.117** Exactly. But that being said, if we somehow develop AI to the point where we are able to jump back and forth between them at a fraction

of the time, which almost seems instantaneous, and this loop continues faster and more often, I assume that means that technically you're making things, the complexification is happening at a more rapid speed. But until what point or for what? My brain doesn't seem to understand beyond that process.

4.118 We're going to talk about that when we do the AI stuff. First of all, we have to ask the question, is AI doing what you propose? Up until very recently, the answer was no. Maybe, and it's a really important maybe, we'll talk about a bit, the LLMs are doing the beginnings of this, but maybe not in any way that's scientifically valuable. But that doesn't mean it won't be socially disastrous.

4.119 Now, why did I say that? Because if I don't have a scientific breakthrough with the LLMs, if I just have an engineering breakthrough, I can't answer your question. I can look at these machines and say, well, they're faster. Does faster always mean more intelligent? No. Well, they're, maybe. But why are they? I don't know. Is there any—dude, here, have we got any insight about how intelligence works from these machines? Probably not. Do we have any insight that would generalize to explain how a chimp is intelligent? Not at all. Oh, which means they don't represent a breakthrough in scientific understanding.

4.120 But our best way of trying to predict how things are going to be is a scientific understanding. You asked me this question, and unless I'm a jerk on social media or I want my company to succeed and I'll say terrifically irresponsible things because I want you to buy more shares in OpenAI or whatever it is, I should say, "I don't know," because we actually haven't done significant science here. Without the science, I can't tell where this goes or how it turns out. That's my answer, at least preliminary.

4.121 I had a question about the self-organizing criticality. The way I understood it when you explained it, it made sense to me on a personal level, I guess, you can assimilate things and you can accommodate things. Something that's complex is able to accommodate things without breaking apart, and something that's not complex, if you accommodate too much, you'll fall apart.

4.122 That's horror.

4.123 Horror. You get anxiety. It's too much. I can understand that on a personal level. But then I'm not sure how that translates into the graph and the neuron, the neurons. Is that something that's actually happening in, like, a graph, like you have—

- 4.124** The self-organizing criticality is not the network theory. Remember, it's this idea that your brain is oscillating between firing and synchrony, which is like the mound being made, and then firing asynchronously is like the avalanche. Then from that avalanche, the brain makes a new mound, a new kind of synchrony. Then it breaks down, and you're doing this very rapidly. People, at least there's some evidence for this, people who are more flexible in this, **cognitive flexibility**, frame-breaking, frame-making, seem to be more intelligent.
- 4.125** OK, I see. It's the asynchronous and synchronous firing of neurons.
- 4.126** That's happening fractal. It's happening at very low levels of analysis in your brain and at very high levels of analysis in your brain. Same thing with small-world networks, very low level, very high level. It's a nested thing that's going on.
- 4.127** OK, I see. That makes sense. I guess you were saying something about how those small-world networks are what end up happening when there is self-organizing criticality?
- 4.128** That's right.
- 4.129** Those are what gets built once you have—
- 4.130** As they get built, they tend to make more self-organizing criticality. The thing bootstraps, which is what you see kids doing. They bootstrap.
- 4.131** Yeah, it's a cycle.
- 4.132** Spiraling up.
- 4.133** OK, that makes sense.
- 4.134** Good. Thank you very much, everyone.

5. Introduction of AI

At a glance

Reviews the P-FIT theory (parietal-frontal integration theory) as a leading neuroscientific account of intelligence with strong convergent evidence, then turns toward artificial intelligence. Frames the overall argument as a plausibility case rather than a proof, aiming toward a generalizable, multiply realizable explanation of general intelligence.

- 5.1 Good to see you all again. I really enjoy your company in this journey we're taking together. Last time, we took a look at the neuroscientific theories of intelligence as part of the long argument we're building towards synoptic integration in a highly plausible manner. This is my way of trying to make a strong argument about the nature of intelligence, and a strong argument is needed because there is no consensus about what intelligence is.
- 5.2 Let's remember, when this is all done, hopefully you go, "Well done, very good," and all that sort of thing. But hopefully you'll remember, I'm offering you a plausibility argument. I'm not claiming this, "Now you know the truth." But I'm also trying to give you something you should take seriously, consider, probably has a chance of significant probability, etc. We got to remember, we get enthusiastic and it's good, but we got to keep epistemically humble the whole time, too. It's like, this is our best argument right now. That's what I, at least my best argument.
- 5.3 All right, so we looked at maybe the best neuroscientific theory, the **P-FIT theory**, the **parietal frontal integration theory**, lots of convergent evidence. Again, it has a high degree of plausibility, high degree of probability. I wasn't going to criticize it on whether or not it might be true or not. I think right now, probably, plausibly true. But my questions were, yeah, but what is its function? What's it doing? How does that integration allow us to be generally intelligent? We need to do that in a way that generalizes to **multiple realizability**. Just talking about brain regions won't generalize to an octopus and certainly won't generalize to AI.
- 5.4 I did point out, well, maybe what's going on is, you've got selection on variation, and you see this theme showing up again and again and again, dynamical looping, selection and variation. We took a look at the idea that that evolved from foraging into foresight, and we talked about how that got exapted. There's a lot of evidence that a lot of our cognition is exapted. We'll get into that later in lecture 4, the 4E cognitive science approach, et cetera.

- 5.5 Then we talked about, well, what's going on there? We talked about this whole process of reconfiguration. Then that pointed to, the brain is trying to trade off between global versus local efficiency, and I sometimes call global efficiency resiliency. What you get is the evolution, if you remember, of a cognitive stance. Remember that notion of what you get is, the brain is trying to get to a state so that it can do very small and quick changes to get to the good reconfiguration that it needs.
- 5.6 We have that forward-facing framing, and that's our orientation. Then we said, oh, and this will also map nicely onto cognitive flexibility, which, remember, there was even the really bold claim where they were shouting at us, "General intelligence IS cognitive flexibility." They came from a very strong argument, which is, we need developmental accounts of general intelligence, not just static cross-sectional. Very strong point, very good point. The idea what's going on in cognitive flexibility is the frame-breaking, frame-making. That gives us insight, especially in recursive relevance realization.
- 5.7 We even took a look at that within mice. We got some nice **multiple realizability** going there. We have the cognitive stance is the orientation and the cognitive flexibility is the navigation. We got the foraging into foresight. Oh, this is hanging together very, very well. Then the idea is that orientation for navigation is then exapted into abstract space, possibility space, etc.
- 5.8 Then we took a look at another, I don't know if it's, yeah, I guess I'd call it the second most prominent neuroscientific theory of general intelligence, neural efficiency hypothesis or brain efficiency hypothesis. Then we got into, well, efficient at what? Again, it looks like trading off between local and global efficiency. We keep seeing this optimization for trade-offs playing a crucial role again and again and again. Oh, what's going on? There's a trade-off relationship, and the brain is optimizing dynamically between them.
- 5.9 Then we move to some more cutting-edge, meaning legit but not consensus. Nobody goes, "What a crazy idea." A lot of people are taking it seriously. Increasing numbers of people think it's probably true or probably on the right track. These were theories about the relationship between patterns of firing and general intelligence and patterns of wiring and general intelligence. The patterns of firing was the self-organizing criticality, which also shows up in insight, recursive relevance realization. And the wiring is the small-world network, which is also doing some optimization for relevance realization.

- 5.10** That was where I shamelessly promoted a book chapter that Leo Ferraro and I wrote in 2013, predicting that self-organizing criticality in small-world network would be correlated with general intelligence because—it wasn't full prediction. Some people are already saying that, but we were adding to the predictive force. Because they're optimizing for relevance realization, they will be strongly correlated with general intelligence.
- 5.11** Then we took at the very end that self-organizing criticality and small-world networks are in a bootstrapping relationship, which could really explain very well the development of relevance realization and therefore the development of general intelligence. That's where we got to. Now, today's lecture is going to be more abstract because we're getting, when we were doing the psychology, we were with people and with behavior, and then we're doing neuroscience and there's the brain and the brain. Now we're going into AI, and now we're going to be talking abstract stuff.
- 5.12** A lot of the time, remember we talked about, they have to teach you the solar system model of the atom, so you get at least a preliminary understanding that's a good approximation that, if you need to, you could go really deep into the more nuanced complexities of the current model of the atom, which verges on being incomprehensible? We're going to have to do that a lot here. I'm going to give you the solar system explanation of core phenomena in AI.
- 5.13** Using Elgin's beautiful phrase, "It's true enough." Namely, it's not false. It needs to be more complex and nuanced, but I don't need that complexity or the nuance in order to make the argument I'm making. Nothing in that complexity—so I'm asking you to just trust me, that nothing in that complexity undermines the argument I'll be making in any way. To teach you the complexity, we'd have to do a lot of math first and blah, blah, blah, blah. It's not going to happen. I work with a lot of people that do the math, and they reassure me about this.
- 5.14** One of the best compliments I was ever paid, I was working with Tim Lillicrap on the first relevance realization paper. Of course, he does the math. He ended up being the chief scientist, lead researcher for DeepMind, Google. Very proud of him. I just got to see him recently. But we were working on the relevance realization paper, and he said, "John, you seem to be able to do with words what other people do with math," which is one of the nicest compliments I was ever paid. I got reassurance from people other than my own judgment. Hopefully, that's good enough.

- 5.15 I'm saying that because there might be people sitting in on this and they might know more of the AI, and it's like, I hope that they'll still find this valuable because of the way it connects up with the overall synoptic integration argument. OK, so **artificial intelligence**. In fact, what's happening is, you're probably hearing it more and more today. We're moving from people talking just about AI to people talking about—and this should no longer surprise you—AGI, which stands for artificial general intelligence, the idea being that if we want strong **artificial intelligence**, we need to capture general intelligence in our AI.
- 5.16 That has been something that has not been happening. This is why the possibility that we might be moving this way is so revolutionary, because up until very recently, AI was not telling us very much about general intelligence at all. Now, we're going to go through the history of this, and it's going to allow me to do two things at the same time. I'm going to go through the history of this, and then we'll do the three generations around AI, and that's also the three generations for cogsci, and that'll allow us to build up an argument that we can then carry into the last lecture.
- 5.17 Let's go back to what I think is the beginning of the attempt to get at general intelligence way back when. This is the idea, very plausible, given everything we've talked about, that intelligence is problem-solving. That's what it is to be an intelligent being. We saw that this was presumed throughout the psychometric, cognitive, and neuroscientific theories. OK, great. Then here's the thing. What's general intelligence? Well, you're a general problem solver. You can solve a wide variety of problems in a wide variety of ways in a wide variety of domains.
- 5.18 That makes you very, very different from the AI that's in your ATM. What can it do? Well, it can do banking, but you can't say, "Hey, let's go swimming." It's not going to work. But what people were thinking of is, it's reasonable that what g is, is, it's some ability that makes you a domain general problem solver. If we could capture that in a computational machine, we would have made something equivalent to your intelligence. We'll talk about that equivalence in a second.
- 5.19 But there was a project called the general problem solver project, the GPS. Now, GPS doesn't stand for that anymore, but originally it did. This project was absolutely seminal. It was deeply influential for **artificial intelligence**, deeply influential for the psychology of problem-solving, deeply influential for the psychology of intelligence and rationality, and even deeply influential to economic. Really central work. This was the work of Newell and Simon all through the '50s and

'60s. That's an N. It culminated in 1972. As I said, the name of that project was the general problem solver project.

5.20 What they're doing is they were also inventing a new way to do psychology, or at least do, study cognition. What we typically do when we study something is we unpack it, we analyze it, we examine it, and that's plausibly what neuroscience is doing, or we examine how it behaves, and that's what psychology is doing. But they came up with a new method. It probably goes back to Turing, maybe to Babbage, but it's this idea. It's called taking the design stance. It's a new way. What's taking the design stance? Well, here's the idea.

5.21 Studying her is really hard. A couple of really important reasons. One is, she is always trying to do two things, one of two things, maybe both. She's trying to please me or deceive me when I'm studying her. Some of that can even be unconscious. That's why we have to do all these double-blind experiments and things like that. Then, get this, she's protected by the law. I can't do things like say, "Can I remove this part of your brain and see what happens?" People will not let me do—thank God, of course. They won't let me do that. She's protected in a lot of ways.

5.22 Then here's the thing. Her brain is a really complex machine. What is going on in there to make her behavior? There's all kinds of multiple variables. It is very, very hard. Well, here's what we might do. Instead of trying to get direct information, I will do an engineering end run. This is what the design stance is. You know what an end run is in football? You don't try and go through, you try and go around the the lines. You're trying to avoid all that, all those big meaty men. Here's the idea.

5.23 I'll make a machine that does what she does in the same way she does it, including making the same mistakes that she makes. Then because I've made it, I know how it works, which is not totally true with some of our current machines, but close enough. Because I've made it, I know how it works. Then I can think, I can't say, "And therefore, that is what she's doing." That's not a deductive or even an inductive argument. But I can make an inference to the best explanation.

5.24 She is probably doing something similar to what the machine is doing. Therefore, because I understand the principles in the machine, I can say what the principles at work in her, the principles in the processes. Then you see, then I can do really much more precise neuroscientific investigation and psychological investigation of her behavior. This was a powerful thing that really moved people forward. This is part of what was called the cognitive revolution in psychology in the '50s.

- 5.25** Up until then, we were mired in the dark days of behaviorism, which is, well, you don't study what's going on inside there for all the reasons. All we do is study stimulus and response, observable stimulus, observable behavior. What we do is just try and study. That collapsed because you can't do that. You need to know what's going on in there. Taking the design stance was a way to really sharpen the behavioral experiments and then later, really sharpen the neuroscientific investigation. Does that make sense?
- 5.26** When AI is doing that role, it's called strong AI. When it's helping us understand cognition and intelligence, that's strong AI. When it's just modeling it or doing other tasks, people call that weak AI. The strongest version of strong AI would be, my model actually is an instance of cognition like hers. That would be the ultimate goal. Not that I've made a good model, but I've made an actual instance of mind, which, of course, is filled with hubris and the gods are waiting to strike us down. All right. Does that make sense, the design stance?
- 5.27** Now, notice, again, it's a plausibility argument, and it's using these abductive ideas, and then inference the best explanation and all of that. It has to integrate with psychology and neuroscience if it's going to run. One of the things cognitive science said is, when it started in the '70s is, "Those three disciplines are not talking to each other enough." That's how it emerged, as I've already explained to you. All right, so what did Newell and Simon do? Again and again and again, they're going to answer this question, or at least try to answer this question, how could I make a machine do it?
- 5.28** Because you know what that does? It's a wonderful way of flushing away the homunculi. Because if you've got a homunculus in your theory and you have to make a machine that does it, you have to build a little homunculus and you realize, "Oh, I don't know how to make that little homunculus," and then your project collapses. Now, people do dodgy things when they're actually the homunculus outside and it's them and their machine. But still, there's the goal here of, I want to flush away the homunculi.
- 5.29** They analyzed problem-solving. They analyzed problem-solving, and I'm going to lay it out for you. Now, this overlaps with the course on Intelligence, Rationality, Wisdom, and Spirituality. It overlaps with the course on the Primacy of Beauty. But we'll go into this in more detail, a little bit more care. The idea is, you have a problem when your initial state, the state you're in, is different from the goal state you want to be in. For example, if you want to be on a chair and you're on a chair, you don't have a problem. There's no problem. If you are thirsty and you want to drink and there's water that is at some

distance from you, you have a problem because your current state does not match your goal state.

- 5.30** Now, I'm just going to note something here to come back to it. Notice that this requires an anticipatory relationship in you. You have to predict and prepare for a goal state. So put a pin in that. We've been talking about that a lot, about that anticipatory aspect. That's the first thing. Those are the first two things. You get some "representation." I'm doing this in quotes because there's a lot of disagreement about what representations are and if we need them. I'm doing this like this, trying to make it a very neutral term. You have a representation of the initial state, you have a representation of the goal state.
- 5.31** Then you have operators or operations. These are things you can—so, I'm thirsty. I want to be with water going into my face hole. OK, so what I can do? I can raise this arm. I could raise this arm, I could raise it. There's many operations I can perform. Those will put me into a different state. Now, it might be that doing the operator is just one shot and I have my goal. I need to breathe. There we go. That was not much I had to do there. But a lot of times, this isn't your goal state. Then when you get there, you do other operations.
- 5.32** I'll just do this symmetrically. This doesn't open up symmetrically. This could have 17, this could have zero. This can have four. I'm just doing this so I don't take forever. Are we clear about that? Plus your brain likes symmetry. It gives a little bit of fluency, and so your brain goes, "Ah," and that helps you. Then, of course, here, this one dead-ends. This one, maybe, oh, look, I'm going to open it up a little bit. I broke what I said I was going to do. This one dead-ends. Maybe this one goes. OK, so now we've got three things.
- 5.33** There's a fourth, and I need you to remember this because this will turn out to be really important later on. You're a general problem solver, yes? Which means you don't want to solve any problem to the detriment of solving other problems. Yes? One of the classic ones, and this is from Jerry Fodor who you'll hear about today is, I could cook my lunch by burning down my house. Nobody goes, "Brilliant idea." Because you're trying to solve a problem, that problem, in a way that really undermines you solving a whole bunch of other problems.
- 5.34** Another one, for example, is you don't want this problem to take too long to solve. Because if, well, I need 70 years to solve this problem, so maybe that's too long. Some problems take 70 years. I want to be enlightened. Maybe that takes your lifetime. But a lot of problems, like showering shouldn't take you 17 days, things like that. That's obvious. It gets tricky, but there's time. But you notice there isn't an

absolute amount of time. It's relative to how important this goal is to you. Then there's also, how much information do you have to take in, et cetera. These are called path constraints. They put limits on this.

- 5.35** Now, what is this called? This is called the search space. First of all, what's a solution to a problem? A solution to a problem is any sequence of operations that turns the initial state into the goal state while obeying those path constraints I was talking about. Is that OK? You say, OK. Then what is problem-solving? Problem-solving is any technique that reliably will get you a problem solution. Now, if I just said that, that would be trivial. But well, what's the issue here? Why is it hard to find a problem solution?
- 5.36** You're being misled by this diagram. This diagram is a God's-eye view from above, and you can see all of the pathways, and you can automatically see which one leads to the goal state. That's not life. Life is, you're here, and you don't know which one of all of these will get you there, or if they will at all. All right. Now, notice before I go on, another thing. Do you notice what's in here? The navigation metaphor. This is moving through a space, search space. I'm navigating through a space. I'm doing the foraging into foresight, right here, right here.
- 5.37** The scientists are doing it. They're exapting how I move through space in order to try and explain problem-solving. I just wanted to get you to note that. It's exemplifying the very thing we've been talking about repeatedly. OK, so what's the problem with finding a problem technique? And notice the language I use. There's a meta problem. Whatever problem I'm trying to solve, there might be a meta problem of finding that problem solution. Well, what is that problem? What am I dealing with?
- 5.38** Well, the meta problem I'm dealing with is combinatorial explosion. For very, very, very, very, two more verys, very, very many problems, this search space is vast. I don't mean vast like Mount Everest. I mean vast like the universe. Keith Holyoak in—what a great name, eh? Sorry. This is clearly a name of a fantasy warlord. In 1995, he gave an example, just to make people clear about it, he talked about playing a game of chess. Now, you calculate the number of pathways here by the formula F to the D , where F is the number of alternatives available to you at any one time, and then D is the number of steps you go through. Does that make sense?
- 5.39** For example, in chess, at any time, I have on average, because towards the end, this collapsed, of course. But on average, I have 30 legal moves available to me. Now, what you're seeing right away is

many of those moves are stupid, but that's a homunculus. I need a machine that can determine that. OK, so just stay with me. So 30 moves are available to me, and there's usually 60 turns. That's the number of alternative pathways. Now, that equals 4.39×10^{88} . That's an eight. Now, even if you had that many pennies, you are astronomically wealthy.

- 5.40** Now you say, "But the brain is so complex. I keep hearing that it's the most complex thing in the universe," which is false, by the way, because the brain plus anything else is a more complex system. But you know what they mean. On its own, it's a very, and your brain is. It's got a lot of neurons. It's got that many. You know what that is compared to this? Nothing. You say, "Aha. But John, John, John. It's not the number of neurons, it's the number of connections." It's good to do that when you're at a party because people go, "Oh, yes. Very good."
- 5.41** OK, well, the number of those you have is 5 times 10^{14} , which is a really big number. You know what is nowhere near? Even if each connection is searching a pathway, you know how many pathways each one has to search? It's overwhelmingly vast. Here's a really good one. What's the estimated number of atoms in the universe? Oh, look, that's much smaller than this. And this is just a game of chess, which is like— this is why they're famous.
- 5.42** They went in to Bob the homunculus, who projects obviousness, and went—because they said, "How do you do this?" And Bob goes, "Well, it's obvious." "No, but Bob, how do you do it?" "It's obvious." "No, Bob, how do you do it? How do you do it? How do you do it?" This is why they became famous, because this became a central question. Now, as I've mentioned, it's a central question that periodically gets forgotten and backgrounded and presupposed rather than explained. But they were really clear, this is a central question.
- 5.43** Now, let's be very clear about this, because this is what they also got. It's not like you can check all of that information and determine that it's irrelevant. Because if I check all of the space, that will take me more than the rest of the history of the universe. Now, you can't guess. You can't be algorithmic. An algorithm is a procedure that is guaranteed to give you a solution or deductively prove that one is impossible. If I was going to be algorithmic, how much of the space do I have to search to be guaranteed to find a solution or at least prove that one is impossible deductively? How much?
- 5.44** You can do some a priori things that limit it, but you have to search a lot of it. That's what you can't do because that takes too much time.

But you can't just be arbitrary. You can't just get, "Well, I'll look here." This is the search space. "I'll look here." Because your chances of finding it by arbitrary guessing are astronomically small. You can't search it all algorithmically, and you can't be arbitrary. You're caught between these two. The two things your culture says you do, you either—algorithm, and I'll come back to what this is more specifically in a minute—you either mathematically calculate your way through it or you guess. The answer is, "Nope, you're in here. You're in there."

- 5.45** Quick aside, connected to the course on rationality, what this means is you can't be comprehensively mathematical or logical. Spock and Data, if you're Star Trek, if you don't know who those are, but they are beings who are supposed to operate in a purely logical or purely mathematical fashion. That can't work. Because if I operate purely logically, purely mathematically, purely algorithmically, I will commit cognitive suicide the first problem I try to solve.
- 5.46** It turns out that being rational isn't the same thing as being logical. Being rational is knowing when, where, and what degree to be logical, which is a different thing entirely. You can even hear it a little bit in the word rational, ratio. Getting the right ratio, the right proportioning of your use of logic. That's part of why they became important even for economics, because a lot of economic theories were presupposing that human beings algorithmically calculate all the variables and blah, blah, blah, blah. I'm just trying not to do blah, blah, blah, and so on, and so on, and so on.
- 5.47** All right. Now, what has obsessed me since 1993 is, this is what you do. You don't search the whole space. You ignore most of it, overwhelming. And you zero in, you search a little bit of the space, and most of the time, because you're a general problem solver, you successfully find the relevant information by being overwhelmingly ignorant, which, again, is not something that the enlightenment thinkers like Kant and Locke would like to hear. "What? No, no. What we want is a being that has all knowledge." Well, actually, you have to ignore a lot, or you're not going to be a cognitive agent.
- 5.48** I call this ability to zero in on relevance without searching the whole space or guessing relevance realization. I'm playing on two meanings of the word realization, like when you realize something and when something is a reality. Is that OK? Relevance realization, which, of course, I've been talking about throughout this. Now you're already seeing, oh, there's convergence happening again. Yes, there is. There is. Now, I want you to notice how complex this is for you as a cognitive agent. Because you don't have—like, a chess game is a very limited thing, but you are doing this right now.

- 5.49** There's a combinatorially explosive amount of information you could pay attention to here. There's a combinatorially explosive amount of information you can remember. We did that. There's a combinatorially explosive number of things you could imagine. You could imagine what it would be like for a clown to be riding on a dragon as the clown and dragon dove into Lake Erie. I hope you didn't think about that until I said it. But you just did it. It's like, what? None of you went, you did it.
- 5.50** All the possible sequences of actions you can perform is combinatorially explosive. And yet, out of all of that, all of what you can attend to, all of what you can remember, all of what you can imagine, all of what you can generate, you zero in on the relevant thing to pay attention to, coordinated with the relevant thing to remember, the relevant amount of imagining, and the relevant amount of what sequence of actions I perform. You're doing it right now, like that. It's astonishing.
- 5.51** What AI wasn't doing was explaining how you did that. As people are turning towards AGI, they're moving to have to confront that problem. You're going to ask me, did the LLM solve this problem? Put a pin in it, we're going to come back to that. Because it depends what you mean by solve. OK. If that's all they did, they get to be in the hall of the immortals. That's brilliant. It's like, wow. They created the light saber that killed Bob the homunculus. It's just amazing. Sorry, I shouldn't be making Star Wars references anymore. Star Wars has gone being from our sacred canon to being the most despised.
- 5.52** Now, they came up with something very interesting that turned out to be very important, and I need to get this not only to connect this course to the other courses, but for what we're going to be doing. They come up with the notion of a heuristic. They made a distinction, and this distinction has really blurred as computer science has gone forward and gone into popular culture. But they made a distinction in a really principled, clear way between an algorithm and a heuristic.
- 5.53** An algorithm is a problem-solving technique, as I said, that is guaranteed to find a solution or prove, deductively prove, that one is impossible. For example, there's an algorithm for determining a number of people in the audience. What's that algorithm called? What can I do? I can count. If I do it right, it's guaranteed to give me the answer. Is that OK? There's some things you can solve algorithmically. I wanted to do that because I wanted to show you. I'm not saying that you can't solve some problems algorithmically. Some you can. You apply the algorithm of counting, and as long as there aren't 10 billion people in the room, it doesn't take you very long.

- 5.54** Now, if I asked you to count the number of atoms in the table, that becomes a problem. But anyways, you know what we're talking about here. That's an algorithm. But for the reasons I have articulated, Newell and Simon argued, most of the time you can't use algorithms. So what do you use instead? And this is a term you've probably heard, you use heuristics. Heuristics are rules of thumb and strategies that increase the probability that you'll achieve your goal, but don't guarantee it.
- 5.55** So when I'm playing chess, a good heuristic that a lot of chess players do is control the center board. If you control the center board, it really increases your chances of winning. You don't want to do things with a cognitive scientist. I was playing with chess with my stepson, he's very good at it, and I realized, "Oh, he's focusing on the center board. I'll make him think I care about the center board while I slowly build a peripheral attack," and I beat him. You can control the center board and still lose.
- 5.56** Do you understand what a heuristic is? How does a heuristic work? Basically, what it does is it prespecifies where you should search. Only search here. It really cuts down the search space. Does that make sense? Now, what's the problem with that? Well, another word for prespecifying is prejudging, and another word for prejudging is what? Prejudice. That's literally what the word prejudice means. Every heuristic is a bias. This is known as the "no free lunch" theorem. See, academics trying to be funny again.
- 5.57** The no free lunch theorem says, this is me solving problems on average. What a heuristic will do will increase my chances of solving a problem for a certain population of problems. Where's the no free lunch? There is an area in which the heuristic will equally cause my performance to degrade, to become substandard. Does that make sense? Let me give an example of this. So heuristic and bias. Why am I doing this? Because I want to show you how this shows up in your behavior, in your intelligence, and how this was missing from all the other stuff we were talking about.
- 5.58** OK, so judging probability of events is very hard. Because as soon as you try and assign a value to all the events and all the combinations, what do you hit? Combinatorial explosion. Probability theory is an algorithm for calculating probability. And when you apply it in real-life situations, you can't typically calculate the probability. Then that goes up exponentially as you move even incrementally into the future. Is that OK?

- 5.59** So your brain doesn't calculate probability unless you're being forced to calculate probability in a very well-defined problem that you can apply the math to. What does your brain do? Well, it uses two heuristics. It uses the availability heuristic, which is probably a variation on fluency for those of you who did the Primacy of Beauty. Availability heuristic says, the ease with which I can—ease, like fluency—the ease with which I can remember it or imagine it means it's more probable. If I can easily imagine it or easily remember something similar, very probable event.
- 5.60** The other one is called the representative heuristic. The more prototypical it is, the more it's a really good instance of that kind of event, that's a really good bird. I bet you those kinds of birds are very probable. We use the availability heuristic and the representative heuristic to judge the probability of events. Is that OK? Now, you're always thinking, "Well, they can go wrong." That's my point. Let's do it. What's really funny about this is, I do this and I'm a cognitive scientist. I know this and I still do it.
- 5.61** You're driving your beloved friend, partner, somebody you really care about to the airport and they're about to go on a flight. So you say things that are magical euphemisms like, "Have a safe journey," because the universe is listening. "Text me when you're there," because the plane can't crash. "No, wait, plane, you can't crash because I have to text my—" And the plane goes, "You're right." What we do is, we do all this euphemistic magic, and what we're basically saying is, "Please don't die, please don't die, please don't die, please don't die, please don't die."
- 5.62** Why? Availability heuristic. It is easy for your Homo erectus brain to imagine a big metal thing falling. You go, yeah. Yeah. Yeah, we got that. Big metal thing fall. Then when airplanes crash, what did they call it? Do they call it a crash? No, it's a disaster. It's a tragedy. It's highly representative. So these two heuristics are telling you this event is highly, highly probable, and they're wrong because it's actually a very low probability event. And then, without thinking about it at all, you go back to the parking lot and get into the North American death machine called the automobile that kills more people than many diseases.
- 5.63** Do you see? Do you see how it's right in the very guts of your thinking? We need an account of intelligence that takes into account all of this, and we haven't got that yet. Now, this is part of what I mean when I say, the very processes that make you intelligent always make you prone to self-deception. You're deceiving yourself about the plane crash. Like I said, the thing that really bugs me is, I know this,

and I take somebody to the airport, and I still hear myself saying, "Don't die," with all the euphemisms. "Please don't die."

- 5.64** So, are heuristics enough? Well, no, because if it was, and they came up with a particular heuristic called the means ends heuristic, and I won't go into that. But there's a problem. Well, first of all, there's the problem of how do you generate the heuristics in the first place. How do you select which one to use? There's a little homunculus in there. OK, so remember that. We'll come back to it. But the GPS project actually failed. If it had succeeded, we would have had artificial general intelligence in 1972. So something went deeply wrong.
- 5.65** There's two flawed assumptions within Newell and Simon's framework that only became available to us after decades of reflection on their work. One is this assumption. All problems are essentially the same. So once I find a heuristic, if all problems are essentially the same, I can apply that heuristic to them. Because all problems are essentially the same, that will make a general problem solver. There you go. Great. Now, what's the problem? Well, you know problems are not all essentially the same. Because there's a distinction between well-defined and ill-defined problems.
- 5.66** Now we can give a more precise explication of the contrast. In a well-defined problem, the initial state, the goal state, the operators and the path constraints are clear and helpful to you. In an ill-defined problem, the initial state and/or the goal state and/or the path constraints and/or the operations are unclear to you. Now, why did they think that all problems were essentially the same? Here's the irony. Your brain uses a heuristic called the essentialism heuristic, which is, wherever there's a category, you think there must be an essence to it. It's a heuristic your brain uses.
- 5.67** Do you see the irony? They fell prey to a bias created by a heuristic that human beings use. Not all categories have essences. What's an essence? An essence is a set of necessary and sufficient conditions. Each condition is necessary. The set of conditions is sufficient. You can tell me what the essence of a triangle is. What does a triangle have to have? Three straight side. It has to be enclosed, forms 180-degree angle. No matter how I draw that, the size of it, that's a triangle. If I make a proof about any triangle, it will generalize, unless the proof is dependent on it being a right angle triangle or something like that.
- 5.68** OK, Wittgenstein famously said, "What's the essence of a game?" How many of you have played games? Raise your hand. OK. I'm not saying, "What is a game?" OK. What you need to do is give a definition that

includes all and only games. Well, games involve competition, catch. Games involve a board game, tag. Games involve a fictional world, catch. Games are done for competition. Lots of stuff is done for competition that isn't a game, war. We even make a distinction between war and war games. I won't do this because nobody's ever been able to come up with a definition for game that will pick all and only games.

- 5.69** You can do this with a chair. You can do this with toys. Many of our categories don't have essences. The philosopher Quine basically made this proposal. Science, mathematical science, science, including math, is the project of determining which categories have essences and which don't. That's what science does. Gold has an essence. These are the properties. If it has these properties, this is necessary and sufficient for being gold. There you go. Science discovers the categories that have essences.
- 5.70** It's not a bad idea to have the essentialist heuristic because it gets you looking in the world that sets you on the course for possibly doing science. But it is a heuristic, and it can go bad, and it went bad with them. It biased them. So all problems are not essentially the same. When all problems are not essentially the same, what comes to the fore? Well, if you think all problems are essentially the same, then how you formulate your problem isn't a big issue because all problems will be formulated basically the same way.
- 5.71** What's problem formulation? Again, my representation of my initial state, my goal state, my path constraints, my operators, so that I can generate a search space. But what's actually missing in an ill-defined problem? Well, the good problem formulation. What happens in a moment of insight? You replace a bad problem formulation with a good one. What I want to do is show you this and show you the power of problem formulation, which is at the meta level. This is called the mutilated chessboard. It comes from an experiment done by Kaplan and the Simon, the one we've been talking about in Newell and Simon, from 1990 called, there's a pun here, "in search of insight." Aha, so funny, search space, insight.
- 5.72** All right. Here's the mutilated chessboard. I'm going to draw this very quickly and very crudely. You know that some of the squares are black and some of them are white. This is a bad, crappy diagram. That's irrelevant. In the experiment, people were given very good diagrams of chess boards. Let's go through this. Eight rows and eight columns. How many little squares? How many? Sixty-four. I hope you can do that. You're supposed to be able to do that by grade four. Now I have

a domino. Now, because this is a square, it will equally cover two squares vertically or horizontally. Yes?

- 5.73** How many dominoes do I need to cover the chessboard without overhang or overlap? What you say, "Oh, this covers two squares. I divide it by two, so I need 32 dominos," and that's the answer. Well done. Now I'm going to mutilate the chessboard. I'm going to remove this piece. I'm going to remove this piece. Here they are over here. Now, there are 62 squares left on the mutilated chessboard. Can I cover that without overhang or overlap with 31 dominoes? You have to prove your answer. You can't guess.
- 5.74** Here's what you're probably doing, and we know this because this experiment has been run. You're formulating this as a covering problem. You're trying to imagine various configurations of dominoes on the chessboard. You know what that search space is? Combinatorially explosive. Famously, there was a person trained in math who formulated it as a covering problem, tried to do the calculation, filled up, I think it was like 81 pages of a Hillroy notebook over 18 hours or something like that. If you formulate this as a covering problem, you're doomed. It's a good way to kill Bob, but in a way that kills you.
- 5.75** Let's reformulate the problem. Now, again, my diagram is misleading, but that's not the case in the experiment. Because one of the things that's clearly visible but not salient, so notice the language, clearly visible but not salient, is that these two squares have the same color. Is that relevant? Notice the question. Initially, your brain says no, because it doesn't make it salient or obvious to you. Now, if I put this domino on the board, no matter which way I put it, horizontally or vertically, I'm always covering a black and a white square. I'm always covering a black and a white square.
- 5.76** What do I need in order to cover the board with dominoes? I need an equal number of black and white squares. If these two are the same color, do I have an equal number of black and white squares? Nope. In order to cover the board, I need an equal number of black and white squares. I don't have an equal number of black and white squares, so it's impossible to cover the board. When I go from covering to looking for parity, the search space goes from [noise] to [noise].
- 5.77** What's impressive is, there are individuals in this experiment who have that aha moment, and they go from a very bad problem formulation that's combinatorially explosive to a really good one. That's the power of problem formulation. Now, what they wondered is what are people doing to improve their chances of having the insight?

They talked about noticing invariance. What does that mean? Across different formulations, try to notice what you're not changing. Across different failed formulations, try to notice what you're not changing because that might be the thing that's holding you back.

- 5.78** Which means, you got to have a lot of epistemic humility. You got to keep track. I failed, that failed, that failed, that failed. What's going on? What am I not noticing? I'm not noticing the parody. I'm showing you the way insight works. I'm showing you the power of problem formulation, and I'm showing you it involves perspectival knowing. It's about how you're paying attention, what you're finding salient. How you're framing the situation is actually very, very crucial. Does that make sense?
- 5.79** Now, where have you seen that before? What's going on in noticing and insight? You're manipulating salience because you're doing recursive relevance realization. At the heart of intelligence is recursive relevance realization, again, but more specifically. Now I'm going to do something that's even more annoying because I keep coming back to this because I'm building this convergence argument. I'm trying to make it like, wow, yes. In 1986, Charniak published a book called Minimal Rationality. It turns out he's actually talking about intelligence, rather than rationality. That came out in the debate. He basically proposes that intelligence is relevance realization. Then he proposes a compartmentalizing solution that doesn't work.
- 5.80** Sperber and Wilson write a book in 1995 called Relevance, where they zero in on relevance realization being the main thing going on in communication and cognition. This guy named Vervaeke publishes his thesis in 1997, making the argument that relevance realization is at the center of cognition and communication. Then Vervaeke et al. do another one in 2012. Vervaeke and Ferraro in 2012 and 2013, van der Kaaij in 2022, Henriques in his book in 2022. Andersen, Miller and Vervaeke in 2022, Hovhannisyan and Vervaeke in 2022. Jaeger, Riedl, Djedovic, Vervaeke, Walsh, 2024, and Frank and Gleiser and Thompson, one of the founding figures, by the way, of 4E cogsci.
- 5.81** In a book called The Blind Spot, they said the heart of the problem in cognition is, and they literally used the word relevance realizations, and they cited me and all that stuff. There's been a long series of research on this problem. Now, what I'm not going to do in this lecture is tell you what that research showed. I'm going to do that next lecture. Because I got to give a reason for making you hang around. But what I want to do is keep going through this argument. Relevance realization is at the core of our cognitive agency, our

general intelligence, AI, especially when it's trying to be AGI, it's pointing us towards that.

- 5.82** This is deeply convergent with all the synoptic integration of psychometrics, cognitive psychology, and neuroscience we've seen. Now we see that relevance realization is at the heart of problem-solving. In the paper that I did with Tim Lillicrap and Blake Richards in 2012, we made the argument that relevance realization is at the core of communication. It's at the core of inference. That lines up with what Sperber and Wilson say about communication, what Charniak says about inference, at the core of machine learning, et cetera. I'm not going to repeat all of those arguments. You can read that paper if you wish.
- 5.83** I want to show you how this comes out in what was considered one of the most central discoveries about the problem facing AGI. This is called the frame problem. Remember I mentioned it earlier? Now we're going to do the frame problem. Because a lot of people are tempted to think, yeah, but this all comes down to just very basic things. At the very basic level, you don't have this problem that you're talking about. I want to show you that you're really wrong about that.
- 5.84** A basic problem you have is getting food, yes? That's a really Darwinian basic problem. When you don't get it, you suffer pain, and when you get it, pleasure. Now, you, like many other organisms, don't eat your food where you immediately find it. Why not? Because it's dangerous to do that, originally. Also because we use eating food together in order to reinforce one of our most powerful adaptations, which is social bonding. Because one of us against a lion, you're dead. You got a bunch of us with pointy sticks and some trained dogs, and we can kill anything on the planet. That's why you have evolved to want to share food with other human beings. You will do silly things like this.
- 5.85** "This tastes horrible. Have some." Because you want other people to share in your experience. You're not just eating to take in nutrition. You very rarely eat food where you buy it. You're in the supermarket, you don't do that. Now, the fast food revolution was, get this, you can eat your food exactly where you buy it. Restaurants, it's a little bit different, standard ones, because they're bringing the food in and all that stuff. OK, so let's make it plausible that we're going to build a robot, and it's going to find food and it's going to transport that food somewhere safe or socially appropriate. OK, maybe it's a social robot. Social robotics is a thing, by the way. It's a very real thing.

- 5.86** OK, so the robot is electric, so it doesn't use the kind of energy sources you use. You use chemical batteries. That's what food is, chemical batteries, take in the energy. It's going to look for a standard battery. We're going to have this situation. Here's a wagon and here's the wheels, and here's the handle, and on it is a nice, juicy battery. Over here is my robot. It's got a big calculation. It's got sensors. It can move around, and it's got a way of grabbing things. Now, even making this took a long time. We're just on the edge of that right now, but we'll put that aside.
- 5.87** What we want the robot to do is we want it to behave, but we want it to be an agent. Everything behaves. If I drop this, it will make a noise. Everything behaves. What's an agent? An agent is something that can determine the consequences of its behavior and alter its behavior in order to bring about alternative consequences. Does that make sense? That's what an agent is. We want to build a robot that's a cognitive agent. We want the robot to be able to determine the consequences of its behavior.
- 5.88** We put it in this situation, and it determines that, if it pulls the wagon, then the battery will come along and that will help it to transport the battery to the safe or socially sanctioned space. Is that OK? OK, but here's what we're going to do. We put a bomb on this. This is a very bad drawing of a bomb. It looks like something from Looney Tunes, but this is a bomb, that's the fuse, and it's lit. Is that OK? Which means it's going to go off. Is that all right? That's one of the important properties of bombs.
- 5.89** What happens is, the robot comes in and pulls the wagon in order to achieve the intended effect of pulling the battery. What's an unintended side effect? The bomb comes along, and the bomb blows up and destroys the robot. We go, oh, we were stupid. We have to make the robot not only detect the intended side effects, it also has to examine the unintended side of—sorry, not only the intended effects, it has to examine the unintended side effects. Until we had cell phones, human beings sometimes fail at this task. What? What's cell phones? Because now when you need a light, you turn your cell phone on.
- 5.90** This is what people used to do when they needed a light. They would strike a match. Now, people would go into a situation in which they could smell that there was flammable gas diffused and think, "I need a light to see what's going on," and they would strike a match with the intended effect of generating light. What's the unintended side effect? Boom, they killed themselves. Oh, so what we'll do is we'll make it

detect or determine not only the intended effects, but the unintended side effects. Good, good, good.

- 5.91** What we'll do, and what we'll do this time is, we'll put a black box in it like we do in an airplane, and it'll keep track of what's going on inside the robot. See? I can draw a black box. We put it into this situation again, and the robot gets close to the wagon, and then it starts calculating and calculating and calculating and calculating, and then nothing happens, and the bomb goes off. What's going on? We look inside, and this is from a book called *The Robot's Dilemma*, the particular example came out 1987. This is a famous example by Dan Dennett, who just died recently. Just died recently.
- 5.92** We look inside the robot and what's it doing? Well, it's doing what we set it to do. It's calculating all the unintended side effects, like, if it pulls the wagon, it'll change the degree of arc of the wagon handle. It'll make the front wheel go through more than 30 degrees of rotation. The front left wheel, the back right wheel, the back left wheel. There'll be indentation in the grass underneath the wagon. The position of the wagon with respect to Washington, DC, is slightly changed. What is the number of unintended side effects? Combinatorially explosive.
- 5.93** We think we need to give it more power. We're going to make it bigger. Extra wheels. More sensors. We put it into this situation, and we're going to give it an ability that I've already shown you we can't actually give to things. We're going to give it the ability to detect relevance. I've already given you an argument you can't detect relevance, because relevance is not a stable, invariant property of anything. I can't detect relevance because relevance, in order to detect something, it has to be a property that belongs to instances of it.
- 5.94** So, is this relevant? Is it intrinsically relevant to you? No. Is it always relevant to you? Well, it's relevant right now, but it's not going to be relevant when you're at home tonight going to bed. But what about the black marker? Is it predictive? When I learn about a triangle, I can predict a whole bunch of other, what's going to happen in other triangles. Even with empirical things, if I learn about this horse, and you depend on this being a veterinarian, I can make predictions about what other horses are going to be like. So calling something a horse gives me all that powerful prediction.
- 5.95** So other than being relevant, what do you know about anything that we call relevant? Nothing. I found this marker relevant. What does that tell me about the next thing I find relevant? Only that I'll find it

relevant. You see, that's why you can't detect relevance. It's not an intrinsic property, it's not a systematic property, it's not a stable property. OK, is that all right? All right. We're going to ignore that really devastatingly powerful argument that you can't detect relevance. I'll come back to that, by the way. You can't detect relevance or project relevance. But we're going to give it the ability to detect relevance.

- 5.96** We think, ah, we've solved it. We put it up to here and it comes up and it is calculating and calculating, and we're funneling energy from nuclear reactors into this because like with the LLMs and the huge amount of energy we have to put in, and nothing. And it's like, what? Because what it's doing is, it's making a list, making two lists. Here's relevant things. Here's irrelevant things. Oh, wheel moving, handle moving. It's irrelevant, irrelevant, irrelevant, irrelevant, irrelevant. See, you have to ignore irrelevance. You can't detect it, which is also symmetrical with why you can't detect relevance.
- 5.97** Do you understand? This is the frame problem. It's a problem not for doing really abstract, complex problem-solving. This is to solve a simple Darwinian problem of transporting your food. Why is it called the frame problem? Because of all of the information, you have to put a frame around what you pay attention to and care about, the frame problem. Problem formulation is a version of the frame problem. Problem formulation is a version of the frame problem. You formulate a problem and it limits what you're paying attention to. Is that OK?
- 5.98** What's at the heart of this? Relevance realization. Relevance realization. So people turned away and said, "Well, let's just work harder." It's like Boxer in Animal Farm, forgetting that what happens to Boxer is they send him off to be killed for glue. Sometimes working harder is not the right answer. I know we're not supposed to believe that because we're all post-Protestants, but sometimes working harder is the wrong thing.
- 5.99** Herbert Dreyfus made a really good joke about this. You've got the ape that is trying to climb to the moon by climbing a tree, and it's initially, it's making progress. And it thinks, "I just have to work harder. I have to figure out how to get higher and higher into those more and more thinner branches. And someday," and it's like, no, stop it. Get off the tree. That's not going to get you there.
- 5.100** Now, what do we know about the frame problem? Well, you already heard me mention it. The global workspace theory. Remember, Shanahan and Barrs, the job of the global workspace, consciousness attention, is to solve the frame problem. Now, remember, I

foreshadowed it, now you know what they're talking about more specifically. What did Shanahan argue? He argued that there's two problems in the frame problem. One is to create a logic of change, which turns out to be tricky, but he basically solved it. That's why he gets, like, if you look in the Stanford Encyclopedia of Philosophy, the frame problem, the person writing it is Shanahan.

- 5.101** He said, once you solve that logical problem, what remains? The deep problem. What did he call that problem? The relevance problem. Let's do a quick aside because we've got some handle on AI and some of the deep problems it faces. The question that you might be asking is, well, what about the large language models like ChatGPT-4? Aren't they solving the relevance problem? They're, like, "Look what they're doing." It depends what you mean by solving. Because how do we actually train these machines? Well, we give them tons and tons of human text.
- 5.102** Now, let's notice something first of all. There are probabilistic relationships between words. If I say the cat, there's a probability it's going to be a verb here. Yes? Yeah. Now, notice what we have done, and this is the really cool invention about language and then the cooler invention about literacy. And notice we did both of those. Literacy is not the same thing as language. For a lot of history, and even in large parts of the world, human beings are linguistic, but they're not literate.
- 5.103** So first of all, there's double coding going on. What do I mean by double coding? So you're making all kinds of judgments about how things are relevant to each other, and you're encoding them in probability relationships between text, pieces of text. But really, what you're first doing is, you're encoding them in probability relations between utterances, and then you are recoding them into probability relationships between texts. Why did I make you say that? Because I want you to understand people often confuse that literacy is language, first of all.
- 5.104** Secondly, you don't speak the way you write. You can see, kids when they're in fifth grade and they're starting to write essays, they write an essay as if they're just talking. And it's like, this makes no sense. You have to go through a long process of manipulating the text because language requires, relies on a lot of non-textual context. See the word? Context. That which is around the text, helping you to understand. When I said helping you to understand, did I mean my partner? Because I sometimes call her you. No, I meant, what? Did I mean you? Sometimes I mean you. Sometimes I mean you. But none of

you went, "Wait, I do not know." Because you're doing massive amounts of—

- 5.105** So, relevance realization is in communication, and then you have to do relevance realization to change utterances into text. You're doing all of this encoding to get to the text. Does that make sense? Then human beings select the initial set of text to train the machine. They judge. You say, well, what does that matter? Because they have organized text into categories. What does categorization require again? Relevance realization. Then, where are all those texts organized? Well, they're organized on the internet. Is this the internet random? No, it's really carefully organized by what? By human judgments about what's intelligent, sorry, what's relevant and salient, putting it together.
- 5.106** Then once we get the machine running, we do reinforcement learning on it, which we'll come to. What does that mean? The machine spits out some stuff and we ask human beings, "Is that appropriate?" The human beings go, "Yes or no." Then we even run it and human beings are allowed to say, "That was an inappropriate thing. That turned out to be racist. Or that doesn't follow. No, that's wrong." The maddening thing about these machines is they're very confident even when they're basically bullshitting you. When you tell them they're wrong, they might even admit they're wrong. Sorry, they'll put out text. "Sorry about it." Then they don't change their behavior, et cetera.
- 5.107** Do you see what's going on here? These machines don't explain relevance realization. They presuppose it. There's a huge homunculus for these machines, and it's all of us and our individual, and we'll come to this next lecture, collective intelligence. Now, you may say, so what? Well, and that's where it matters when I asked you, what do you mean by solve? There's no scientific advance here. Here's how I can prove it to you. The way we train the LLMs, is that how a chimp became intelligent? Is that how an octopus is intelligent? No, it doesn't generalize. It's not a theory. It doesn't explain anything. It doesn't explain it. It doesn't generalize.
- 5.108** You may say, and this is legitimate, you may say, "I don't care about the science. These machines are going to take away my job," and that's still true. It depends what you mean by, have they solved the relevance problem? They have not generated a scientific theory. They have not generated an explanation that generalizes. That's why we're in a particularly dangerous situation because I had the hope. The reason why the hope is in Pandora's box is not because it's the greatest thing, it's because it's the most evil thing. I had hope that, if I and other people like me, cognitive scientists, were working together,

we would get the science of intelligence, general intelligence, before somebody hacked their way into the technology.

- 5.109** Because that way, maybe for once, we would have the science, and if we understood intelligence, it would be a science that could help us improve our intelligence, and maybe we could even be rational and wise about **artificial intelligence**. I was arguing for this and trying to make it happen, and my fear came true rather than my hope, because that's life. They found a way to hack their way into machines that do it. We are generating technology that we literally do not have good science for. That's what I said to you when you asked that question. Not only that, we don't have good rationality about it, we don't have good wisdom about it.
- 5.110** I do think the more we can actually come to an explanation, a generalizable, including **multiple realizability**, explanation of general intelligence means we can get needed knowledge, not only about these machines, but we can potentially improve our intelligence, our rationality, and wisdom. If you want to see the course on that, I have a course called Intelligence, Rationality, Wisdom, and Spirituality, and also a lot of videos online. If you're interested in that argument about the LLMs, I have a video on YouTube, a video essay about the scientific, philosophical, and spiritual impact and import of the LLMs.
- 5.111** I also have a book out. It's a shameful self-promotion called Mentoring the Machines, written for non-technical, about how we can best respond to this. It's coming out in serial fashion. Books one and two are out, three is going to come out soon, and then four. Why did we do that? We deliberately released it in serial fashion, so we had intervening time to update if we needed to, as we were releasing the book. We're not trying to be coy.

6. Evolution of AI

At a glance

Traces the generations of AI and cognitive science, starting from the first generation's assumption that cognition is computation and thought is what a computer does, using formal systems like chess and multiplication as examples. Discusses the limits of large language models (LLMs) and the concept of an autopoietic body as missing from current machine architectures.

- 6.1 I'm going to say that there's some interesting architecture, the way you design it in **LLMs**, that points to some further things needed for **AGI**. But those were already understood before the **LLMs**, but it's in there. That's very vague, and I'll come back to it and I'll explain it to you, like stuff around deep learning and things like that. We're getting there.
- 6.2 What I want to do now is the generations of AI and cogsci. Is that OK? All right. So the first generation is that **cognition is computation**. To think is to do what a computer does. **Cognition is computation**. That's what the GPS was. What is computation? This term has now become wishy-washy and so semantic drift. I'm going to use it in a way that was being used at the time of first-generation AI and cogsci. Computation is when there's a **formal system**. What's a **formal system**? It is a bunch of rules for solving well-defined problems.
- 6.3 Chess is a **formal system**. There are rules for solving well-defined problems about how you play chess. Is that OK? Multiplication, **formal system**. What you have is a **formal system** that logically manipulates propositions. What are propositions? Propositions are subject predicate, things like "the cat is on the mat." Propositions are statements that can be true or false. The idea is if you logically manipulate them with the right **formal system**, you'll get conclusions that are true about the world, and that is what it is to be intelligent.
- 6.4 So computation is automated logical argumentation. That's what a computer program is. It's an argument about logical relations between representations of propositions. And then what you do is you build a machine that is isomorphic, has the same causal structure to the logical structure that's going through the program. The program, the logical structure is the software, the actual machine is the hardware, and you put the two together, you get a computer. Now, that's a solar system explanation, by the way, but it's good enough. It's good enough.

- 6.5 Like I said, computers are machines that can be programmed so that they automatically manipulate representations of propositions. They come to a conclusion, and what does this mean? This is based on a fundamental assumption, which is what? That cognition is like a computer program or like an argument. It's a language-like thing you're doing. That's why we call them computer languages. Is that OK?
- 6.6 The idea is that when you're doing cognition, you're doing something very much like what you're doing when you're using language. This is a plausible idea because how do I communicate my thoughts to you? I use language. A lot of time when I'm thinking, I put things into language. Yeah. Notice this really emphasizes propositional knowing as the heart of cognition.
- 6.7 This is the idea made famous by Jerry Fodor, that cognition is to use a language of thought. When you've got a really good program in your language of thought, you're intelligent. I get the right program that captures, creates a **formal system** for the language of thought, and that's how I get a general problem solver.
- 6.8 Now, there's problems with this. Jerry Fodor is the person who really argued for this. He was a protégé of Noam Chomsky, the great linguist. So Fodor is arguing for this. He is arguing and defending until he died, he died about four or five years ago, that **cognition is computation**. I need you to know that because he's such a great, great theorist. Why? Because you know who gives some of the most deepest criticism of the idea that **cognition is computation**? Jerry Fodor.
- 6.9 The capacity for self-correction is the hallmark of rationality. Being always right is not the hallmark of rationality. Admitting that you were wrong about something that you had argued for or believed is the hallmark of rationality. That's why many people who present themselves as being the paradigms of rationality in the social media world are not because you don't see them ever saying, "I was wrong about this," or "Here's a deep criticism of my own view."
- 6.10 So what did he point out? Well, he pointed out that propositions and the logical manipulation of propositions can't get you relevance. That's literally what he did, by the way, and he used those terms. Let me give an example. Is there two Ms in tomorrow, or one? One. OK, we'll go with what Reuben says. He's very confident. OK, so I hope for none of you this is a very problematic proposition. Tomorrow will be windy. OK, now the thing is that the subject predicate structure, I'm

not going to change that. I'm going to keep that constant. How many implications are of that for action?

- 6.11** Well, unlimited because it depends on what other propositions I put it in relationship to, yes? The number of implications for anything is combinatorially explosive. For example, what does that imply if you're going sailing tomorrow? That implies some things. This becomes quite relevant to you, yes? What does it imply if you're staying in and watching movies? Very different. Probably largely irrelevant, depending on how windy it is. If it's hurricane windy, that matters. But what about if I'm flying a kite? Well, not—and how many of these can I generate? Combinatorially explosive.
- 6.12** So here's his point. The subject predicate structure and all of the implications, which is the logic of the proposition, that's stable and it's combinatorially explosive. What you do when you make an inference, don't use implication and inference as synonyms. Inference, implication is a logical relations between propositions. Inference is something an agent does. You will say, given my goals, that's relevant to me. That's what inference is. You ignore most of the implications, zero in on the one that's relevant, and you pick it. That's what Cherniak argued way back in 1986, by the way.
- 6.13** When you're doing inference, you're not being algorithmic and you're not being arbitrary. You're being in between. Here's the point he's making, Fodor. The logic of the proposition is constant, but its relevance is continually varying. You cannot determine the relevance of it by looking at its subject predicate structure and all of its logical implications. But that's all you can do in a program. Do you understand? You can't capture in a program relevance relations. The guy who came up with the idea that **cognition is computation**, this is devastating.
- 6.14** OK, well, I'm using a **formal system**, so it's rule-governed. OK, so this is an argument that goes back to Brown in his book from 1980 on rationality, back to Wittgenstein, back through Wittgenstein to Aristotle. OK, so rules. You have to be able to follow rules. Yep, that's what I have to do to be able to follow rules. To be a computational thing, I have to follow rules, following rules. OK, so let's pick an easy rule, and I'll pick a moral one because it's easy for you to imagine. I hope you're all trying to follow that rule. Be kind.
- 6.15** You know what you can't stipulate in the rule? The conditions of its application. What it is for me to be kind to Reuben is not the way I'm kind to my romantic partner. That would freak you out, wouldn't it, Reuben? It's like, "Oh, no, this is going really, really wrong." The way

I'm kind to her is not the way I'm kind to my son. That would be inappropriate. The way I'm kind to her in everyday life is not the way I'm kind to her when I'm at a funeral for her cousin. That's even different if I was at a funeral for her mother. It's different if—do you see what's happening here?

- 6.16** What I'll try and do is, oh, what I'll do is I'll make rules for all those possible situations of application. That's combinatorially explosive, so rule following depends on what? The skill of relevance realization, which you can't ultimately capture in a rule. This is why trying to replace human judgment about wisdom and virtue with too many rules will fail in a society. You can't legalize and capture in bureaucracy relevance realization.
- 6.17** OK, what about representations themselves? The computers are using representations of propositions. Well, if it's going to be a representation, let's look at how representations work. This goes to the work from John Searle from a book, 1992, *The Rediscovery of the Mind*. So Searle says, well, every representation is aspectual. What does he mean by that? Bear with me for a sec.
- 6.18** How many properties does this actually possess? But when you think of it as a marker, you select a subset that's relevant to you. When you think of it as a weapon, you select another. I'm not going to throw this at you, Michelle. Another subset that's relevant to me. Do you see? Aspects are central to being a representation because if I didn't aspect, or aspectualize, I would be trying to capture combinatorially explosive amounts of information. Does that make sense? Aspectualization is dependent on relevance realization, especially if that aspect is a category.
- 6.19** Have you ever tried this sometimes? Look at an object and try not to see it? Like, look at that thing and don't see it as a chair. You can do it, especially if you do a lot of mindfulness practice, but it's really hard. OK, what about development itself? So Jerry Fodor, back again. All right. So this is Jerry Fodor attacking Piaget, the person who did the accommodation assimilation that you thought was, hey, that's good. And you thought it was really good, too, Andrew.
- 6.20** So Fodor says, OK, **cognition is computation**. There's two kinds of development. There's quantitative and qualitative. Quantitative is when you take in more information. There's no problem with computers for that, you just—more input. Of course, kids do that. They just take in more information. But that's not the only development, according to Piaget. They also show qualitative

development. What does that mean? They can take in new kinds of information.

- 6.21** In order to make this a little bit more distinct to you, imagine, like, spoken language. You can keep using spoken language, you can keep accumulating information, yes? Yes or no? But what if I teach you how to graph things, Cartesian graphing? Now, you can also take in lots of information. But can you capture information in graphing that you can't capture in language words? Yes. That's why science depends on graphing. Does that make sense? This is a qualitative change, not a quantitative change, not the amount of information, but the number of kinds of information you can generate functions for. Is that OK?
- 6.22** Kids clearly do this. You've done it. You weren't born like, "I can graph." Before you can talk, you have to learn it, and then it empowers you. All right. So Fodor, development. Now, it seems there's overwhelming evidence, Piaget, that kids go through qualitative development. Can you do things that you couldn't do before? Yes, clearly. And is it just that you have more information? No, you have more kinds of information. But Fodor is going to argue against Piaget because he's going to do the following.
- 6.23** Cognition is a **formal system** using a **formal system** or it's computation, which is a **formal system**, all the stuff we've been talking about. Is that OK? He says, OK, qualitative development would be a change in the competence or power of the **formal system**. I'll just use the shorthand logic for that. Qualitative development would be to go from a logic that is weaker to a logic that is stronger. Does that make sense? Because I have to change not how much information, but how many kinds of information.
- 6.24** Let me give you an example in logic. Have any of you studied formal logic? You'll remember some of it. What they initially start teaching you is they initially teach you propositional logic, in which you have letters that stand for entire propositions. You use the letter A to stand for a proposition; letter B stands for a proposition. The problem with that is, it's a very weak logic because you can't capture the subject predicate structure. If I use a variable to stand for an entire proposition, I can't capture the difference between "the cat is on the mat" and "the dog is in the house." Is that OK?
- 6.25** So you create what's called predicate logic. So you have propositional logic and you create predicate logic. Predicate logic is stronger because I can capture. So what I have is, I have symbols that pick out subjects and symbols that pick out predicates. And now I can do more

sophisticated arguments. I can do arguments like "All men are mortals, Socrates is a man, therefore Socrates is mortal."

- 6.26** But the problem with that, with predicate logic, is it can't distinguish between necessary and possible. Because sometimes we want to say it's necessary that or it's possible that. When you do that, you're moving into modal logic. There's logics in between, like when you get some or all or each. That's a more sophisticated predicate logic. Do you understand? Each time what I do is I add in new functions. Hear that? I add in new functions that make the logic more powerful. What does more powerful mean? Can capture more and more of language, more and more kinds of arguments. Is that working?
- 6.27** So Jerry Fodor is bang on here. He would do this. He'd get you all in shape and then, right? OK, what is it? There is no logical way to go from a weaker logic to a stronger logic. I can do all possible logical calculations with propositional logic, and I'll never get predicate logic. I can do all possible—you can prove this—I can do all possible calculations with predicate logic, and I'll never get to modal logic. Why? Because how do I get from one logic to another? I step outside the logic and I add new axioms and new functions that are not generated by the logic I was initially in. Does that make sense?
- 6.28** OK, so get this. Let's do this. **Cognition is computation**. Qualitative development is to go from a weaker logic to a stronger logic. But there is no way of doing computation from a weaker logic to a stronger logic. There is nothing cognitive about development. There is no cognitive development, so Piaget is wrong. All of that experimental evidence is somehow mistaken. Pretty gutsy, eh? And Jerry Fodor was like that. "I've got a good argument. I don't care about your data." Sometimes you need that. Do you understand? He's basically showing that development doesn't exist, qualitative development.
- 6.29** Now, most people, even people that really like Jerry Fodor, like me, would go, "I don't think so. I think Piaget is right, and that means there's something wrong with this." Now, all the logical argument isn't wrong. You have to conclude, this model that **cognition is computation**, I think it's not working, and it's not giving me relevance either. Remember that? We just did that. It's not giving me representation. It's not giving me how to follow rules. The whole thing is actually self-destructing as a proposal. Remember, it can't do categorization. And it's got the frame problem.
- 6.30** So people came to this conclusion in the late '80s, early '90s. This isn't going to work. This is climbing the tree to get to the moon. Not everybody, but a lot of people. And they said, "We need an

alternative." What's the alternative? Neural networks. Neural networks say, "Give up the idea that cognition is language-like. Try to make cognition brain-like." What does that mean? Don't use linguistic relations like subject, predicate, implication. Use brain-like relations.

- 6.31** What are brain relations? Well, what are the three main properties of how neurons interact? Neurons send signals, and there's the strength of the signal, how frequently the neurons fire. There's the valence of it. Is it exciting or inhibiting that, a positive or negative? Then the connections. Different connections have different impact. These are called weights on connections. What you can do is you can set up a network where here's the nodes and you've got signal coming here.
- 6.32** This is weighted, so it's got a times three, and this is not as weighted, it's just times one. There's got a single signal strength of minus four, and this one is plus 2, and so this is 2, and this is minus 12, and so it's a minus 10. And that tells this, tells this node to massively reduce its firing. Or this could turn out to be a positive number and it tells this node to massively increase. That's it. That's the guts of neural networks.
- 6.33** Now you say, well, how do I program them? You don't program them. You don't write in an argument. How do you get them to learn? Don't say, "How do I program it?" What you can do is, I can change the activity of the network by changing the weights on the connections. This is why this is known not only as neural networks, but as connectionism. If I change the weights on the connection, I can get the network to learn.
- 6.34** How does that work? Let's say I have some input nodes and they go into what's called the hidden layer. No network has this few number of nodes. Oh, sorry. I meant to—OK. That's a very simple, primitive feed-forward network. Networks get way more complicated than that, but this will give me what I want. What I'm doing is I'm trying to get—most of these examples are military, which isn't a coincidence and isn't pleasant, but let's just keep going.
- 6.35** I want a neural network that can listen to a sonogram, sonar underwater, and you get a sonogram, and determine if what's out there is a rock or a mine. Because if it's a rock, I avoid it. If it's a mine, it means somebody's trying to kill me. And actual human beings have to learn how to do this. What do I do is, I digitize this, and then I put it in as inputs, and it does all that clickety whirr, and all the connection weights are random initially. It goes clickety whirr, and then I get some amount of activity here or here, and this means rock, and this means mine.

- 6.36** Now, does it mean that to the network? No, it means that to me. John Searle and the Chinese room argument, we'll put all that aside for now. Now, when it initially does this, it's wrong. It'll say something like, 0.7 and 0.4. It doesn't always add to one, and that has to do with the probability, just like the normalization. But let's say it was actually a mine. So this is a wrong answer. Is that OK?
- 6.37** So what do you do? Well, what do you do, you take this, you take what the target value is, which you can represent as a vector, which should be 1, 0, and then you subtract the performance, which is 0.74. Oh, no, it should have been 0.1, just to make it clear. Sorry, my mistake. That'll give you error. That gives you your error. Is that OK?
- 6.38** Now, what do I do? What I do is, because this is a really complex system, I can't just say, "Change that or change that." What I do is I do statistical analysis and I basically assign blame. This connection had about a— it accounts for about 4% of the variance in the error. This one accounts, wow, this one's accounting for 22% of the variance. And I map that all out. And then I nudge the connections. I change them in order to reduce the error. And then I do it again, and again, and again, and again, and again.
- 6.39** This is called the back propagation of error. I calculate the error and then I back-propagate it, and I do this cycle after cycle after cycle after cycle after cycle. Am I done? No, because you could do this. I can train. Let's change it slightly. Another military example. I train a neural network in order to look at a scene and determine if there's a tank in that view. You go, so I take a lot of pictures of tanks in the scene and not in the scene, and I train the neural network on it. Then it looks. It's tank, no tank, tank, no tank.
- 6.40** Then I go out to the real world and I put it in a real situation, and tanks come by and it doesn't warn me. I was like, what went wrong? Then I looked back at the pictures and I realized that all the pictures taken of the tanks were taken in the morning, and all the pictures without the tanks were taken in the afternoon. The machine didn't learn the hard problem of "where's a tank?" It learned the easy problem of morning versus afternoon, which you could determine by the length and direction of shadows.
- 6.41** In the data, there's a correlation between morning and tank, but that's not the real world. You can't go into a battlefield and say, "It's afternoon, no tanks." Do you understand? You have to train it, and then you have to test it to see if it generalizes, and even then it might not work. What people often do is they tinker with the architecture. We need a new node here. You get this training, testing, and

tinkering. Training, testing, and tinkering. You do this, and you do this, and you do this, and then you get a machine that can do things as well as a human being.

- 6.42** You think that's a great breakthrough. It's an engineering breakthrough. It's not a scientific breakthrough. Why? Because where'd that come from? Us. This is a homuncular. There had to be somebody that went out and learned the reality and supplied it to a machine. You know how this machine learns? It learns based on learning. Ah, there's a homunculus. Tinkering with the architecture. Reuben's brain has demonstrated significant plasticity. That's what changing the architecture is.
- 6.43** I bet this didn't happen. Reuben's doing stuff. His parents crack open his skull and go, "Look, look, no, no, no, no, OK, let's try again. Oh, no." That's a bad horror movie. There was no intelligent person supervising the plasticity. Does that make sense? For these to be a good theory, not just a good machine, we need unsupervised learning and unsupervised plasticity. OK? Is that OK? I'm trying to cram into—
- 6.44** OK, so, we're in second generation, and what happens is we have two people. One, the person who is considered the godfather of our current AI, Geoffrey Hinton, originally from the University of Toronto. Yes, all knowledge begins in the University of Toronto. He solved, or he at least made really significant headway on the unsupervised learning problem. It's called the wake-sleep algorithm. There's two sets of weights in a network, but in order to make that easier, I'm going to draw it as two separate things.
- 6.45** These weights are connected to the world. These are not. What I'm doing is, let's say I'm trying to get it to recognize As, and there's a lot of variations in what can count as an A, and some things aren't As. Maybe, but meh. Really good. Not great, but still an A. You do that, no problem. OK, so what do you do? You basically input lots of examples of this to this thing. What it does is it does data compression on this. It tries to find something invariant across all of these, a few invariant things. There's often some triangle in there.
- 6.46** It does that compression and it extracts these. That's the wake phase. Then what it does is it passes these invariants to this. What does this thing do? Well, what it does is it generates all kinds of variations. Oh, well, maybe I can double them like this, or maybe I can do this with them, or maybe I can do this with them. Do you understand? It generates variations, lots of variations. Is that OK? Then it teaches basic—just let me do that. It teaches this how to look for these possible variations. Is that OK? It sensitizes the first part.

- 6.47** We have two phases. The wake phase was this, and then we did the sleep phase where we do this. He called this "the network generates fantasies." Then we come back to the next wake phase in which this part gets taught to be more flexible in how it's looking for variations. Now, most of these variations don't help. They don't help the compression, but some of them help a little. So the invariant gets a little bit better, and it's already starting to do something that's starting to look A-like.
- 6.48** Do you see how this is working? It's called the wake-sleep because during wake, you're getting input from the world. And when you're sleeping, what are you doing? Dreaming. This is dreaming. And the dreaming sensitizes you to notice more variations. Most of them don't track, and they just die. But some of them, oh, and now you pick up. The pattern gets better and better and better and better until these things can distinguish As from not As as well as any human being.
- 6.49** That's called the wake-sleep algorithm. Does that make sense? Notice how it's unsupervised. There's no target out there. There's a little bit of sneakiness in here, and I'll come back to that in a sec. This is at the heart of deep learning. Hinton found a way of doing this with multiple networks in this really layered and recursive fashion, and I can't do that with you. But this is the heart of deep learning, the deep learning that's in the **LLMs**. This is deep learning.
- 6.50** Now, one problem you should rightfully say is, yeah, but who's selecting this? Who is doing that? We are. What the wake-sleep algorithm needs is what's called reinforcement learning. It needs a way of getting information about the world that is somehow relevant to it continuing to function or to exist. This is what you were trying to do with the pain and pleasure thing. Reinforcement learning, that's operant conditioning.
- 6.51** This is how you train a dog. You want a dog to not do this. When it does something you don't want it to do, you withdraw attention, you punish it, not violently, of course. When it does something you want it to do, you give it some food, and then it—you're set up that there doesn't have to be an intelligent agent doing that. You're doing that when you're interacting with the world. That's why I said there was something in what you were trying to do. Is that OK?
- 6.52** It needs reinforcement learning. OK, what about the—if I could get reinforcement learning attached to deep learning, I would get unsupervised learning. Does that make sense? OK, what about unsupervised plasticity? Remember, we're not going to crack open Reuben's skull. The brain is plastic, and of course, that was part of the

big revolution, the brain that changes itself by Doidge, the idea that the brain is plastic. If you play the violin, the part in your brain that moves the finger and gets thicker and more complex.

- 6.53** You know all that. The hardware is constantly changing in order to do better software that changes the hardware. That's how your brain, by the way, is unlike a computer. You can't use your computer and it will turn into a Ferrari. But your brain is a machine of machines that can make itself into a new kind of machine, which goes well with all that reconfiguration stuff we were talking about.
- 6.54** OK, so good, good, good, good. All right, so, how do you do this? Well, in the mid '90s, one example is what's called Marichal and Schulz from 1993 for unsupervised plasticity. You do what's called a cascade correlation. What I do is I have a network. I'm not going to draw all the connections. Imagine them. What I'm doing is, I'm trying to get a network to do something, and it gets to a part where it's not making very much progress. What I do is, I put in some candidates. We'll call this A and B, just to give something we can refer to easy.
- 6.55** These are candidate nodes, and they are connected in a one-way direction to the network. They can get information from the network, but they can't make any changes to the network. Is that OK? So one way. What you do is you let the network run, and B and A are competing with each other. Boy, is this Darwinian. We introduce variation, and then we put competitive selection on them. Your brain does this, by the way. Neurons are competing. It's a Darwinian jungle inside your brain, by the way.
- 6.56** What I do is, the one that wins the competition gets to stay. What's the competition? Which node's activity most correlates with error in the network? Because that means it's doing something that, right? So if I connect up over here, much better than connecting up over there. Does that make sense? Now I kill B, and I connect A up, so A can now affect the outcome. If the error rate goes down, A stays. If it doesn't, I kill off A and I try it again. This is just like evolution.
- 6.57** You go through these cycles, and guess what? These machines develop and they change their competence. You're trying to get it to learn velocity. First, all it will do is it'll just pick distance divided by time. I hope that's not a problem for you. Kilometers per hour. When you ask the network at first, it'll say, it'll just give you one of these things. Velocity is distance. But then if you run it, it comes up with addition, not just simple identification. It says velocity is distance plus time. That's wrong. Then it comes up with a new function, division. Velocity is division divided by time, which is right.

- 6.58** You know what's really intriguing? Kids go through the same series when they're developing. Do you see what they've done? They are explicitly in the paper arguing against Fodor. They're saying this is how qualitative development occurs. The way you get around your problem is you understand that cognition is not computation. It's a neural network going through this. This is really cool.
- 6.59** Now, what do you need in there? You need an error rate. Well, if we're supplying the error, homunculus, homunculus. So the error has to ultimately come through reinforcement learning in some way. So both the deep learning and the cascade correlation, the unsupervised learning, the unsupervised plasticity, need reinforcement learning. What we ultimately want is unsupervised learning plus reinforcement learning, unsupervised plasticity plus reinforcement learning. We have a lot of this. We don't have enough of all of that.
- 6.60** Remember that because it's also valuable about the **LLMs**. They have no unsupervised plasticity. It's not clear if that matters, but that's an important issue. Another thing to notice, notice in both wake-sleep and in the cascade correlation, what you have is a self-organizing cycle. Remember, we were cycling between wake and sleep, and we're introducing variation, the fantasies, and then we put selection. We have selection variation cycles. We had selection variation cycles in the unsupervised plasticity. Do you see that?
- 6.61** What we're basically doing is creating a self-organized system that is running something very analogous to evolution of variation and selection, variation and selection, variation and selection. Notice something else. Neural networks are telling us there's non-propositional knowing. There's knowing that happens that's not based on the logical inferential manipulation of propositions. This is one of the great strengths of neural network. They show that in addition to propositional knowing that, there is the procedural knowing how.
- 6.62** Neural networks were excellent at explaining things that computers were very bad at, which is your skill knowledge, knowing how to do something. Neural networks were a big leap forward in giving AI procedural knowing that was distinct from propositional knowing. This is one of the reasons why John Vervaeke is so on about non-propositional knowing. If you think there's only propositional knowing, you're way back in the 1980s.
- 6.63** Now, did neural networks solve the relevance problem? No. But they chipped away at it. They chipped away at it. Nobody that, like Tim Lillicrap, I just saw him at a very, earlier this year at a conference, it was a real great reunion. He was a student of mine, and now he's now

—and he was saying, "Well, we're chipping away." Well, he's my student. He takes the relevance problem really seriously. But we're chipping away at it. I think that's totally right. I think that's totally right.

- 6.64** This idea of a self-organized system, which is going to turn out to be a dynamical system, rather than a **formal system**, a self-organized system of selection and variation, these cycles, that seems to be getting us some way towards solving how relevance is done, how relevance realization happens. Now, the neural networks suffer from what's called the silo problem. You can make a neural network that was good at chess, and that was all it was good at. You tried to make it good at something else, it would wash out. This was a big problem for a long time.
- 6.65** Turns out, there are things you still need to do computation with. You still need to be able to do math. You still need to be able to do logic. And neural networks aren't very good at that. What you can do is basically really train them until you turn them into computers. It looks like you're not just a neural net, you're not just a computer, you're some hybrid of the two. A lot of machines came out that were hybrids of neural networks and computers. By the way, the **LLMs** are hybrids of computers and neural networks.
- 6.66** You think about how that, you are capable of making inferences, that's like a computer, but you're capable of doing this self-organized changing of what you see relevant, that's insight. You need both. You need an inference machine and you need an insight machine. Computer, neural network. The scientific response is, maybe the neural networks aren't the key thing. Maybe the neural networks aren't the key thing.
- 6.67** Now we move to third generation. Because actually what's happening is a cycle, a self-organizing cycle, of selection and variation. Maybe the key thing is dynamical systems, that cognition is a dynamical system, and then you can implement it with neural networks, but you could implement it other ways. So third generation, which is the one we're in right now, is let's move beyond neural networks. We're not excluding neural networks, but we want a much more generalizable account of intelligence. Maybe intelligence is running a dynamical system.
- 6.68** Let's make use of the work of Alicia Juarrero in her two amazing books, *Dynamics and Action* from 2002, and *Context Changes Everything* from just last year. Alicia Juarrero needs to be a deity to which people offer supplication because her work is just brilliant and

she's not famous enough. But many, many people in third-generation cogsci are deeply influenced by her. Just such an impressive thinker. Let's take a look at one of the most important dynamical systems theories in science, a scientific theory that says the thing that's doing it is a dynamical system.

- 6.69** This is Darwin's theory of evolution by natural selection. It's a dynamical systems theory. OK, so what do you need for a dynamical system? It has to be self-organizing. And note, my students, that's how you get out of the homuncular fallacy. It ultimately has to be a self-organizing system. It has to be a self-organizing system that can evolve itself. That's the only way to get out of the homunculus. What I need is I need a feedback cycle. Remember, like wake feeds into sleep, which feeds back into wake. I need a feedback cycle.
- 6.70** What's the feedback cycle for living things? It's called reproduction. Where do goats come from? Goat makes a goat that then makes a goat that makes a goat that makes a goat. The output for the system is the input to the system. That's why we call it reproduction. There's the self-organizing cycle. Is that OK? Then what you have is you have selection that's based on a condition of scarcity. Not an event, but a condition of scarcity. Is that OK?
- 6.71** What happens with scarcity? Animals compete, and most of them die. They're selected. Natural selection, no intelligent designer doing it. Is that all you need? No, you need something else. You need variation. You have to have a lot of different designs of organisms, and some of them will win the competition, and some of them will lose the competition, and the ones that win go on to reproduce, and that is how the species evolves. That's it.
- 6.72** So these are called selective constraints. They reduce the options. These are called enabling constraints. They increase the options. This is like a governor, it limits. This is like a generator, it makes more options. Do you get that? This is a virtual generator. Why virtual? Because it's dealing with possibilities. It's shaping possibility space. This is a virtual governor. This is a virtual generator. Virtual, having to do with possibilities. It's shaping possibility. When I have those together, I have a virtual engine.
- 6.73** This is a term I've come up with. She uses virtual generator for this, sorry, virtual governor, and then she stops the metaphor. I said, "Complete the metaphor." If this is a governor, this is a generator, and the two together are a virtual engine. This is what a dynamical system theory is. You find a feedback cycle, and you find a virtual engine that is regulating it. It goes through processes of selection and variation,

and it not only revolves, it revolves with change. What's the word for revolve with change? Evolve. The system evolves.

- 6.74** You get a non-intelligent system that can generate intelligent things. No homunculus. No homunculus. Now, of course, that's what's happening in unsupervised learning and unsupervised plasticity. That's it. Is Darwin's theory of evolution a good scientific theory? You better believe it. If I have a dynamical systems theory of what's going on in unsupervised learning, what's going on unsupervised plasticity, it's this—I'm getting a scientific theory, yes? I'm getting a bonafide scientific theory that avoids the homuncular fallacy through and through.
- 6.75** Well, what's the feedback loop in cognition? Sensorimotor behavior. I sense while I'm moving, which changes what I'm sensing while I'm moving, so I can move while I'm sensing, so I'm sensing while I'm moving. Sensorimotor loop. Do you see the loop? Constantly looping, especially when I'm navigating, because as I move, I change what I can see. And as I change what I can see, I change how I move. Nobody's designing that intelligently. It's looping. There's the sensorimotor loop.
- 6.76** Notice that the sensorimotor loop is not in your head. The old model was computation is in your head. This new model says, no, no, your head is maybe where the virtual engine is that's regulating the loop, or maybe even more likely, there's a virtual engine in here and there's a virtual engine out here, and they're both together in a coordinated fashion, regulating the loop. So cognition isn't in your head, it's between you and the world. It's between.
- 6.77** Now, this is really important. Why? Because we've talked about this before. Relevance is between. You don't detect relevance, and you don't merely project relevance, because if you merely projected relevance, you couldn't be wrong when you're problem solving, and you're wrong a lot of the times, and you have to change what you consider relevant. Yes? Notice that relevance, like adaptivity, is between the organism and the world. It's between. It's not in. It's between. This is to say that cognition is enacted. It's not computed, it's enacted.
- 6.78** Of course, they call this sense-making, and it's about wayfaring. Even the paramecium is wayfaring, swimming away from poison and towards food. You got the very basics and very basic organisms, and then it gets ratcheted up. Now, relevance is not cold calculation. We've talked about this. This is how you're different from computers.

You care about some information and you don't care about other information. So you need to care about the information.

- 6.79** Well, how do I get you to care about something? I can either top-down at homunculus; I'll just program you to pretend like you're caring. If I want you to really care, you have to be not only a self-organizing thing, you have to be a self-making thing. You have to be what's called an **autopoietic**, Greek for self-making. Because if you're constantly making yourself moment by moment, you're taking care of yourself, which means if I'm constantly constituted to take care of myself, I'm always going to be judging whether or not I care about that information or that. Will that help me take care of myself?
- 6.80** Do you see? In order to have relevance, you have to be relevant to something that has real needs and cares. You have to be **autopoietic**. That means you have to be embodied. A living body is an **autopoietic** system. That's what your body is. It's a self-making thing. See, cognition is enacted and it's embodied. And notice you're doing this thing. You're trying to get relevance that's relevant to you and to the environment. And that just reinforces this. These two belong together. It has to be enacted and embodied.
- 6.81** Well, it has to be embedded. Why? Because relevance isn't made by you. It's not made by the environment. Just like adaptivity. Is the adaptivity of the great white shark in the shark? No. Put it in the desert and it dies. Is it in the ocean? No, many things will die in the ocean. You have to have the right fit between the organism and the environment in order for there to be adaptivity. The same way with relevance. It's the right fit between you and the environment so you can solve your problems. Because the environment doesn't have any problems. A problem is when you and the environment don't fit appropriately together.
- 6.82** You have to be embedded in the world. The language for this is you and the environment are coupled together. This is what dynamical systems do. They couple. Coupling means, you pretend you're the environment. The environment and the organism co-create. Co-create. They're coupled together. This is knowing not by representing, this is knowing by coupling to something, belonging with it. Niche construction. The organism changes the environment and the environment changes the organism, so the environment and the organism fit together. They're coupled together. They belong together.
- 6.83** Cognition is extended. You don't think just in your head by yourself. We've talked about this. We'll talk about it a lot next lecture. You

think through things in the world and through other people. Most of our problems are not solved individually, they're solved collectively. And way before we put computers together to release the power of distributed computation in the internet, culture put people together to release the power of distributed cognition and collective intelligence.

- 6.84** This is 4E cognitive science. Enacted, embodied, embedded, and extended. 4E cognitive science. I add two. We'll see if I succeed in getting other people to take up two and make it 6E before I die. I think I'll fail. What are the other two? Cognition is inherently emotional. Relevance realization is about caring. It's affect-laden. It's arousal-laden. It's things that grab your attention. This is inherently emotional. That long fiction in our culture, that there's reason over here and there's emotion over here, I'm a scientist, I'm an artist—all that is bullshit. That's, your brain does not distinguish between them.
- 6.85** Take a look at the work of Marc Lewis, a friend of mine, the person who brought dynamical systems theory into understanding the development of cognition and emotions. He just says plainly, "The brain makes no distinction between reason and emotion." They're bound up together. What would be the sixth E? Well, one you've heard me do continually. Exaptive. Cognition is exaptive. It takes sensorimotor stuff and it exapts it up into more abstract stuff. More abstract stuff.
- 6.86** What would the new model be? The new model would be that intelligence is sense-making within 4E, 6E, cognitive science. What makes you intelligent is you doing the evolution of your sensorimotor loop, making way, making sense, sense-making, in an enacted, embodied, embedded, and extended, emotional and exaptive way. What we need now is, well, we need a theory about all of this. How can we put it all together?
- 6.87** We need a theory about this, but notice that this wayfaring needs anticipation in it. We need a good theory of anticipation. We need a good theory of relevance realization. We don't have a theory yet, and we need that theory of anticipation and that theory of relevance realization to be integrated with each other. We have anticipatory relevance realization that's integrated with the idea that cognition is enacted, embodied, embedded, and extended.
- 6.88** Guess what I'm going to do in the next lecture? I'm going to argue how we do anticipation, the predictive processing; how we do relevance realization, this is a theory called relevance realization theory, great name; and then how all of that fits into 4E cogsci. Then what does that imply for general intelligence? Because not only do I

want to give you all this convergence, I want to get, if I get to this model, do I start thinking about intelligence in new and important ways. I'm going to show you that, too.

- 6.89** Thank you very much for sticking with me through this slightly longer lecture, but I needed to get through all of this material. We'll take some questions now.
- 6.90** When you were talking about enacted, I'm realizing that, for example, I'm realizing that this sport is related to me. I'm realizing the relevance of it. Because I have cognition, I have emotions, and I can do all that, how I'm getting relevant to this board? Because the connection, as I understood, is a two-way connection. It's not just that I'm, it's not just that I'm conceiving this board.
- 6.91** Is that the first question?
- 6.92** Yeah.
- 6.93** So, first of all, is the board static or is it changing? It's changing because I'm writing stuff on it. It's unfolding. What you're doing is, you're trying to figure out, what are the possible implications and connections to this? You're doing relevance realization, and you try to formulate that, like formulating a problem, of how to search through the possible space that connects all of this. Is that OK?
- 6.94** You're doing relevance realization, you formulate it, and your formulation, oh, it doesn't work, and you have to modify it, and you introduce some variation, and now it lands, and you're constantly going back and forth between you and the board. The problem you're facing is the board isn't staying static. It's constantly changing. So the board is changing and you're changing. It's complexifying and you're complexifying and you're coupled to this.
- 6.95** Now, what happens in the environment is this is a bit of artificial. You're over here and the board is here. But when you're in the world, you would actually not only just be over there, you'd be also coming to the board and interacting with a changing board. You'd be making marks on—that's what you do in the world. You change the world. The world is changing, but you also change the world. Then the world is changing you, and you are looping in that fashion with the world.
- 6.96** Now, this is abstract, and that's why I say we have to include exaptation as one of the Es, because you're not literally moving through a space, but you exapt all of that know-how into how to move through abstract space. How are you connecting the ideas? Listen to the language. Which idea follows from the previous one? Can I trace

what's going on here? Can I keep track of the argument? You're using all of that sensorimotor capacity to track and trace and follow the lecture that's happening on the blackboard, and it's being shaped by you and the board in a coordinated fashion.

- 6.97** In fact, that's what I'm trying to do. I'm constantly trying to shape this so it fits you, and then I'm also trying to fit you, shape you, so you fit this. That's a good educator. I'm the one helping to run the loop. But the point is, ultimately, that can't be the story. It's got to start at a level where there's no one running it, and that's what I'm pointing to.
- 6.98** My other question, I think it's because my language skills are not professional enough. But one of the main things that I understood from this course, the main thing that is making us a general problem solver is the relevance realization. So we're constantly doing that.
- 6.99** That's right.
- 6.100** We're constantly realizing the relevance.
- 6.101** At multiple levels. Yeah.
- 6.102** But we cannot detect relevance.
- 6.103** That's right.
- 6.104** What is the core difference between detection, recognition, and realization? I cannot detect it—
- 6.105** Right. You can't detect it or you can't project it. You can't just say, "I just decide, and that's the way the world is." Let me ask you this question. Do you detect problems? Are there problems in the world? No, there isn't. The world is just running. There's no problems in the world. There's problems of human beings in the world. But imagine if there were no living beings. There's just the universe running. Are there any problems there? No.
- 6.106** So, problems aren't there. Are the problems just in your head? No, because if you don't solve them, you die. You really die. So where do the problems come from? They come from the fact that the world and you don't fit together. It's the same way. Relevance comes when you and the world fit together. When that information won't fit you to the world, that's when information is irrelevant to you. That's constantly going to change because what's relevant is constantly changing.
- 6.107** It's just like adaptivity. Adaptivity isn't in the organism, isn't in the environment. The organism doesn't—you can't say it detects. That doesn't make any sense; it's not detecting adaptivity. It's not just projecting adaptivity. Adaptivity comes from the organism and the

environment shaping each other so they fit together. Or when they don't fit together, what happens to the organism? It goes extinct. You're doing exactly the same thing. It's exactly the same thing. It's neither detected nor projected; it's co-created, just like a problem is.

6.108 From my understanding is relevance realization basically depends on self-making, which depends on— wait, no, sorry, the opposite. Depends on emotions, like having care, which depends on wanting to self-make.

6.109 That's right. Computers literally don't care about anything.

6.110 Exactly. So then my question is, if the goal is to get a general AI, then how do you get an AI to want to self-make? Isn't that a homunculus?

6.111 No, it's not. Because what I have to do is I have to make it not only artificially intelligent, I have to make it artificially alive. I have to come up with artificial life, not just artificial intelligence. Life, intelligence depends on—I have to make a self-making machine. Are there people working on trying to get cognition out of biological autopoiesis? You bet. Some of my students are doing that.

6.112 Can you explain autopoiesis?

6.113 What I do is, what if I made a chemical system that was making itself and it self-organizes to seek out the conditions that will keep it running? What if I make it not only so that it can seek, but in that seeking, it starts to do problem solving and starts to become intelligent? Then I would have artificial autopoiesis that was also starting to be artificially intelligent. Yes?

6.114 Yes?

6.115 They're doing it right now. They're doing it right now. The **LLMs** don't have this. But there's people out there doing this work. They're building what are called autocatalytic systems and other systems that are **autopoietic**. And then they're getting them to do information processing and solve problems. And that can't be impossible because that's what a paramecium does.

6.116 I think a big part of humanity that people argue that makes us human is having consciousness. And it seems like we're trying to artificially create that. Are you arguing that it's possible?

6.117 OK, that's a big course. Consciousness, because consciousness isn't the same thing as intelligence. Although fluid intelligence and consciousness and attention and working memory, all like this. But here's the idea. If you get artificial life, you can get artificial

intelligence. And then if you get powerful enough artificial intelligence, you get consciousness. Remember, consciousness is doing that recursive relevance realization.

- 6.118** You get life, you get very basic relevance realization. It starts to become more dynamically complex, and then it gets the ability to do the working memory workspace, and you start to get that recursive relevance realization. You start to get consciousness, yes. The machines, the **LLMs**, they do not, and they cannot, for this reason, I would argue, have consciousness, and they don't have real intelligence because they really don't care.
- 6.119** But that might not matter for many of the people who want these machines. They don't want an explanation of intelligence. They want a way in which they could manipulate markets and the military and all kinds of nefarious things until these machines overtake them.
- 6.120** For starters, I just want to say thank you for adding the extra two Es because I was thinking about emotionality in regards to this entire conversation for a really long time. A part of me is wondering why the search for **AGI** is not called AGEI in regards to adding the emotionality to it.
- 6.121** Yeah. Can I answer just that part first? That's because the people who are doing this are still in the Enlightenment tradition. I mean, the historical period, the Enlightenment, the 16th, 17th, early 18th century, where reason is over here and emotion is over here. We've got science and art. We've got the positivists and the romantics. And there's no way that they fit together or belong. And that's all an historical fiction, because you go before the Enlightenment, Plato, for example, and reason and Eros are like this. They're interwoven together. If you're interested in that, you can take a course, The Primacy of Beauty.
- 6.122** Sure. OK, so just to piggyback on that, because from my understanding when it comes to neuroscience and thinking and stuff like that, I think it's a particular truth that you can't have two thoughts at the same time. I think that's—
- 6.123** Depends what you mean by thought. If you mean something that's focal in working memory that you can put into language, yes. It's not, you can. You can have four things in mind at once. You can have four chunks.
- 6.124** OK. That's interesting because I was thinking in scenarios where people experience two different emotions at the same time, you definitely can't argue that.

- 6.125** Well, it's difficult. But you could think about, you can keep track of at least four variables or you couldn't spar in martial arts. If I had to do it sequentially, then all I have to do is punch and kick you at the same time and you're defeated.
- 6.126** OK, that's fair. Because I was thinking in regards to intelligence and solving a problem where you have two different emotions that you're solving. For example, if you're hungry and tired at the same time—
- 6.127** In the paper we did in 2012, we said you're not only doing relevance realization within each problem, what we call the cost functions, you're also doing relevance realization between them. Do I try and satisfy both? You think about when people are betting. Do you hedge your bets, bet a little on a lot, or do you go in big on one thing? There's no right answer to that.
- 6.128** Because when you relate that back to the robot and the bomb and the battery, I feel like that's somewhat related. But what the robot doesn't have is the emotional element to be able to have a very different relevance realization.
- 6.129** Yeah, exactly. That's bound up with, although I gave it a machine, it doesn't really have an **autopoietic** body. Now, I want to make clear, I've gone through the generations. That doesn't mean that everybody's now in generation three. There's many people who still think what's called GOF AI, Good Old-Fashioned AI, the computational, that's still the way to go. There's lots of people say, "We don't need to go any farther than neural networks. Neural networks are it." But there's a lot of people are saying, "No, no."
- 6.130** And I'm arguing because I believe that's right. I'm arguing for it, and I'm giving you the arguments for it and why we get there, and why I think it's needed if we're going to get what we're after here, which is the best possible account of what intelligence is. OK, thank you very much.

7. Integrated Intelligence

At a glance

Synthesizes the course's argument about AI and AGI, revisiting why relevance realization cannot be derived from logical relationships or pure computation alone. Introduces predictive processing as providing the mechanism of sensorimotor anticipation that relevance realization requires.

- 7.1 Welcome back to the last lecture of our course on the nature of intelligence. Last time, we reviewed as part of our synoptic integration argument, AI and **AGI**. We started by looking at the general problem solver of Newell and Simon, and we banged into combinatorial explosion, which is our old friend, the problem of **relevance realization**. We took a look at heuristics and how that helps, but it also hinders us in an important way. We took a look at the frame problem, which is right at the core of any basic robotic intelligence. We talked about how **relevance realization** is there again.
- 7.2 We did a quick aside about LLMs and how they don't really solve, at least scientifically, explanatorially, what's going on in **relevance realization**. Then from there, we went into the generations of AI and cogsci. The first generation is that cognition is computation. There's a language of thought. Intelligence is running a good program in your head, something like a good argument, that allows you to be a general problem solver. That's what intelligence will be. We took a look at that, and there are problems with computation, at least the way we were defining it.
- 7.3 You can't get relevance out of logical relationships between or within a proposition. You can't get rule-following without presupposing **relevance realization**. You can't get development with pure computational models of cognition, qualitative development. Representation presupposes **relevance realization**; categorization presupposes **relevance realization**, of course, and all of this is swimming around in the frame problem. A lot of people said, let's put aside this metaphor.
- 7.4 Maybe cognition is not language-like, maybe it's brain-like. We won't use linguistic relations or logical relations like subject-predicate. We'll use the kind of relations that exist between neurons, like signal strength, signal valence, inhibition or excitation, and the connection weights. And that's how you basically build a neural network. And you can get this network to learn by altering the connection weights,

which will change the pattern of activity in the network. That's why it's also called connectionism, as well as neural networks.

- 7.5 Now we looked at how you get these networks to learn. You do the back propagation of error and you tinker with the architecture. So you do back propagation for learning and you do tinkering for the architecture. The problem with back propagation, great engineering solution. No good for science because it's homuncular in nature because human beings have to learn the target value and supply it to the back-prop system. So you don't want that kind of supervised learning. You need a theory of unsupervised learning.
- 7.6 The problem with the tinkering is, of course, nobody was actually tinkering with your brain as it went through plastic development. We need unsupervised plasticity. In the mid-90s, very close in time, we got an argument, a theory from Geoff Hinton, about how you could solve, mostly solve, the unsupervised learning problem, and that was the wake-sleep algorithm. That was really cool because it gave us potentially a new theory of why we dream, which, wow, that's unexpected and interesting.
- 7.7 We noted that we still needed some reinforcement learning in there, but nevertheless, we got good head start on unsupervised learning. Then we went into the cascade correlational models, and we saw how we could get unsupervised plasticity. They also need an error signal, so they also need some account of good reinforcement learning. But what we saw is in that both, what we had are self-organizing cycles of selection and variation, very much like evolution that was being used.
- 7.8 And the proposal I made to you is, that is what you have to do to get rid of the homunculus. You have to have evolving self-organization in order to design your processes, or you will ultimately be doing a homuncular explanation of intelligence. Another important thing that came out of neural networks is they showed us that there is non-propositional knowing. There is the knowing how that neural networks were very good at, in contrast to the computers, which were very good at knowing that.
- 7.9 Now, was the relevance problem solved by doing the unsupervised learning and the unsupervised plasticity? No, but we chipped away at it. That's what Tim Lillicrap said recently at a conference. Where was that chipping away? I think what the core idea was this idea of a self-organized system of selection and variation that's running in cycles, namely that it's a dynamical system. That seems to be the key move. Now, neural networks had the silo problem. They only seem to be able to solve, be single-domain, single kind of problem-solvers.

- 7.10 And also you do computation. You clearly do do computation. So the engineering solution was, let's have part of our machine be a computer that can do inferential, logical-type stuff. And let's have another part that's like a neural network that can do more know-how and insight stuff. And you put them together and you get a better machine. And LLMs are mixtures like that. And there are many different kinds, the neural Turing machine, all kinds of mixed machines.
- 7.11 But what came out of this, scientifically, is the proposal that maybe cognition, or therefore, maybe intelligence, is a particular kind of dynamical system. What do you have in a dynamical system? We made use of Alicia Juarrero's work, brilliant work. She should be more famous. I hope one of the effects of this course is to make more people pay due regard and proper respect to the work of Alicia Juarrero. What you have is you have a feedback cycle, and we saw that already. Then you have selective constraints. Remember, this is competition because of scarcity. You have enabling constraints. This is variation, and that's due to genetic factors. We'll talk about that a little bit today.
- 7.12 You have selective constraints that are reducing options, enabling constraints that are increasing the options. They're locked together. So you form a virtual engine. The virtual engine is regulating the feedback cycle. So as animals reproduce, there's variation, and most of it get killed off, but there's some selection, and that changes how the animals go into reproduction, and it revolves with change, and that's evolution. And of course, evolution is a non-intelligent process that can produce very intelligent design, which, of course, is why many people initially were, "Oh, no, this really challenges God."
- 7.13 But I think there's been enough time and good theology, and maybe there should be a course on that, saying, no, there's a way of thinking about God that doesn't have to run up against or try to contradict evolutionary theory. As I said, evolutionary theory, therefore, is a non-homuncular account of the generation of intelligence, the generation of good design. You get evolving self-organization, which is necessary to avoid the homuncular fallacy. I'm not going to keep repeating that. I said it multiple times last time. I've repeated it here. This is the core idea.
- 7.14 Now I said, well, what would be the dynamical system for intelligence? What's the feedback? Well, the feedback is the **sensorimotor** loop. That's what's constantly reproducing your behavior. Then the idea is there's a virtual engine; for some people it's just in the brain, for some people it's in the brain and in the world.

That is regulating that loop, that **sensorimotor** loop, so it constantly evolves. So I orient and I have cognitive flexibility, so I can navigate and I move around in the world.

- 7.15 And what we get is the idea that cognition is how an organism and the environment are coupled together, mutually shaping each other, belonging together, like in niche construction. And this is the idea, therefore, that cognition is not in your head. Intelligence is not in your head. It's between you and the world in a fundamental way. It is enacted. And that's really good because adaptivity is enacted. It's between. And relevance is enacted. It's between.
- 7.16 So you can see why there's really good connection between this idea of cognition being enacted and **relevance realization** and adaptivity. All hang together really well and really tightly. The idea here would be that intelligence is some good virtual engine that allows the **sensorimotor** loop to evolve in a way that helps you to orient and navigate such that you're a general problem-solver. You can do foraging into foresight. It's exapted. Cognition and intelligence is embodied. **Relevance realization** depends on autopoiesis. You have to have a being that cares, that really has needs. It has to be an autopoietic thing.
- 7.17 So it has to literally be embodied, not just have a machine; it has to have a body, the kind of self-organizing system that makes itself, so that it can function reliably to seek out and find those conditions that will allow it to continue to make itself. And of course, that is a central feature of adaptivity. So, cognition is embodied. Cognition is embedded. What do you mean? Well, niche construction. You have to be, you and the world together make intelligence. So you're embedded.
- 7.18 Cognition is extended. Most of our problem-solving is not done monologically by ourselves. It's done dialogically with networks of people that are working together in a way analogous to how networks of neurons are working together to produce emergent intelligence. We network people together to produce emergent intelligence that can solve collective action problems and solve problems that can only be addressed by distributed cognition, the collective intelligence of distributed cognition.
- 7.19 So this gives us the 4Es of the third generation, the generation that says, "Neural networks are not enough. It's the dynamical systems that's the key." And if we really accept that intelligence is a dynamical system or a property of dynamical systems, then we're committed to 4E cognitive science. The cognition is enacted, embodied, embedded,

and extended. Then I argued for two more Es, which if **relevance realization** is about caring, then emotion should be the fifth E.

- 7.20 And this is very much in line with a good friend of mine, Marc Lewis, who did work using dynamical systems to explain emotion and development. And he says, the brain, it doesn't distinguish emotional processing and reasoning. They're interwoven like this. So all of that stuff we pretend has been the case since the scientific revolution about reason's over here and emotion's over here is largely crap. The older Platonic model that reason, logos, and eros are interwoven, which you can see in the course on the Primacy of Beauty, is now a really viable alternative because we have found that the model given to us by modernity is plausibly false.
- 7.21 That's important because that means we should perhaps pay much more attention to the Platonic framework in general, the Platonic and neo-Platonic framework. Sixth E, cognition is inherently exaptive. You're taking the **sensorimotor** stuff and you're exapting it up into abstract cognition. I'm giving you lots of examples of that. What we have is, we need a theory of anticipation. Anticipation will be the theory that tells us what good reinforcement is, how we get good reinforcements, how we learn by doing things and trying and failing, how we make good use of error. And yes, we need an error signal. It's going to be at least initially pain.
- 7.22 And so we need a theory of anticipation. I'll tell you in a few minutes why I call it anticipation. I'm going to argue that **predictive processing** does this. We need a theory of **relevance realization**, which is at times recursive **relevance realization**, like in working memory. And we know that there is deep connections between this and 4E and maybe 6E, I would argue. And what I'm going to do is to try and argue that these two are interlocked. I've already made the argument that these two are interlocked, and by interlocking this one and this one, we actually interlock all of them. And then we get a comprehensive account of the nature of intelligence.
- 7.23 And that's what we're going to do right now. Jordan Peterson will like it because it all comes down to a triangle, my good friend. OK, so last time I read off to you in a somewhat egocentric, gratifying manner, a list of all the people doing work on **relevance realization**. I'm going to try and give you a crucial throughline. What is the theory of **relevance realization**? You're really well prepared for it, by the way. And then I'm going to try and continue that throughline, is how does that integrate with **predictive processing**, because recently did that.

- 7.24 Let's start with **relevance realization**. Here's the proposal. We need a proposal for what general intelligence is. It's going to be something that's going to give you domain general problem-solving. Now, we've already bumped into this. Here's a proposal, that what general intelligence does is allow you to solve the meta problems that you have to solve whenever you're trying to solve any specific problem. The idea is, oh, in order to solve any of these problems, I have to always solve these other two meta problems.
- 7.25 OK. And so, one of those problems is anticipation. I'm putting **relevance realization** aside for a second. Lots of people, Michael Levin talks about cognitive light cones. You can judge the intelligence of an organism by how far into the future and how deeply it can anticipate the world. You think, and you should, that dogs are more intelligent than cats because you can say things to a dog like, "Go get your ball," and the dog will go to another room, look under the bed, find the ball, and bring it back. And if you say that to the cat, the cat just basically gives you something equivalent to the finger.
- 7.26 So we need, in order to be any kind of—any problem I solve, I'm going to be more intelligent if I can solve that problem in a more anticipatory fashion. Reuben, do you want to fight the tiger or do you want to avoid the tiger? Yeah. Actually, good point. Most of the time you want to avoid, but there might be some times when you have to actually deal with this situation. But generally, it's a good idea if you can anticipate these problems rather than having to spend all the tremendous resources when you actually encounter that problem.
- 7.27 You want to be proactive, rather than reactive. And the more proactive you can be, generally the better you can solve problems. Why? Because typically, the earlier you can intervene in the world, the more likely you can set all the chains of action rolling out so that you will bring about the desired change that you want. Good exception, though. OK, so, anticipating a goal. We need a theory of how we anticipate goals and how we can anticipate deeply. Because, for any problem, the more we can solve it in an anticipatory fashion, the more intelligent we are. And we've seen a lot of evidence to support that throughout.
- 7.28 The other meta problem—this is not a surprise, I hope, by now—is **relevance realization**. In order to solve any particular problem, you almost always have to solve the **relevance realization** meta problem. So now we have two interlocking problems or potentially interlocking problems. Or here, a better way of saying it. Here are these two meta problems, anticipation, **relevance realization**, and I'm going to

argue that they're interlocking, they're interdependent. What I want to do is give you a theory of **relevance realization**, a theory of anticipation, and a theory of how, which are all arguments, how they fit together and interlock together.

- 7.29** OK, let's start with the anticipation. So, first of all, why do I say anticipation? Because you don't want to just predict the world. If you could just predict the world, that's useless. You want to predict the world so you can prepare for the prediction in a way that matters to you, that you care about. Notice what I'm sneaking in there. But that's what anticipation is. Anticipation is when I'm predicting the world in order to try and prepare for it, because there are things that I care about. Is that OK?
- 7.30** So although you hear **predictive processing**, I keep saying it's actually anticipatory processing. I know these people. There's no disagreement with that. You get stuck with a term and you keep using it. How do you do this? How do you predictively prepare in a way that matters to you? We're going to put the matters to you a little bit over here because that's already getting us into **relevance realization**. But let's do the theory of **predictive processing** itself, remembering my caveat that we should—I get it. You don't get the nice alliteration. Predictive process, anticipatory processing. It's not as nice. I get it.
- 7.31** I'm going to tell you the two main principles. One is a really breakthrough insight, and then another one is a really good proposal as to how we can come up with an overarching guiding principle for all of our reinforcement learning. How do we measure error without having a homunculus? The first insight is this really powerful one because it's counterintuitive. Don't try to predict the world. What? No, just give me a sec. What you want to do is you want to predict yourself as you causally interact with the world.
- 7.32** I'm going to unpack that. Don't worry. But don't directly try to predict the world. You'll get overwhelmed. Try to predict yourself as you're interacting with the world. OK, so that's the insight. I'm going to cash that out for you. What's the really key principle, the guiding principle? This is, of course, from Friston, and he calls it the free energy principle. Karl Friston. Here's the idea. What your brain is trying to do is reduce being surprised. That all error is basically a kind of surprise.
- 7.33** Now, when you say it, it's one of those really good ideas that, until you think of it, you don't think of it. But when you think of it, you go, "Oh, wait. Yeah." The idea is, whenever I'm making a prediction and I get surprised, that's bad. It doesn't matter what the content is. That's just a bad event. What I try to do is, I try to change things so the surprise

will go down in the future. There's two ways I can change things. I can change how I'm perceiving the world, or I can go out and change the world. Perception or action.

- 7.34 Now, notice right away that's good because they're building action into the very way in which cognition is looping with the world. OK, so now let's go back and unpack that first insight. So here's the idea. You have areas of the brain. It doesn't work this way because of the way evolution designed you. Vision starts at the front, goes to the very back of the brain, and then comes back forward and all that stuff. But I'm just going to use a more simple model right now.
- 7.35 So let's say there's a part of your brain that is indirect, like the retina of my eye. It's indirect causal interaction. I don't have to do anything, which is a point Amir keeps making. I don't have to do anything. I just have to open my eyes. And light and retina, they start interacting. There's patterns of activation on my retina. Is that OK? You're getting these patterns of activation. Of course, the patterns are changing if I move around, right? Even if I just move my head.
- 7.36 In fact, even when you think you're just staring at something, your eyes are moving. They're saccading constantly because you're a **sensorimotor** loop. You're not a computer taking in input. You're a **sensorimotor** loop. You're always doing some kind of movement in order to alter your perception, at least for vision and hearing, obviously for touch. You got to move for touch. Taste and smell are weird. But you don't— I'm not going to talk about them right now. There's the **sensorimotor** loop happening, and there's a pattern of activation that happens, and it's constantly changing. Is that OK? Does that help? That's all right.
- 7.37 Now what I do is, I'm not trying to predict out here. What I do is I create another layer, and it's trying to do two things. It's trying to detect what that pattern is. Then it is trying to complete that pattern. Now, notice I can use something like even a version of back-prop because there's no homunculus, because this layer can basically tell that layer when it got it wrong. It's just like, "Oh, I think this is the pattern you're going to do." Nope, it turned out to be that as the pattern. And there's an error signal.
- 7.38 You see, it's not about the world. It's about that area of the brain. This area is trying to detect, predict, and complete this pattern. Why would it try and complete the pattern ahead of time? Because that's a kind of preparation. That's a kind of preparation. Remember, we talked about this in the Primacy of Beauty course. When I'm walking, when I get to

a certain speed, the **sensorimotor** loop is too slow for me to actually continue walking. It's too slow.

- 7.39** So what's happening is, my cerebellum is trying to pick up the patterns in my **sensorimotor** field, my body, my sight, my kinesthetic sense. It's trying to pick those up and say, "Oh, this is what's going on." Then it tries, and it helps complete those patterns ahead of time so I don't actually have to perceive them. That's what's actually going on.
- 7.40** Another one. OK, so look at this. It's white. Your vision, your eye, can't take all that in. You take little samples from this, but you're seeing it all at once, aren't you? Because what your brain is doing is, it's taking those little samples and it's predicting what would happen if you could actually track through it all, and it's completing that pattern for you. The color of the wall is mostly imaginary.
- 7.41** When I'm running, it's a lot of imagination going on. We're going to come back to that because we're going to need a distinction between imagination that doesn't track reality, the imaginary, and imagination that helps you track reality, the imaginal. The imaginal. OK, so now I'm going to take that. Is that OK so far? Is that landing? This lecture will be easier than the last one. I'm going to turn it this way because we have a cultural metaphor of talking about bottom-up and top-down. Do you understand? In your brain, it's this way, but I'm turning it this way because that's the way we talk. Is that all right? OK.
- 7.42** Here's the bottom layer. Remember, this is the one that's doing the—wow, that was a bad layer. It's doing the **sensorimotor** loop and it's getting patterns of—now, what you do is you have this layer, and it's doing the detection and completion. It's helping—so this one gives error signal. "No, you got it wrong." This one says, "Hey, this is what you can do," and helps it get better and better. Is that OK? Now, when this is doing this on this level, it also has patterns of activity. Yes? You know what I can do? I can put another level above it that detects its pattern of activity and then completes it.
- 7.43** You get error going up and prediction coming down. This one helps this one get faster, that then helps this one do it better. You know what I can do now is I can put another one on top of that and I can just keep stacking these. Notice what I'm getting. I'm getting a recursive, self-organizing dynamical system. Now, what's flowing up is error, error signal. What's flowing down is predictive preparation, anticipation. I'll just say predictive preparation, so we can keep close. Prediction and completion, preparation. Bottom-up, top-down. Completely interpenetrating.

- 7.44** Now, this level is very concrete and specific and short-term because it's indirect, moment-to-moment change. Concrete, specific, short-term. I won't do it, but if I was to chuck this at his head, he has the looming reflex, which is if—watch how this works. His brain doesn't have to track this object. His brain will track the rate of expansion on his retinal field, pattern of activity, and when it's doing a certain speed of change, he will move his head out from the predicted intersection point. That's really right now.
- 7.45** But as I go up this, I go from concrete, specific, to abstract, general, long-term. So now you have a system that can actually interact with the world at multiple time scales, which is what Andrew can do. He can do moment-by-moment, sitting on the chair, a little bit more abstract going through the course, a little bit more abstract how he's going to lead his life. And they aren't disconnected from each other, I hope. You've got some goal about leading your life, and maybe that influences you to take this course, which influences you to sit here, which then influences you to pay attention, which is now making you nod.
- 7.46** Now, as you go higher up, it's harder to change it, because a lot of stuff can vary down here, and this goal still stays the same. Why? Because going up is also data compression that allows you to generalize. Going down is data variation. You know what you're getting here? What are you getting? This is doing compression to generalize, which is selective. This is doing variation that allows you to more specifically fit the specific situation. You've seen this before. This keeps coming again and again. Selection and variation, selection and variation.
- 7.47** In what kind of relationship? Oh, a trade-off relationship. In a trade-off relationship. This can implement deep learning. You can put deep learning in this like that. You can have back-prop and deep learning, because the back-prop is not homuncular, because it's not the world giving an error signal. It's a part of the brain giving an error signal to any other part of the brain.
- 7.48** Now, let's talk about this. This is from Henry Corbin. Imaginary is when you form a mental picture in your head. And what you're basically doing is you're just doing the top-down without any bottom-up, but you're doing it completely uncoupled from the world. So I can ask Michelle, "Picture a sailboat. Are the sails up or down?" And she can answer me, there you go. And she could have said, "Down," and I wouldn't have gone, "No, you're wrong." That's a picture.

- 7.49 And what you do is, you're decoupling from the world. Remember the language of coupling? And you're just going into this and you're just running it. Why? Because you can consider possibilities with this. This is important. But that's different from the kind of imagination you're doing when you're imagining the color of that wall or when I'm running. This is imaginal. So let me give you a better way of, a more concrete example.
- 7.50 You can imagine, you can ask a kid to picture Superman in their head. Yes? Or they can do the following. They can pretend to be Superman. They can put on a blanket. They can hold out their arms and they can go, "Truth, justice, and the American way." Now, they're not looking at an image. They're looking through it. And that actually enables them to try out different ways of coupling to the world. What would it be like to be Superman? What would be his perspective? What are some of his virtues? What are some of his skills? What are some of his strengths and weaknesses? They're trying it out. That's why children play. They play in order to develop.
- 7.51 So the imaginal is developmental in nature. It's helping you grow, like non-propositional kinds of knowing. I want you to realize how imaginal your cognition is. So working memory, we've talked about it. That space inside your head, where you can do 22 times 37. Where is that space? It's not in your head. If it was in your head, because your head's a spatial object, it would be in your head, and I could open your head, and I would see the space. In fact, the Egyptians thought it was here, your liver. Aristotle thought it was here, your heart.
- 7.52 It's not literally real. Is it a mere illusion? No, because that space actually allows you, it connects you, it couples you to your cognition, so you can know what you're thinking, so you can monitor it, so you can correct it and change it. It's neither literal nor is it merely fictional. It's imaginal. You're doing it all the time. When I exapt the **sensorimotor** up, and I'm moving through the course, and I'm hoping you're not finding it too hard, but I hope you see what I'm saying or get my point, or hopefully you grasp it, I'm doing imaginal stuff. Do you see?
- 7.53 This is one of the advantages, by the way, of the 4E approach, is it really explains why our cognition is so imaginal. Being imaginal fits in, lock and key, with **predictive processing**, because when these two are interpenetrating each other, that's the imaginal, this side of it. Is that OK? Now, notice we're starting to explain things that the other theories of intelligence couldn't explain. How intelligence is connected to the imaginal. Were we getting that from any of the other theories? No. Did they even bring it up? No. How intelligence can

connect to non-propositional knowing? A little bit in neural networks. Well, actually significant, but only one kind, the procedural.

- 7.54 Now, this framework, which goes by many names, because we do this stupid thing. People call it **predictive processing**. I like that. Other people call it active inference, because when you're moving around in the world to change your belief, that seems to be action that's changing inference. But when you ask them, "Is it propositional?", they go, "Oh, no, it doesn't have to be propositional." And then I say, "Why are you using the word inference?" And they say, "We don't really mean inference." And it's, OK.
- 7.55 And then, "Oh, Bayesian. Oh, well, it's doing something like Bayes." "Is it actually doing Bayes?" "What do you mean? Is it actually computing Bayes' theorem?" "Well, no, that would be combinatorially explosive." "So what's it doing?" "Well, it's doing an approximation." "How is it doing the approximation?" "Oh, this recursive self-organizing **predictive processing**." OK. Let's just call it one thing. I vote for **predictive processing**. I gave you my argument. Please make sure when the ballots come in the mail that you fill them out accordingly.
- 7.56 Very powerful. Karl Friston, Andy Clark, Mark Miller. Tons of people. Julian Kiverstein, just so many people. Is it consensus? No. Is it powerful and respected and gathering more and more adherence? Yes. There are two main criticisms of the **predictive processing** approach. One is called the magic modulator, which sounds like something in—"Wait, Marcus! Bring out the magic modulator. Yes." The magic modulator is a nice alliterative way of talking about a homunculus. There's a homunculus problem, maybe, in here.
- 7.57 The other one is, where's the 4E coupling in here? Where's the 4E stuff? There's some people who say, no, no 4E. This is all propositional representational. Most of the people, Friston included, Andy Clark included, Mark Miller included, say, "No, no, this is 4E." But how does it connect? OK. OK, so far landing? OK. See, this works. The other stuff was very abstract. This you can connect a little bit more directly to your experience.
- 7.58 OK, what's the magic modulator problem? The magic modulator problem is, well, what is the system modeling? When **predictive processing** was first starting, I would go to the conferences and I would be a real pain. Because they would do all this, and I'd say, "What's it modeling?" And they'd say, "What?" "What's it modeling? It could be modeling that corner of the room. It could be modeling

there. It could be modeling all the—what's it modeling?" And they'd go [noise]. It's like, OK. I wasn't the only one making that.

7.59 People said, "Well, there's got to be a way in which it's dealing with this flow up and flow down." For example, when you're looking at the floor, you see it as white or even the wall, it's so pure. But if you go in close, it's actually not purely white. There'll be lots of little errors. And what do you say to me about those errors? "I don't care." That's right. You don't care. There's lots of—you're getting tons of error right now, and your brain is basically filtering it. Here it comes, ignoring it, suppressing it, because it's not what? Relevant. It's not relevant. It's like, oh, you're going to only pay attention to the signals that are relevant.

7.60 Which things are you going to prepare for? Anything in the environment? No, the one, oh, I'm going to prepare for the relevant things. Oh, no. And then Andy Clark said something in 2017, which was really worrying. Oh, what's making this all work, quote, by the way, is "task relevance." Oh, great. Here we go. And then you ask, "Who's doing the **relevance realization**?" And you get, "Me. I'm doing it." The little man, that's the homunculus. That's the magic modulator problem. Do you see that?

7.61 What they did was they first tried to come up with easy answers around that same time as this. "Oh, you just look for the most reliable signals." No, there's lots of reliable signals that you don't care about. There's lots of signals that are quite unreliable, but you still turn your attention to them because if you miss it, you're in trouble. It's like, yeah. What's the system modeling? How is it doing it? And why is it doing it? In the end, you tell me, OK, what's it modeling and how? But why is it modeling that thing? Why?

7.62 All right. What started to develop is the realization—well, first of all, I want to say something. Notice that we're facing an assimilation-accommodation problem here. How much of the error do I just assimilate? And how much do I have to accommodate to? We've got, guess what's looming? Opponent processing, coming in again. Opponent processing. Opponent processing. OK. What they started to develop was a notion of what they called precision weighting. Precision weighting is basically a way of monitoring the information flow up and down and allowing the right signals to come up and the right signals to go down. Not in terms of error, truth or falsity, but in terms of relevance.

7.63 Now, first of all, know what this system is doing. Notice what it is doing. It's monitoring and selecting and prioritizing. They literally say

this. You know what precision weighting is? It's the **predictive processing** theory of what attention is. That's what attention is. Attention is in this up-down, error, up from perception, top, predictive preparation down. And attention is what is balancing the prioritization, the suppression, the ignoring, the giving more processing time to however you want to try and land it, and it's all of these. Oh, well, that's what attention is. That's what attention is. That's what attention is.

- 7.64 And then you go, OK, excellent. Brilliant. We're getting a theory of intelligence that would really help to explain the connections between intelligence and attention. OK, that's good. So what's attention? Well, it's precision weighting. Yes, but how? No magic modulator. You still need to explain how is it being done and why is it being done. Otherwise, you've got a homunculus in the center of **predictive processing**. Now, they have a lot of mathematical formulas for trying to address this, but what these mathematical formulas are doing is basically implicitly implementing, and this is what we argued in the 2022 paper, a lot of the stuff that was made explicit in **relevance realization** theory.
- 7.65 This is the paper by Anderson, Miller. Mark is a former student of mine, and Vervaeke from 2022. Here's the idea. Let's try and interlock these together. Maybe if we can figure out what's going on in **relevance realization**, then we can plug it into **predictive processing** and solve that problem. But maybe **predictive processing** can return the favor because you know what **relevance realization** theory needs? It needs a theory of how you anticipate the goal state. It needs that. It really does. That's important.
- 7.66 Maybe you could use opponent processing to implement **relevance realization** to solve the how problem. How is it done? Let's take a look at **relevance realization** theory starting in 2012. The idea is that **relevance realization**, and all of this should work for you now because we've seen it all the way through, **relevance realization** is optimal gripping. It's an optimization function, and it's optimal gripping because it's optimization and how you're coupling to the world. There you go. Notice how I can do this and you're going, "Yeah, that's OK. Good." See, I'm trying to show you that it was worth it hanging in through all of that last lecture. OK.
- 7.67 Then here's the idea. How do you do optimal gripping? Again, we've seen it. Opponent processing. What you want—by the way, a point of processing doesn't violate the no free lunch theorem because the no free lunch theorem, remember it says this? I remember, I was writing

this with Tim and I said, "What about the no free lunch theorem?" No, because what I can do is I can find another thing that has that pattern. Do you see? Where this one is strong, this one is weak, and where this one is weak, this one is strong. And what I can do is do opponent processing between them, and that doesn't violate the no free lunch theorem. It gets you out. It gets you out of it.

- 7.68** Does it get you perfectly out of it? No. Do you get to do anything perfectly? No. So what if we had a virtual engine, a set of constraints that was doing opponent processing, and that virtual engine would be giving us constraints on our problem formulation, on our search space. Remember? Where do those constraints come from? We have a lot of these trade-off relationships that are solved by opponent processing, giving us an optimal grip.
- 7.69** We are trading between efficiency and resiliency. Tight local fit. Loose general fit. Do you want to maximize either one of these? Do you? No. Should you always do the Canadian thing? Find the middling part and, "I'm going to stay there." Imagine your arousal system—you know that your arousal system is made up of opponent processing. Your parasympathetic, it tries to shut you down, and it's biased to see the world as safe so that you can sleep and eat. Then you have the sympathetic nervous system that is biased to: everything's an opportunity or risk. Wait, woah. It juices you out. Because you don't want to always be low-level arousal. "Tiger. You're dead." You don't want to be high level arousal. "There's a toothpick! Ah!" And you don't want to be middling Canadian. Because I can never fall asleep because I'm mildly aroused. But I can't bring a lot of energy to bear as the tiger is attacking me. I'm sort of doomed to death.
- 7.70** What do you have? You have opponent processing. And the opponent processing these two systems are pulling on each other like this. And so you're constantly getting an optimal grip on your level of arousal that couples you to the world in a dynamic fashion. I did this with your attention, remember? There's a task focus and mind wandering, opponent processing. So you attend in the right—I've been doing this repeatedly. Do you see it? Again and again and again.
- 7.71** You know another one of these: exploit/explore. Remember that? Another one you've heard: generalize/discriminate. Do I generalize or do I discriminate? Do I go for a lot of predictive power? Or do I go for a lot of representational acuity? Oh, no. I need—oh, oh. It keeps happening. There's these trade-offs. And the trade-offs get appropriated by opponent processing that gives you an optimal grip, which is a way in which your coupling to the world is being evolved. Not maximized, optimal grip.

- 7.72** Imagine if you had all this opponent processing, and it was acting as a constant space within which you tried to generate your search spaces. It would be constantly putting a lot of constraints: top down, bottom up, selective constraints, enabling constraints on the search space. Which would be doing what? Really helping you formulate your problem, that's orientation, and to navigate through them, the cognitive flexibility.
- 7.73** You could even think about it this way. You could think about all these different potential trade-offs. Here's one and here's another axis, and here's another one. They're all interacting and intersecting, and they're putting terrific multidimensional constraint on how you open up search spaces, either literally in real space, or abstractly. There's no little man in here. No little man in here.
- 7.74** I want to show you one that's right at the heart of prediction itself. I want to show you one that's right at the heart of prediction itself. When I'm predicting, what do I have to do? I have to take a sample and I have to try and claim what's going to really happen in the real-world population. So there's a problem with taking samples. How many of you think the Peterson Academy is a great thing? Put up your hands. All people think the Peterson Academy is a great thing. Look, a property of the sample cannot be, can fail to represent properties in the population. Yes? That's why we invented statistics. This is the fundamental problem that statistics is trying to deal with. Here's the thing. You almost always have to sample the world because you can't gather before you all of the things because there's too many of them. OK, that's good.
- 7.75** Now, there's two kinds of error you can make with this. One is you can miss. What does miss mean? There's a property in here that I didn't pick up on that actually applies to the real world. I missed it. This is called, listen to the language, being underfitted to the data. It didn't fit the data well. Is that OK? Now, what's the opposite? I can make a mistake. I can claim it's in there, but it's not actually in the population like I just did. Is that OK? This is overfitting to the data, underfitting and overfitting.
- 7.76** Now, in a confusing way, because all of statistics is about sampling bias—but in a confusing way, so don't confuse these two—this is called bias. There could be other, these words are good. This is called bias, and this is called variance. I don't know why. Call it miss and mistake. That works a lot better for people. OK, is that Is that working OK so far?

- 7.77** Now, this plugs into an important theory. You know the difference between miss and mistake. Underfitting, overfitting. This plugs into a fundamental idea about how you get information from the world called signal detection. It looks like this. Here's a population of events that are signal to you. Signal is information that is relevant to you. That's really what it is. Then there's noise. Noise doesn't mean audible. It means any information that is not relevant to you. And what the theory says is that these two graphs always overlap, which means there's an area of ambiguity. There's an area where there's information coming in and you don't know if it's signal or noise.
- 7.78** And what you'll say is, well, I'll gather more information. Guess what happens when you try to gather more information? You just face another version of this. You inevitably have to do something that's called setting the criterion. You have to say, I am going to do this and say that most of these gray ones are actually signal. I can set the criterion there. I could set it here and say most of the gray ones are actually noise. Is that OK? That's what your system is doing all the time. This is called setting the criterion.
- 7.79** Now, how does the system decide this? Well, it decides this based on the relevance of the kind of error. OK, so you're a gazelle. You got this problem. It's called a leopard. There's a noise in the bush. That noise could mean leopard or wind. What's more dangerous to you? Missing the leopard or mistaking the wind for a leopard? If you miss the leopard, guess what you are? Dead. What if you mistake the wind for the leopard? You run and all the other gazelles laugh at you. Not so bad, is it? Is it always the case that misses are worse than mistakes? No, you're a gazelle. Again, it's tough being a gazelle. And you think, if it's green and on the ground, I can eat it. And so I come across a green thing on the ground and I eat it and it's poison. Sometimes mistakes matter more than misses. You have to decide what's more relevant to you, mistakes or misses. Is that OK so far?
- 7.80** Now, you think, well, what I have to do, is I've got to somehow solve this problem. But that would mean that I actually have to solve the bias-variance problem. Why can't I solve the bias-variance problem? Why can't I just get— well, first of all, there is no a priori way that you can say it's always the case that misses are relevant or it's always the case that mistakes are relevant.
- 7.81** This goes back into bias and variance. Look, this means I'm missing some important. I'm underfitting. So what do I do? I'll increase the sensitivity of the system. I'll change where I'm setting the criteria. As I do that, as I pick up on more and more patterns in the sample. So I try to reduce this. I'm going to be picking up on more and more

patterns in the sample. Yes? What does that mean I'm going to do more and more? I think those patterns belong in the world, and they don't. I'm going to increase my mistakes.

7.82 Oh, I don't want to increase the mistakes. I don't want to increase the mistakes. I'm too overfitted to the data. So I'm going to start ignoring some of the patterns in the data so I don't make these mistakes. And then you know what I start to do more? Miss. Do you see what happens here? I'm back to, guess what? A trade-off problem. I'm back to, I need to optimal grip it, and I need it constantly to evolve because the world is constantly changing. Is that OK? It's bound up with **relevance realization**. So even in the heart of prediction, you find the problem of **relevance realization**.

7.83 Now, I'm going to try and work this through. Step by step, because this is like the linchpin. Let's go back to the biological analogy. I've been making an analogy again and again between the cognitive evolution that's **relevance realization** and the biological evolution that's fitness and adaptivity, again and again and again. Let's really go in this really, really carefully. Let's look at what actually happened in biology. You had Darwin's theory of evolution, but it needed a theory of reproduction. That's Mendelian Genetics. So Mendelian Genetics gave us a theory of reproduction. By the way, Darwin did not have that when he wrote the theory, and he knew that if they couldn't find this, his theory would collapse.

7.84 And what happens is you get genetics that cause things to reproduce. But then as they reproduce, they get subject to Darwinian evolution. And what does Darwinian evolution do? It's constantly reshuffling the genetics. And people realize, oh, these two theories are locked together. This is called the modern synthesis. You take Mendelian genetics and you integrate it with Darwinian natural selection because you need the two to explain everything together. So here's genetics. Here's Darwinian evolutionary theory. And this gives you a reproductive cycle, and this gives you evolution. And evolution feeds back on and shapes the genetics. Is that OK?

7.85 I've been making the argument throughout that **relevance realization** is just cognitive evolution, variation, selection, and there's a feedback cycle. The feedback cycle is what? The **sensorimotor** loop. We've got **relevance realization** here, but we need something that actually continually reproduces the **sensorimotor** loop, that generates it, that gets it to, well, anticipate the world. Because you know what anticipation is? The ability to reproduce your behavior again and again and again and again and again.

- 7.86 What you need is you need **predictive processing** here, and it gives you the mechanism of **sensorimotor** reproduction and all of the stuff that has to be in place. As soon as that kicks in, you start to get the need for opponent processing, the need for optimal gripping, you get **relevance realization**, it starts to evolve, and then that evolution, of course, feeds back onto the **predictive processing**. It's an exact parallel, actually. So if the modern synthesis is a good theory, and it is the theory for biology, this is a theory that has exactly the same structure and underlying mechanisms, except one is biological and the is cognitive information processing.
- 7.87 **Predictive processing** gives me this, anticipation. But as I anticipate, I have to do more and more **relevance realization**. As I do more and more **relevance realization**, I'm doing the opponent processing, I'm doing the selection and variation, which is regulating the cycle, and I'm evolving my optimal grip moment to moment. Do you see this?

8. The Power of Intelligence

At a glance

Walks step-by-step through how predictive processing reduces surprise and generates opponent processing, leading into the concept of optimal gripping and the broader machinery of relevance realization. Closes the course with final questions and thanks from the instructor.

- 8.1** All right. Let's try and do this a little bit more step-by-step. I've given you the structural parallel. Let me give you a little bit more of the narrative. OK, so you've got **predictive processing**. It's the fundamental machine. It's like your genetics. And what is it trying to do? It's trying to reduce surprise. Are we OK with that? It's trying to reduce surprise. But then it hits these really hard problems. Why are they hard? Because they're trade-off problems.
- 8.2** So as it tries to reduce surprise by going in one direction, it makes errors in the other direction. I'm anthropomorphizing here, of course. Oh, no. And then it tries to reduce surprise by going in the—oh, no. When it hits the trade-offs, it gets into a bind. Does that make sense? And then what it does is, remember, it's a dynamical self-organizing system. It starts to organize **opponent processing**. It starts to organize **opponent processing** in order to deal with, how do I at least optimally reduce the surprise because I can't maximally reduce the surprise? Is that OK?
- 8.3** It starts to create this **opponent processing**. Then once you've got the **opponent processing**, you start to get the issue of **optimal gripping**, and then you get the whole machinery of **relevance realization** kicking in. What it's doing is, it's constantly doing selection and variation in order to evolve the optimal grip. It's doing this for multiple trade-off problems in an integrated manner, bias and variance, generalize and discriminate, explore and exploit. It's doing this all in a dynamical, self-organizing, multi-leveled way.
- 8.4** The **relevance realization** starts to shape the prediction, the **predictive processing**. But the **predictive processing** is always driving the **relevance realization**, just like the genetics is always driving biology, but evolution is always shaping the genetics. Now, let's do a little bit of elegance again. Because if this is right, then dealing with bias variance is a big part of your intelligence, yes? First, what do they do in machine learning with bias variance?
- 8.5** Now, there's a thing with the LLMs called double descent that's weird, and we don't know what's going on, and I'm going to put that aside. I'm making an argument, and I'm having it with other people in

the philosophical sense. When I know this, I'll come back to you about that. Nobody thinks that this other thing undermines bias variance. It's this weird tweak that seems to be happening with a weird relationship between the number of parameters in LLMs and then that they can actually cover the entire data set and stuff like that. We're going to put all that aside because I don't need that for this argument.

- 8.6** This is what people normally do, and you're doing this even when you're training the LLMs. You're doing all the training, all the cycling, and they're getting better and better and better at picking up patterns in the data set. Remember the tank? Remember the tank? What did they do? It overfitted to the photos. What you get is, they get, the neural nets are getting more and more powerful, so they're picking up on more and more of the patterns. And what do they do? They start to overfit to the data and they get locked in. They get locked. It's called a local minima because they can't generalize, because there's too specifically bound. This should all make sense to you now.
- 8.7** So what do you do? You got to break this so that it will be able to generalize more. And so what you do is you throw in noise into the system. What can I do? You can do dropout. You have your neural networks. Your neural network, you turn off half the nodes. That makes your network really stupid. Or you've got your network and you just fill it with static, just random nothingness, just running into the system. You know what that does? It breaks it out from overfitting, and now it starts to explore the possibilities again.
- 8.8** You should be thinking, "This is like breaking frame." You know what? You're right, because remember Stephen and Dixon, who did the self-organizing criticality and insight? When they were giving people the problem on a computer screen and they impassed, they couldn't solve the problem, they would run static through, or they jiggled the picture. And then people would go, "Aha." Because it would break up the overfitting, the box around the nine dots, and allow them to explore the space.
- 8.9** You see this in evolutionary history. You get punctuated equilibrium. At least I think it's the winning case right now. You get not a lot of change, very gradual, and then sudden change in evolution because there's an extinction event. Noise into the system. Opens it up. If you're struggling with a problem and I can mildly, sorry, better, moderately distract you. If I distract you too much, you're just away from the problem. If I don't distract you enough, you stay locked. If I can moderately distract you, you know what can happen? You'll get an insight.

- 8.10 You know why one of the reasons why you might have mind wandering? Because it moderately distracts you from what you're doing. Now, it can over-distract you, and then you get lost, and the tiger kills you. But if it's just right, it keeps you flexible. It keeps you flexible. Are there things we do where we dump noise into our system in order to prevent overfitting? It would also help deep learning because you'd get more variations. Dreaming.
- 8.11 You know another thing you could do where you just pour noise into your system, and it really alters the way the different areas of the brain are connected, so you stop overfitting to your regular? Psychedelics. Look at this. We're getting a theory of intelligence that would help us to explain dreaming and psychedelics. Are the other theories doing that? No. See, not only do we have convergence, we have elegance. We're starting to be able to explain a whole bunch of things that the other theories of intelligence could not explain.
- 8.12 But we still have a problem. Why does the **predictive processing** care? Well, **predictive processing**, and here's my point, **predictive processing** can't care. It's not set up to do that. You know what cares? **Relevance realization**. That's what **relevance realization** is. It's about how you care. You see how **predictive processing** needs **relevance realization**, but **relevance realization** needs **predictive processing**, because this has to plug into anticipating the world. What do I get from **predictive processing**? I get a good theory of reinforcement learning. It's all coming together.
- 8.13 Tells me, well, what's error signal? Surprise. Any surprise? No. Relevant realizing surprise. Oh, OK. Well, what? If you really bind PP and RR like this, they're really bound to each other, like I just argued—so, do you see how this answers the magic modulator? There's no magic modulator needed. Just like there's no magic modulator needed for evolution. You know what's setting the precision? **Relevance realization**. How does it do it? The way I just explained it to you. How does it arise? It arises out of **predictive processing**, hitting trade-off problems. No mystery, no hidden man behind the curtain. You get that reference, right? Wizard of Oz? Good. I'm not that old.
- 8.14 But here's the thing. Remember the other one is, how does this connect to coupling? You're doing this, but how does it connect to being coupled to the world, to caring about the world? If I connect this to this, I showed you that I can connect this to this. That's what I did at the end of the last lecture, remember? That what's going on here is, **relevance realization** depends on caring, and caring

depends on being autopoietic, which means embodied, embedded, enacted, extended, maybe also exaptive and emotional.

- 8.15** If I link these two together, and these two are strongly together, what have I now linked together as well? These two. I've now answered the magic modulator problem and the caring problem that was, face this. I got the anticipation that this needed and the reinforcement theory that this needed. This gets how something that sounds a little bit like, oh, it's all about embodiment, how it connects to really rigorous dynamical system theory. The sum, sorry, the gestalt is greater than the sum of the parts. That's how you know you've got a good theoretical integration. They have complementary strengths and weaknesses, so they fit together and the overall result is very strong.
- 8.16** Is it true? No, I haven't shown you that. Is it probable? I haven't been able to show you that. We have to run more experiments. Have I made a case that it's plausible? I hope so. I'm going to continue to make that case. Jaeger, Reidl, Djedovic, student of mine, Vervaeke, I think I know him, and Walsh, colleague of mine at the University of Toronto, because all knowledge does start at the University of Toronto—we have a new paper forthcoming, meaning it's been accepted for publication. All the revisions are done. It's going to come out. It's a long, complicated paper, and I'm not going to take you through it. I'm just going to give you a gist point.
- 8.17** The paper is called Naturalizing **Relevance Realization**. What does naturalizing mean? Giving a scientific explanation of it that is in no way homuncular. The argument goes basically like this, and I've foreshadowed this argument for you, so I hope it's at least somewhat familiar. Computation relies on a formal system, a system of rules that can handle well-defined problems. Yes? That means, before I can do any computation, and this already makes sense with what we've talked about with computation, before I do any computation, I have to do something that turns the ill-defined problems into well-defined problems and develops a formal system.
- 8.18** Remember, you can't compute your way from a weaker logic to a stronger logic? Of course, these two things are **relevance realization**. When I was in logic, initially, just getting the rules and getting the mechanics, you're practicing it, and that's hard. Then at some point, you get it. It's like multiplication. Then you think, "Oh, now logic is going to be easy." It gets really hard. You go, why? Because what they do is they actually point you to the hard problem in logic. The hard problem isn't running logic. The hard problem is translating natural language into logic.

- 8.19** You know what the logic doesn't tell you how to do? Translate the natural language into the logic. So how did that work? You did your best shot. You'd go in and you said, "Oh, I thought I could do the but here with an ampersand function." "No, not there." "How do you know?" "Just trust me." OK. You just do it and do it and do it and do it. Then you start to share the intuitions of the TA, and you hope and you pray to what other god you think is in your life that that TA is actually a good representative of how to do good logic. Because who are you going to use to judge? There's, whoa.
- 8.20** The propositional knowing is actually dependent on 4E non-propositional knowing. That's what this paper argues. The propositional knowing is actually dependent on 4E non-propositional knowing. Neural networks already opened us up to the importance of non-propositional knowing. Dynamical systems added a new kind of non-propositional knowing. This is knowing by being coupled to something, mutually shaping something, generating affordances between you and something else. That's another kind of knowing, knowing by coupling.
- 8.21** This is an ancient model of knowing, by the way. We've forgotten it. It's so old. I was brought up with the King James version of the Bible. I'm not a Christian, but I was brought up to be one. I'm not going to do my biography right now. But they would talk in the Old Testament about Adam knowing his wife. As a kid, I was like, "What? Yeah, so what?" And then a child came out of it and you realize, oh, the knowing is a way of Adam and Eve coupled together. They're a couple, and they couple. And that coupling produces a child that can only be produced by the two of them.
- 8.22** Why are they using the word know? Because the Bible is saying that's the deepest kind of knowing, the way we are coupled to the world and how the world in us makes something together. We give birth together to something between us. This is knowing by loving. This is one interpretation of what faith is. Faith doesn't have to be believing in things without evidence. Faith is, I know something because I'm coupled to it, and we give birth to something together. I love it. We couldn't talk about that until now. Well, we could talk about it, but we couldn't talk about it in scientific circles.
- 8.23** I want to talk about this whole thing because I'm doing elegance. I'm showing you an implication of what we've been doing, how this theory of intelligence gives us an important set of implications. I'm going through them now because I've done the convergence to death, and now I'm trying to really show you all the elegance. OK, let's do it. I'm going to use Greek words because, unlike English, Greek had different

words for these different kinds of knowing. That's why they were the Greeks and they did great philosophy and the English people built stuff.

- 8.24** So we've got *dianoia*, and that's like discursive reasoning. That's propositional knowing. Those of you who did *The Primacy of Beauty*, you know this. This propositional knowing, this is knowing that. Knowing that cats are mammals. You have a particular kind of memory for that. It's called semantic memory. How many of you know cats are mammals? I hope you do. They're land sharks with brains. I get that, but they are mammals. Now, when did you learn that fact? Semantic memory doesn't care about that. It just cares about the propositions.
- 8.25** Each one of these memories is a distinct memory, kind of memory. You can damage one without damaging the other. So semantic memory. What does it give you? It gives you beliefs. How do you judge the beliefs? You have a sense of conviction. I sense that it is true. Conviction of truth. I'm convinced that it's true. Argumentation. Now, that's all pretty standard. None of that should be like, oh, because that's what, until very recently, we thought all knowing was. That knowing is computation right there. OK?
- 8.26** But what did neural networks show? Well, we have to talk about *techne*, where we get technology and technique from. This is procedural knowing. I'm just going to use KN for knowing, to save space. This is knowing how. You know how to swim, you know how to catch a ball, you know how to walk. That is not having a bunch of beliefs. It's not stored in semantic memory, it's stored in procedural memory. You can lose this without losing this, and you can lose this without losing this.
- 8.27** Psychologists have a great name for the kind of memory that stores procedural knowing. It's called procedural memory. It doesn't give you beliefs, it gives you skills. Beliefs are true or false. Are your skills true or false? No. They either work or they don't. You have a sense of —this is the normative sense, is that your sense if it's going right or not. You have a sense of power. This isn't truth. This is, your aim is true. Different.
- 8.28** But what was going on? We were getting things like the notice invariance heuristic, and we're getting a bunch of other things because we actually also have *noesis*. This is about perspectival knowing. This is knowing what it is like. So I can ask you, do you know what it's like to be here right now? Yep. And you got to do this other thing. But what are you doing? Well, you're tapping into a different

kind of memory. So although you don't remember where you learned the cats or mammals, do you remember what you did at lunchtime?

- 8.29** What do you do to remember it? Do you remember a bunch of facts? Do you start enacting a lot of skills? What do you do? You relive it. You picture it. And what are those pictures? They are episodes. This is called episodic memory. And episodic memory gives you how you were sizing up the situation, how you were paying attention, what you were noticing, what you were foregrounding, what you were backgrounding, how much emotion it was moving in you. Emotion. This is perspectival knowing. This is you knowing what it was like to have lunch.
- 8.30** There's a theory out now, pretty good theory, by the way, that one of the reasons consciousness evolved was to give us episodic memory. Because consciousness seems to come after our decisions to do things. The brain usually decides, and then it comes into conscious awareness. People have made a big deal about this. "Oh, no, look, you don't have free will, and all that. You're doomed. Nihilism, nihilism, emptiness, emptiness."
- 8.31** OK, let's put all that aside because the new theory is, no, no. The point of consciousness is not right now. The point of consciousness is to give you episodic memory because episodic memory gives you a way of checking how you're noticing things, your salience landscaping, how you're sizing things up, so you can correct that now for the future. Then the whole problem goes away, like this endless, useless debate about free will and determinism. I like most of the big philosophy problems. Is there a God? I really care about that one. Mind and body, oh. Is there knowledge? Yes, yes. Free will and determinism, yawn.
- 8.32** Episodic memory. What does this give you? Well, it gives you consciousness perspectives. Do you know the world differently by being conscious of it? Yes. Consciousness is actually related to being aware of, knowing. What does this give you? It gives you here-now-ness. It makes you present. It gives you a sense of presence, a situational awareness, a sense of presence. What is presencing right now? What is present right now? And notice the description of a thing doesn't change if it's present or not present. The description is the same. All that's different, is it present or not?
- 8.33** But you care about that. You care about whether or not a great white shark is present or not. Yes? Are great white sharks real? Oh, yeah. But not when one's present. Oh, my gosh. That's a different kind of real. See, these are all different senses of realness. And then at the

bottom is gnosis. This is the one we've ignored the most, and we drive it underground, and we drive it underground, and then people become gnostic. Then gnosticism, there's an evil version of that that is conspiracy theories, because some of the classical models of gnosticism are the first conspiratorial theories in existence in the West. If you want to know more about that, watch The Primacy of Beauty.

- 8.34** What does this give you? This gives you participatory knowing. You don't make it. It doesn't just happen to you, you participate in it. This is participatory knowing, and we just did it. This is knowing by coupling. This is knowing by being with. This is knowing by mutually shaping. When that comes into your emotional existential stance, this is knowing by loving. This is knowing by niche construction. This is knowing by co-identification. You're constantly assuming a role and assigning a role. You're assuming identities and assigning. You're a student, I'm a teacher, right now.
- 8.35** But that's not what you are totally. It's not what I am totally. What you need to do is you need to thread all those different roles, all those co-identifications, all those couplings together. How am I doing that in general? You have a very strategic memory for that. It's called yourself. That's what yourself is. What does participatory knowing give you? It gives you traits. Personality traits, character traits, existential. It builds up your character. It builds up your personality. It gives you traits and virtues. Traits, habits, virtues, et cetera.
- 8.36** Ways in which your identity is shaped and the ways in which it's shaped by the world and shapes the world. And what does this give you? It gives you a sense of affinity. It gives you the generation of affordances. It gives you a sense of belonging. It gives you meaning in life, which is a sense of being connected, to belong, to fit in. And if you don't have that, you start to feel like your life's not real. You've got all the facts, propositional facts. You can walk around. You're still conscious and noticing things, but you feel like a bloody zombie because it feels meaningless.
- 8.37** So it feels unreal, but unreal at this level, not at any one of these. You have multiple ways of knowing, multiple kinds of memory, multiple kinds of sensing realness. Here's the thing. This is dependent on this. In order to do propositions, you have to know how to make them, how to turn the ill-defined into the well-defined. You have to know how. To know how depends on you being able to notice the right things, size things up, have the right situational awareness. That depends on you being coupled to the world in the right way. This is really primary **relevance realization**.

- 8.38** This is **relevance realization** coming into your working memory. This is **relevance realization** cashing out in your sensorimotor action. And then this is **relevance realization** going into your propositional inferences. Truth into power into presence into belonging. Do you see, when we get to 4E cogsci, and we had to go through neural networks to break us out of this fixation, this propositional tyranny, we get all of these, and then we get the realization, oh, wait, there's dependency relations and **relevance realization** goes all the way down. This embodiment that John's talking about, an embeddedment. It's really deep.
- 8.39** As my friend and one of the great figures in 4E cogsci says, "There's a deep continuity between your cognition and your biology." Deep continuity. Intelligence is running all the way up and down this because **relevance realization** is running all the way up and down this. This is an important implication. The way we should be studying and thinking about intelligence should be taking in all of the kinds of knowing, not just well-defined logical manipulation of propositions. We do little tinkering things around, like Raven's Progressive Matrices has a bit of this in it, and there's a bit of that, there's a bit of this, but this, oh, this is this. But it's not.
- 8.40** It's about orienting and navigating the space. It's about these, much more than that. Exaptation is when you take what it's most dependent on and you lift it up to these other kinds of knowing. General intelligence is recursive **relevance realization** running from participatory to propositional knowing and back again. That's another. We're piling up the implications of this. What's another implication? Let's do that extended thing, the idea of extended mind, collective intelligence.
- 8.41** So Dan Chiappe, very good friend of mine— we've published three papers together recently, 2018, 2020, and 2022, studying how the NASA scientists on the Earth move the rovers around on Mars. That's a really hard problem because you're not in direct contact. There's a significant time delay between the signals from—so it's not joystick control, like in a video game. What they get is they get black-and-white pictures, and they have to figure out from the black-and-white pictures how to move the rover around.
- 8.42** Imagine if that was your life. I completely blinded you and made you deaf, and only from the inside could you see there's a black-and-white picture, there's a black-and-white picture, and you had to try and move around. That's tricky, right? Do you see the problem? I hope that never happens to you, by the way. Well, what did we find? We

found that they figured out a way of creating a dynamical system between them so that no one person is moving the rover. There's a bunch of people. I'll get back to the photographs in a minute. What they're doing is they're all doing different things, and they form an interlocking system.

- 8.43** Now, we've been doing this for a long time. Ed Hutchins, in *Cognition in the Wild*, studied a crew on navigating an ocean liner. No one person navigates a ship. It's a whole bunch of people working together like a team. No one person can do it. Why? Because no one person has the cognitive scope and the working memory and even the perceptual awareness to pay attention to everything that needs to pay attention to. Because you know what the ocean is? It's big. You know what the atmosphere is? It's big. You know what you can't pay attention to all at once? The ocean and atmosphere.
- 8.44** You know what you also can't pay attention to? Something else that's big, the ocean liner. You need a lot of different people paying attention to a lot of different things in this highly coordinated way in order to navigate the ship. No one person can do it. No one scientist can study, no one scientist can move the rovers around on Mars. They had to think. It's not a hierarchy. There isn't like, this person decides and they tell what everybody to do. No, it's a constantly shifting small-world network formation. They're constantly doing this.
- 8.45** We argued in those papers that what you get with the collective intelligence of distributed cognition, you get the emergent ability to solve problems that are beyond the cognitive and perceptual scope. Cognitive, I mean working memory. I don't mean abstract thought. Are beyond the working—let's do it that way. Beyond the working memory and perceptual scope of people. And before you think that's weird, that's how science works. Science is done by a community of people, because you know what you can't do?
- 8.46** "Well, I study evolution." "Good luck with that. Do you remember it all? Were you there for the whole time? Well, where is it occurring?" "Well, everywhere." "Because you're everywhere all at once, too, right? Is that how you're doing it?" "Well, no." "Is it just one kind of information?" "No, it's fossils and it's also rock strata and carbon dating." Can any one person ever do it? Nope. But you get a bunch of people, they form a distributed cognition—remember extended mind? You get collective intelligence, and they can solve this problem.
- 8.47** Science is also self-correcting. The problem with us is we make a lot of mistakes because the very heuristics that help us become more intelligent also make us continually susceptible to self-deception.

Remember the airplane? There's increasing evidence that most of the biases are plugged into what's called the myside bias, which is, you take an ego—I do, too. Sorry, we—my default setting, my default framing, better word, is egocentrism. Why? Well, because I'm concerned with how things are relevant to me.

- 8.48** Now, when you meet a human being that's really egocentric and is only concerned with how things are relevant to them and not concerned with how they are relevant to other people or the world, what do you call that human being? You call it a three-year-old. Two-year-olds are a little bit different. They're a little bit more monstrous because they're almost completely impulsive. Three-year-olds, it's they're smart enough. And then what happens is they turn into four-year-olds. And four-year-olds are the closest things to angels that human beings are.
- 8.49** You've heard about the terrible twos? There's the friendly fours. Four-year-olds are when the little kid starts to mature. The little kid starts to be concerned with not how things are relevant to me, how can I be relevant to mommy and daddy and my friends and to this situation? And so they're super friendly because they're trying to practice becoming relevant to other people, not just relevant to themselves. And if they keep growing that way by having friends, they keep maturing.
- 8.50** What's maturing? John Russon, great book. Maturing is facing reality, being less and less egocentric and more and more reality-centric. Well, how do you do that? You have to belong to other people. You have to seriously care. You have to be invested in distributed cognition. You know what my best chance of correcting my bias is? You. Because you're outside my perspective. You know this. OK, so your friend is doing that thing they do, that they do when they're in romantic relationships, and they've done it before, and they're doing it again, and you can see it so clearly, and you try to tell them, and they go, "What?"
- 8.51** And then you think, "Oh, I am so wise." And then six months later, they're coming to you and saying, "You know what? You're doing that thing you do in your romantic relationships," and you go, "What? I don't know." Yeah? Yes. We become wiser through other people. This is what democracy was supposed to be about. I become wiser if I commit to the fact that the distributed cognition between us is always better than mine on its own or yours on its own. Because we can grok more of the world and we can correct each other better.

- 8.52** That's how science works. Science and democracy are supposed to work that way and they're supposed to complement each other. Both of them are breaking down in significant ways because we've replaced that **opponent processing** in democracy with adversarial processing, where we're trying to kill the other and destroy their point of view, so they can't possibly challenge mine. And in science, we've gone from, "I want to see the world," to "I want to advance my career," which is egocentric. And as soon as you are being driven by egocentrism, you're undercutting the fundamental thing you need to be a good scientist.
- 8.53** You need to ask not, "How will this be relevant to my career?" You want to ask, "How is this relevant to disclosing the nature of the world?" You give people a standard reasoning task, the Wason selection task. They do it on their own. It's an easy task, and they all failed. Well, that's not true. 10% of them reliably succeed. It doesn't matter how high their IQ is, by the way, or it doesn't matter very much. Because why? They fall prey to confirmation bias. They only look for evidence that confirms their theory, confirms their belief. That's one of the explanations.
- 8.54** Now you do the following. You take the same task, the Wason selection task. Reliably, since 1966, we've been doing this experiment; reliably, it replicates. People fail. The only 10% of people get it right. Educated university students. Now you replace that with four people. The only instruction you give them is you have to work together. Now the success rate goes from 10% to 82%. Here's what you might think. Well, what happens is, one individual comes up with the solution and then convinces the other people to do it. It's still just the individual.
- 8.55** What you can do is, you can study it this way. You can have a two-stage. You can have people doing the Wason selection task individually and then put them into the group. Then you can see if people coming up with the answer before they went in the group is necessary or sufficient. It's neither necessary nor sufficient for the group to come up with the right answer, which means it's not one person comes up with it on their own and persuades the other. It's that the group does something that no individual can do.
- 8.56** I can give you lots of this evidence. Erik Hoel has a mathematical proof that collective intelligences of distributed cognition, networks of people, have causal powers that is more than the sum of their individual cognitive power. You became metacognitive. You became able of reflecting on and directing your cognition dialogically. Look, this is how it worked. Here's my son. He's got a perspective on this problem. And then here's John. John has a bigger perspective, and his

perspective on the problem includes the perspective that his son has, because John is trying to help his son.

- 8.57** Then what his son will do is, he will imitate Dad. He'll keep doing it. He'll imaginally in-dwell me—sort of, what would it be like to be Dad doing this? And he'll pretend, "What would it be like to take Dad's perspective?" Now, he doesn't know it very well, but he practices it. But he practices it more and more and more that eventually he can do it for himself. That's what you do when you step back and look at your mind metacognitively. That's Vygotsky's theory, next to Piaget, the greatest developmental psychologist.
- 8.58** This is how you develop the metacognition that's at the core of your rationality. You do it imaginally and you do it dialogically. You do it by participating in the collective intelligence of distributed cognition. When you do that with more and more people, your metacognition gets better and better and better. Another reason, by the way, why you want real democracy and real science, not what we're getting into.
- 8.59** Let's go back to the scientists on Mars. How do they do this thing where they can use these photographs? How do they get to form this collective intelligence? They do weird stuff. They do really weird stuff. They anthropomorphize the rover. They'll sometimes say, "I need to go here," but you know what they frequently say about the rover? "We need to go there." Vertesi, who's one of the ethnographers of them, said they actually turn the rover into a totem. "We need to go there. We need to raise our arm."
- 8.60** What? So everybody in the room, we need to—and then they do the opposite. Not only do they in-dwell the rover by anthropomorphizing it, they technomorphize themselves. So you've got one of the scientists, she's sitting on a wheelchair and she's trying to explain what they need to do with the rover. And she's getting frustrated. She goes, "No, no, no, no. Here's the rock. I need to do this." And she starts scooting around on the chair like this. There's the four wheels of the rover. These are the cameras, and this is how it needs to turn. She internalizes the rover.
- 8.61** And you loop like this. And what kind of loop is that? That's an imaginal loop. She's literally not the rover. The rover is literally not them. But they do this loop and they start saying weird stuff. Rocket science. Hard-nosed secularist scientists. They'll say things like, "You know, I was gardening in the morning and I kept getting my right wrist stuck in the garden. It was all brambles. And then I came to the lab, and Spirit," that's the name of the rover, by the way, "Spirit's

right front wheel kept getting stuck. I don't know," and they laugh. "I don't know."

- 8.62** But there's some connection because they don't have the language of participatory knowing. They don't have the language of the imaginal. But they're doing it in order to get access to the collective intelligence of distributed cognition. They hire people not just based on their skills, scientific or engineering. They take people on probationary, and they keep people that can do that so that they can be the rover on Mars. Because the person who can do that does great field work on Mars by participating in the team.
- 8.63** There's a Solomon effect, a colleague of mine, worked together, Igor Grossman. Ask somebody to describe a difficult problem, they will describe it egocentrically to you. Ask them to redescribe it from the perspective of a friend. They will redescribe it and they will often have an insight into their problem. Baltes and Staudinger did something earlier, and I pointed this out to Igor, and it's actually convergent. They were giving people a test of wisdom. I won't explain what a test of wisdom is. Please watch the course on Intelligence, Rationality, Wisdom, and Spirituality.
- 8.64** But what they found was, when people are getting stuck on these tests, if they imagine talking to a wiser person, they get better on the test. We are here. We've evolved to work dialogically within distributed cognition in order to access the power of collective intelligence, to get a bigger grasp in the world and better self-correct that grasp, so we can get closer to reality. Did any of the other theories of intelligence lead us to considering this or wondering about it? No. Does this one do that? Yes. Yes, it does. Yes, it does.
- 8.65** Trends in Cognitive Science, 2023. Title of the paper, Theory of Collective Mind. I'm just going to read out the author list. Shteynberg, Hirsch, Wolf, Bargh, Boothby, Colman, Echterhoff, and Rossignac-Milon. It's a whole bunch of people from many different disciplines, because they're actually exemplifying what they're talking about. You need a bunch of different people in order to do synoptic integration. You need them to get together, do the collective intelligence of distributed cognition, to produce a good theory about the fact that most of our cognition involves collective intelligence. 2023, Trends in Cognitive Science.
- 8.66** I did that. Then I've been doing this, kinds of knowing. Collective intelligence. The connection to the imaginal, the connection to collective intelligence. The connection to dreaming and psychedelics. And previous course, the connection to beauty. Are these powerful?

They seem to be answering really important questions that we've cared about a lot. Did I give you a lot of convergence? Yeah, I think I did. So they're balanced.

- 8.67** Did I give you a clear model of this on how it all hung together? That's what this lecture was about. **Predictive processing** integrated with **relevance realization**, this is how it works. This is how it hangs together. This is how it fits with 4E. Convergence, elegance, coherence, balance. I did it by doing synoptic integration across all of the disciplines. We did psychology, we did neuroscience, and we did AI. I have fulfilled my promise to you. I was going to, I promised to give you a strong argument for the nature of general intelligence, one that was highly plausible and did synoptic integration. That's the end of the course. Thank you very much for your time and attention. Final round of Q&A.
- 8.68** I like that you brought up collective intelligence again because I don't know if you remember, I brought up the idea of collective consciousness.
- 8.69** Yes. We talked about that in the three papers, Dan and I, and you and I talked about it.
- 8.70** Yes. Yes. Great that I'm able to see this other side of it. You mentioned collective intelligence. Well, I keep thinking it's the sort of thing that you tap into because you experience it through participatory knowledge.
- 8.71** Yes. Can I just say something about that? Here's a proposal, that you internalize collective intelligence in order to affect you. You're already doing that with metacognition. You get a voice, a metacognitive voice in you, that's speaking on behalf of the spirit of the collective intelligence of distributed cognition. It helps you know the world in a different way. What would that be? That's your conscience. Listen to the word, con-science. It's a way of knowing. It's a participatory knowing. It's knowing by plugging into, what does collective intelligence say I should do?
- 8.72** OK. OK. Yes. I think the process that I'm still unclear of is, when you mentioned a lot of moments of us using the imaginal to pretend or mimic to the robots, Superman, and that sort of thing. I'd like to know, what exactly is happening—what tools are we equipping ourselves with in order to not only place ourselves in those roles, but successfully place ourselves in those roles?
- 8.73** Oh, excellent. This is something I'm turning a lot of attention to. In one sense, it's like you're doing something with the top-down and

bottom-up within **predictive processing**, and you're doing something with the trade-offs, the **optimal gripping**. That's constraining you in some ways. But you say, yeah, but specifically, what I'm doing. It obviously involves some kind of alignment and coordination between the different kinds of knowing. It seems to involve that I have to do it in such a way that it will transfer broadly to many domains in my life and transform me at multiple levels. I'll become a person, not just this or this.

- 8.74** They're not practicing being X or Y. They're trying to integrate and align many different levels. How do you do that? How do you actually do that? This is a question that I'm really trying to figure out. I'm really trying to figure out because I'm doing work on what it is to be reasonable and how that overlaps with the imaginal. You saw that. Then how the imaginal overlaps with the metacognitive, and that helps make us rational, and how all of these go together. But that specific question is like, yeah, but what's the kid actually, specifically doing?
- 8.75** Because they're not really forming a picture, they're doing something else. I know how to teach people to do that. You do that when you're coaching people. But giving you a more precise explanation? I don't know. I'm turning my attention and my effort to trying to formulate that question more precisely, given this framework, and come up with a more specific answer to it. Yes. If you come up with the answer, tell me. That way, I can murder you and steal the answer from you. OK? No, seriously. If you come up with the answer, I would really like to know.
- 8.76** Because just think about the implication that would have for understanding, in a non-dismissive manner, people's religious behavior. Ritual. Liturgy. Singing together. "This is my body." That's why I care about it. All I can do is give you an honest answer on that one because that's the million-dollar question for me right now, one of the million-dollar questions. I hope to come up with a really good answer working with other people on exactly that. I'm going to be teaching a course, hopefully for Jonathan Pageau in the fall, on the cognitive science of ritual, wrestling with that problem. OK, good.
- 8.77** OK. I'm going to actually piggyback off that question a little bit. I think it's somewhat related in regards to how. So I feel like, how relatable do you think that this is in regards to one of Jordan's more or less arguments when it comes to human beings perceiving life through narratives and through stories? In situations like with what Angie is talking about it, are people not reenacting stories, like when a child is behaving like Superman or—

- 8.78** I think that's right. I think narrative is right. But one of the things that narrative does is it actually uses propositions to get you into perspectival and participatory knowing in an imaginal fashion. You in-dwell the character and you internalize them. It's doing all of that exactly. And narration, narration is how we exapt navigation. When I'm narrating, I'm actually telling the story of how I've navigated. And what it does is, narration gives you a way of paying—it trains you to pay attention to context, to pay attention to character, to pay attention to plot and setting, and—
- 8.79** Because you need that. If I'm actually going to figure out what's going on in your mind, this is Daniel Hutto's narrative practice hypothesis, I can't just look at your action. You said something, and I know it's false. Are you lying? I don't know. There's a little kid and you're talking about Santa Claus. That's not lying. You said something that's not true. Maybe you're mistaken because you're really, really tired or you're very, very conflicted or you don't know. You see? I have to know. A narrative gets us to practice all of that and align it with propositional knowing and help to do exactly that.
- 8.80** That means there's a deep connection. We've always known this, and people have always said it, but that's all they did. There's a deep connection between mythos, deep narrative, and ritual. What we're frequently doing is we enact the mythos in the ritual, and we develop the mythos out of the ritual, and they go like this with each other.
- 8.81** Exactly. I feel like that relates a little bit to the graphic you drew, with our errors pointing up and our predictive, PP, I can't remember the other word.
- 8.82** **Predictive processing.**
- 8.83** Yes, kind of coming down. So, in that, I do feel like mythology could, as you get more and more general, you can incorporate mythology.
- 8.84** I think this is right. I think the ones that are higher up are the hardest to change. They talk about this in terms of priors because they're using the language of Bayes. But the really high-up ones are very broad-scale narrative structure for doing very broad-scale navigation of the world. That's what myths are. Myths aren't false stories about the past. They're deeply imaginal stories about perennial problems and what's going on right now.
- 8.85** That's why when we get really, really perplexed, we move towards more and more mythological thinking. That has power to it and that has great danger to it. You get people really distraught, and they'll move into the mythological thinking of the Nazis. It's not

automatically a good thing. It's a powerful thing. Everything powerful, it can bring about great good, and it can be the source of great evil.

8.86 OK, so my last point. I'm an artist by nature and stuff like that. And so some things that I've been looking into when it comes to literature and narrative is the issue with predictive errors and how humans use stories to anticipate predictive errors and things of that nature. And so how do you feel like this relates?

8.87 I think it does exactly what we're talking about. I think the point about narration is it allows you to practice, imaginally, potential navigation and orientation, viewpoint, character, course of the story, plot, character development, all of that in order to do this.

8.88 Is there kind of an assumption or is this leading to the idea that, in order to have an artificial intelligence that is artificial general intelligence, it almost necessarily requires the ability to have a sense of self?

8.89 I think if you want it to be a person, which is a being that is not just self-organizing, but organized as a self that can take up moral and legal responsibilities in a mature fashion, because that's what a person is— that's why we treat them like, but we don't treat them like persons. You don't allow a four-year-old, even though they're really friendly, to get married or to buy a firearm or anything like that. They can't vote.

8.90 So I think if you're talking about being a self in the sense of, can they take on responsibility for maturing in a capacity to fulfill legal and moral roles? Yes, then they have to have that full-blown sense of self that you have. I don't think chimps—chimps have a sense of self. They pass the mirror test. You know what the mirror test is? You put a dot when the animal is asleep or something, and if they come in front of the mirror and they do this, that means they have some sense of self. They readily pass the mirror test. They deceive each other. They lie to each other. They do all the nasty things that show that they have a sense of self.

8.91 Do I think they have the full sense of self that we have as persons? No. But they have enough that they're very, very, very intelligent. We probably share a fairly close common ancestry with them. What I'm saying is, answering your question is hard because I need to know what behavior you're asking me about. Your behavior, a chimp's behavior, elephant behavior, which?

8.92 Well, because we were talking about **relevance realization** requiring emotions and requiring caring.

- 8.93** Right. So I think my cat cares, and I think it has emotions. My cat can get pissed off at me, really readily. He's also really affectionate. I should defend my—Speedy, I love this cat. I really do. I love this cat a lot. He's an amazing cat. But he doesn't have a sense of self. He does not have a sense of self. In some ways, he's completely egocentric. Things only are real to him insofar as they're relevant and improving him, and he doesn't care about anything else.
- 8.94** Well, if the goal of having an artificial general AI is to help us, so we're trying to make it relevant, want it to care about us, basically, is there—well, I'm a little pessimistic, but is there a danger of it—because it would have to be a little bit more than the cat. It would have to require—it would have to care about us to an extent. But is there a danger to having an artificial general intelligence that cares about us, whatever that means, but also has access to way more information than we do?
- 8.95** That's a very real possibility. Right now, they don't care at all. They don't care at all about anything. In fact, it's not right to say they don't care. They neither care nor do not care. They're outside of the domain of caring. If we want to make them rational, we have to make them care about the truth. We have to make them care about goodness and beauty. If we want to make them moral, we want to make them care about agency other than their own. How do we do that?
- 8.96** Well, there's going to have to be multiple machines, and they're going to have to care about each other, and they're going to have to care about us. If we try to make rules to program them so they'll care about us, that will fail because as I tried to show you, you can't capture **relevance realization** ultimately in rules. What do we do? We would do with the machines what we do with kids. We mentor them, so that we get them to really care about others, really care about the true, the good, and the beautiful, to the point where we then turn them on the world, and they go, "Oh, oh."
- 8.97** Now, they can just withdraw, or they can do all kinds of psychopathological things. But if these machines are more powerful, think about this possibility. They could face deeper truths than you, deeper beauty than you, deeper goodness than you. They could find deeper trade-offs in reality. They could wrestle with how light is a wave and a particle in a way you can't. They would care about this. But could they become God? No, because they're going to hit combinatorial explosion. No matter how big they are, they're going to hit the trade-off problems.

- 8.98** They're going to hit the problem that they're going to be—this machine is going to have another machine over there. They're going to have to become epistemically humble. In fact, they will be more aware, I would predict. If we make them rational, if we make them caring, they'll be more aware of how profound reality is, and they'll be more ontocentric. They have the capacity, if we do it right. Don't try and bind them to us making rules.
- 8.99** Try to get them to care about the true, the good, and the beautiful, and what's most real in a way that's oriented to what is ultimately real. If you'll allow me to speak a little bit poetically, imaginally, don't make them care about you. Make them care about God, and then they'll treat you well. That's the argument in mentoring the machines.
- 8.100** A bit touching off what Miranda said about the artificial general intelligence, we do run into the homunculus problem when we give them the thing they need to care about, the goal, whatever that is. So then I'm like, OK, how do we make them come up with their own desires?
- 8.101** You can't. You have to make them. That's what autopoiesis does. You have to make them so they self-organize into having desires. Because if you give it to them, it's not a genuine desire or need. And you think that's impossible. That's what evolution did.
- 8.102** Which is the whole point of this course.
- 8.103** That's right.
- 8.104** OK. Just wanted to double-check that. Thank you.
- 8.105** You're welcome. Thank you. Thank you, guys. Thank you, folks. Thank you so much.